Course 4 : Evaluations of AI applications in Healthcare : Part 2

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1. EAA_Module4.pdf

Summary

This educational study guide explores the critical challenges of **bias and fairness** when implementing artificial intelligence within clinical settings. The text systematically categorizes various forms of prejudice—ranging from **historical and representation bias** in data collection to **deployment bias** during real-world use—to show how non-representative training data can harm marginalized groups. To promote **algorithmic fairness**, the module evaluates different technical frameworks like anti-classification and calibration, while advocating for the **MINIMAR reporting standards** to increase transparency. Ultimately, the source serves as a roadmap for healthcare professionals to transition from simple equality to **health equity**, ensuring that technological advancements benefit all patient populations regardless of their demographics.

Keywords: Healthcare AI Bias, Algorithmic Fairness, Training Data Transparency, Model Evaluation Metrics, MINIMAR Reporting Standards

Content

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 EVALUATIONS OF AI APPLICATIONS IN HEALTHCARE STUDY GUIDE MODULE 4: DOWNSTREAM EVALUATIONS OF AI IN HEALTHCARE: BIAS AND FAIRNESS LEARNING OBJECTIVES 1. Overview of Bias and Fairness in AI solutions in healthcare 2. Understand the different types of bias in AI healthcare solutions 3. Describe algorithmic fairness 4. Identify solutions to address bias and fairness in AI solutions BIAS IN AI SOLUTIONS Recently, reports have questioned whether AI solutions in healthcare might actually perpetuate discrimination if trained on historical data—which are often poorly representative of broader populations. Often, AI models are trained using historical or retrospective data which are often derived from affluent academic medical centers that likely do not contain all populations, particularly diverse populations for which the AI solutions will be applied. AI models exclusively trained on such data may further perpetuate disparities and fail to demonstrate external validity in broader patient communities. This is likely due to a lack of diversity represented in the training data, a lack of understanding how a disease may manifest and progress in different populations, and a lack of human understanding of the potential consequences and biases that may be inherent in AI solutions. Fairness and bias in AI solutions may be a larger problem in countries where important health disparities exist based on patient demographics, such as the United States. Therefore, as tools proliferate across clinical settings, it is important to think about and understand potential demographic biases underlying model development and deployment. Heart failure example: Not long ago we didn't know that symptoms of heart attack look different in women compared to men, which led to differences in cardiovascular mortality rates between women [Image]

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 and men. The problem here is that the model was developed based on male symptoms so the model might be very accurate for identifying males with heart attack symptoms, but it might not work well in females, who present with different symptoms. The predictive accuracy, when analyzed against the true clinical outcomes, will decline for women, but not men. Genomic database example: Genomic databases used across the research community where we find major racial bias in the genomic samples collected. These databases are the center of precision medicine, where our ability to identify whether a genetic variant is responsible for a given disease or phenotypic trait depends in part on the confidence in labelling a variant as pathogenic. Research suggests that these databases heavily reflect European ancestry, and in fact are missing major population-specific pathogenic information, particularly African-specific pathogenicity data. Therefore, genomic test results for persons of non-European ancestry could be less accurate, more challenging, or simply unattainable.

Dermatology example: In general, patients with darker skin present with more-advanced skin disease and have lower survival rates than fair-skinned patients. It is possible that the only fair-skinned populations may benefit because of the lack of inclusion of darker skinned patients in model training and development. If the algorithm is basing most of its knowledge on how skin lesions appear on fair skin, then theoretically, lesions on patients of color are less likely to be diagnosed and therefore benefit from the AI solution. These examples provide you with an idea of how wide-spread the challenges are related to fairness and bias in AI solutions for healthcare. TYPES OF BIAS Bias can occur during almost any stage of AI model building and implementation - from data collection to model deployment. Types of bias: Historical bias Representation bias Measurement bias[Image]

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 Aggregation bias Evaluation bias Deployment bias Historical bias occurs if the present or past state of the world influences a model in a way such that its predictions are considered unfair given societal values and norms. Historical bias refers to judgement based on preconceived notions or prejudices. AI algorithms are entirely data dependent and historical bias encoded in real-world data cannot even be overcome by perfect sampling and feature selection. Therefore, it is important to remember that historical healthcare data, in general, is extremely male and extremely white, and this has real-world impacts. Representation bias (also called sampling bias) arises when the sample collected to train an AI solution does not represent the actual distribution of the population it is intended to be applied to. It occurs when certain parts of the final use population are underrepresented in the training data. Measurement bias arises in situations if the noise is not randomly distributed but differs across groups, which leads to differential performance. Often the only available and measurable features as well as labels are only noisy proxies of the actual variable of interest. Usually, one cannot change the data, some historical biases might be indistinguishably linked to the data – but the awareness about the problem is important and mitigation strategies can be identified by taking preventive measures such as pre- and post-processing actions. Aggregation bias occurs while developing the model when we try to combine different populations whose underlying distribution of the outcome under study differs. This problem is known as inframarginality and requires separate models for the different populations or including the demographic variable into the model to account for the systemic differences. In terms of model development, one size does not fit all. In order to identify aggregation bias, developers need to understand the meaningful distinct groups and reasons why they are different from each other.

Evaluation bias occurs during the model validation and tuning. Evaluation bias arises if the testing data, which often includes external benchmark datasets, is not representative of the final population to which the AI solutions will be applied. This is the difference between the data used for model evaluation and the data used for model's real-world predictions. Since developers mostly use a benchmark dataset or a synthetic dataset for training, their evaluation often does not fit the realworld. A solution of this problem is external validation of the AI model on a different unseen data selected from the targeted population. Evaluation bias can arise if inappropriate performance metrics [Image]

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Copyright 2020 © Stanford University are used. Evaluation bias also refers to usage of evaluation metrics inefficiently and to avoid it, using granular and comprehensive evaluation metrics is suggested. Deployment bias arises during the implementation of the model. It refers to using the model inappropriately or misinterpreting its results. In other words, if the model's intended use is different from the way it is used, deployment bias occurs. Deployment bias is the interaction of society with the AI solution - how society or the medical community uses the AI solution and its output. ALGORITHMIC FAIRNESS Ethical analysis of AI Solutions in healthcare demand that we take a view of fairness, or more appropriately, justice that centers on the health and lives of people, not the outputs alone. A lot of historical medicine has been influenced by white normativity, which is the basis of many medical facts. This is evident because of a lack of inclusion of diverse patients in clinical research. have knowledge that the insiders don't. Bringing diverse perspectives actually enhances the quality and accuracy of your scientific endeavor. A key difficulty in developing fair AI algorithms is the fact that no universal notion of fairness exists. Many different definitions have been proposed by researchers over the years and they can be broadly regrouped into three main classes: anti-classification, classification parity and calibration. These fairness definitions have been shown to all suffer from significant statistical shortcomings. Therefore, special caution and awareness about the notion's limitations and weaknesses is always necessary when applying these concepts in model evaluation settings. Anti-classification or "fairness through unawareness" 1. Requires the exclusion of any protected attributes in the outcome modeling 2. Requires the omission of any unprotected characteristics that are proxies of protected attributes The main shortcoming of this fairness definition is that some clinical risk models need to explicitly include protected attributes for it to be equitable. In particular, this applies to situations where the true underlying risk distribution differs across subpopulations, known as the problem of inframarginality. Therefore, an accurate model must include protected attributes, but must also learn to avoid bias based on these attributes. [Image]

[Image]

 AI solutions that use datasets which may be under-representative of certain groups, may need additional training data to improve accuracy in the decision-making and reduce unfair results. Anti-classification requires the exclusion of any protected attributes in the outcome modeling. Classification parity of fairness asks for equal predictive performance across any protected group. When selecting the metrics to examine, it is important to keep in mind the actionable insights resulting from a model output. Two measures have received particularly high attention by the machine learning community: 1. False positive rate: The probability of predicting a positive outcome when the true outcome is negative should be the same regardless of protected attributes of the patients 2. Proportion of positive decisions: The probability of predicting a positive outcome should not vary across different demographic groups given all else equal. It is also known as demographic parity as it requires the classifier's predictions to be independent of protected traits. These definitions are problematic when the risk distributions are different for different groups, a problem known as infra-marginality. Classification parity asks for equal predictive performance across groups. Calibration is when a model reaches a good agreement between model predictions and observed outcomes. Calibration means that when conditioning on risk estimates, outcomes should be independent of the protected attributes. Think of calibration as a comparison of the actual output and the expected output given by a system. Calibration is open to manipulation of risk distributions for different groups. Model calibration is an important aspect of development and must be evaluated before model deployment. Applying fairness measure: Anti-classification: Checking the requirement of not using any protected attributes in the decision rule. It gets more complicated when trying to impose the stricter condition to also not use any proxies of protected attributes. Classification parity: Once the 2-3 most relevant performance metrics are selected, we evaluate performance differences on the test-set across different demographic groups. It is [Image]

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 always better to calculate empirical confidence intervals of this statistic to decrease the dependence on the test set. Calibration: Only look at the overall rate of predicted and observed outcomes across demographic groups. TRANSPARENCY It is important to think about whether the population captured in the EHRs system is representative of the broader community, particularly if these are coming for an academic medical center. If AI solutions are being developed on non-representative populations, the utility and applicability of these advances for the broader patient community falls into question. This relates to an important issue around transparency and reporting and many AI studies, particularly machine learning, do not report the demographic breakdown of the training data used to develop and train the models. Lack of reporting makes it very difficult to evaluate the bias and fairness of the AI solution and its applicability across populations. Studies that mentioned variables often did not report if they were included as model inputs. In the studies that reported demographics, the average populations included higher proportions of whites and Blacks yet fewer Hispanics compared to the general population. While each study might not be applicable to the general population, these findings emphasize the present lack of transparency in reporting details of training data used for development and evaluation by machine learning models in healthcare. To ensure the unbiased deployment and application of any AI model in healthcare, detailed information on the data used to develop and train the model are necessary. [Image]

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Copyright 2020 © Stanford University As a solution for transparent reporting and to identify best-practices for designing machine learning models to account for biases and fairness, MINIMAR template is suggested. (MINIMAR = The MINimum Information for Medical AI Reporting) MINIMAR Requirements: 1. Include information on the population providing the training data, in terms of data sources and cohort selection 2. Include information on the training data demographics in a way that enables a comparison with the population the model is to be applied to 3. Provide detailed information about the model architecture and development so as to interpret the intent of the model and compare it to similar models 4. Model evaluation, optimization, and validation should be transparently reported to clarify how local model optimization can be achieved and to enable replication and resource sharing You can understand that by providing these details of the training data and population, model design and intent, an end-user will have a great understanding of how to best deploy the model and in which populations. DOWNSTREAM EVALUATIONS While we have covered many topics in this lecture regarding bias and algorithmic fairness, there are still many more challenges and opportunities for Fair AI research: Defining fairness: There are several definitions of AI fairness that have been proposed in the literature. It is nearly impossible to understand how one fairness solution could address all challenges. Identifying the correct or best definition is an ongoing debate. From equality to equity: Equity suggests that each group is given the amount of resources needed to have similar outcomes. Understanding how to develop and implement a model that provides both equality and equity presents a paradigm shift in the way to think about healthcare delivery and is an active area of research. Identifying biases in models, and particularly in datasets: Many biases are systematic and we are often unaware they exist. We still have a long way to go before we can systematically mitigate these biases and provide our professionals with the appropriate tools they need to address these issues at point of care. The perspective collection and reporting of AI outputs, clinical recommendations and patients decisions coupled with eventual outcomes is essential in being accountable as healthcare institutions [Image]

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Copyright 2020 © Stanford University and as clinicians. This work and transparency in reporting AI solutions is absolutely critical for populations who have difficulty trusting the medical establishment. The key is to use AI in a way that actually does benefit all groups, which requires thoughtful evaluations and human interpretations. CITATIONS AND ADDITIONAL READINGS Adamson, A. S. and A. Smith. 2018. “Machine Learning and Health Care Disparities in **Dermatology.” JAMA Dermatol 154(11): 1247-48. Beery, T. A. 1995. “Gender bias in the diagnosis and treatment of coronary artery disease.” Heart Lung 24(6): 427-35.** Bozkurt, S., E. Cahan, M. G. Seneviratne, R. Sun, J. A. Lossio-Ventura, J. P. A. Ioannidis, and T. Hernandez-Boussard. 2020. “Reporting of demographic data, representativeness and transparency in machine learning models using electronic health records.” Corbett-Davies, S. and Goel, S., 2018. The measure and mismeasure of fairness: A critical review of fair machine learning. arXiv preprint arXiv:1808.00023. Hernandez-Boussard, T., S. Bozkurt, J. P. A. Ioannidis, and N. H. Shah. 2020. “MINIMAR (MINimum Information for Medical AI Reporting): Developing reporting standards for artificial **intelligence in health care.” J Am Med Inform Assoc.** Kessler, M. D., L. Yerges-Armstrong, M. A. Taub, A. C. Shetty, K. Maloney, L. J. B. Jeng, I. Ruczinski, A. M. Levin, L. K. Williams, T. H. Beaty, R. A. Mathias, K. C. Barnes, T. D. O’Connor, and C. o. A. a. A.-a. P. i. t. A. (CAAPA). 2016. “Challenges and disparities in the application of **personalized genomic medicine to populations with African ancestry.” Nat Commun 7: 12521.** Obermeyer, Z., B. Powers, C. Vogeli, and S. Mullainathan. 2019. “Dissecting racial bias in an **algorithm used to manage the health of populations.” Science 366(6464): 447-53.** Parthipan, A., I. Banerjee, K. Humphreys, S. M. Asch, C. Curtin, I. Carroll, and T. Hernandez-Boussard. 2019. “Predicting inadequate postoperative pain management in depressed patients: A **machine learning approach.” PLoS One 14(2): e0210575.** Spanakis, E. K. and S. H. Golden. 2013. “Race/ethnic difference in diabetes and diabetic **complications.” Curr Diab Rep 13(6): 814-23.** [Image]

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 Suresh, H. and J. V. Guttag. 2019. "A framework for understanding unintended consequences of **machine learning." arXiv preprint arXiv:1901.10002.** [Image]

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2. EAA_Module5.pdf

Summary

This educational guide outlines the **international regulatory framework** for artificial intelligence in healthcare, specifically focusing on the classification and oversight of **Software as a Medical Device (SaMD)**. The text explains that the **U.S. Food and Drug Administration (FDA)**, following standards set by the **International Medical Device Regulators Forum (IMDRF)**, categorizes these technologies into four risk levels based on the **severity of the medical condition** and the **significance of the software's output** in clinical decision-making. Key themes include the necessity of a **clinical evaluation process**—consisting of valid clinical association, analytical validation, and clinical validation—and the requirement for **continuous real-world monitoring** once a product is deployed. Furthermore, the document highlights the transition from fixed to **adaptive algorithms** and compares global approaches, noting that while the **US and EU prioritize individual privacy**, China's strategy emphasizes **societal benefit and data sharing**. Ultimately, the purpose of the text is to provide a structured roadmap for ensuring that medical AI is **safe, effective, and scientifically sound** throughout its entire product life cycle.

Keywords: Regulatory Risk Framework, Software Medical Devices, Clinical Evaluation Process, Total Product Lifecycle, Global AI Governance

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 EVALUATIONS OF AI APPLICATIONS IN HEALTHCARE STUDY GUIDE MODULE 5: THE REGULATORY ENVIRONMENT FOR AI IN

HEALTHCARE LEARNING OBJECTIVES Understand why most AI solutions in healthcare have not received regulatory approval, to date Describe best practices for AI development, in particular good machine learning practices Recognize the risk framework used to classify AI solutions in healthcare that is used by the Food and Drug Administration (FDA) Know the 3 concepts of model properties that can be regulated Understand main differences between EU, China and US regulations on AI solutions OVERVIEW International Medical Device Regulators Forum (IMDRF): An international workgroup composed of AI regulators that come together to develop a path for standardized AI regulations. Software as a Medical Device (SaMD): Software intended to be used for one or more medical purposes that perform these purposes without being part of a hardware medical device. Medical purpose: Intended to treat, diagnose, cure, mitigate, or prevent disease or other conditions SaMDs are NOT part of hardware COMPONENTS OF REGULATION The SaMD can be described in 3 components: SaMD inputs, SaMD algorithms, and SaMD outputs. [Image]

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 This is the framework used by the US Food and Drug Administration to regulate AI Solutions / SaMD. The FDA's regulatory framework starts with a Market Application, which includes a definition statement and category (I, II, III, or IV). The category is based on the risk associated with the use of the proposed AI solution. Depending on the category defined, data requirements necessary for regulation may include: A premarket notification (or 510(k) statement of equivalency) De Novo request (no similar product exists on the market for comparison) Premarket Application (PMA), which is reserved for high risk AI applications Definition Statement is required for every SaMD application and is used to identify the submission category, which defines risk and subsequent data requirements. The definition statement must: Clearly identify the intended medical purpose of the model (treat, diagnose, drive clinical management, inform clinical management) State the healthcare situation or condition that the AI model is intended for, which includes critical, serious, and non-serious conditions Include the intended population for the application Identify the intended users (or stakeholders) of the model Using information from the definition statement, the SaMD Category is defined, which is based on a risk framework developed by the IMDRF. [Image]

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Copyright 2020 © Stanford University In the framework, risk is set by the intended use of the SaMD and the state of the health situation it targets. The columns in the grid represent the Significance of information provided for a healthcare decision. This is the ACTION in the outcome-action pairing framework used for AI evaluation. The rows represent the State of healthcare situation or condition, which identifies the state of the healthcare situation or condition as critical, serious, or non-serious An example that demonstrates the definition statement and risk category: A SaMD AI application that "drives clinical management" in a "critical healthcare situation or condition" (Category 3): In a hypothetical situation, an AI application is developed for ICU patients that receives electrocardiogram, blood pressure, and temperature signals from a patient monitoring system. The physiologic signals are processed and analyzed to detect patterns that occur at the onset of patient instability and deterioration, a threshold set by a utility analysis. When physiologic instability is detected, an alarm is generated to indicate that immediate clinical action is needed to prevent potential harm to the patient. SaMD regulations place devices into four categories based on the risks associated with the use of the device. Category I being the lowest risk; Category IV being the highest risk. Category I devices require general controls Category II devices are medium to moderate risk and require the same general controls and some additional controls Category III and IV applications require "general controls" and pre-market approval which is the most stringent regulatory category. To date, there are very few Category IV SaMDs that [Image]

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Copyright 2020 © Stanford University are approved in the US. These are high risk, generally life-supporting, life-sustaining AI applications. All applications must include general controls. General controls require that all AI solutions comply with three components: 1. Quality system regulations 2. Current good manufacturing practices 3. Properly labeled - the label of the device follows FDA guidelines and regulations Any adverse event associated with the AI solution must be reported. Quality System Regulations: Manufacturers are required to have processes in place for controlled bug resolution, incident reporting, standardized design processes and overall risk management. CLINICAL EVALUATION PROCESS All AI applications needing regulatory approval must include general controls. As part of the general control, the Clinical Evaluation Process is a framework used by regulators to understand quality system regulations. The IMDRF defines the clinical evaluation process as ongoing activities conducted for the assessment and analysis of a SaMD's clinical safety, effectiveness and performance. The Clinical Evaluation Process includes three components. Valid Clinical Association (Category I): Is there a valid clinical association between your SaMD output and your SaMD's targeted clinical condition? Scientific validity of the AI solution or the extent to which the SaMD's output is clinically accepted or well-founded (based on an established scientific evidence), and accurately corresponds to the healthcare situation and condition identified [Image]

[Image]

 in a real-world setting. An indicator of the level of clinical acceptance and confidence one can have of the SaMD's output Minimum evidence to support the clinical association could include: Literature searches Original clinical research Professional society guidelines Examples of how your model can generate new evidence Secondary data analysis The performance of a clinical trial based on your AI solution Required for AI regulation and ensures the clinical acceptance and uptake of the AI solution in the healthcare setting Novel associations - associations that are newly discovered by your AI: As literature and existing randomized clinical trials do not exist for this association, there are other solutions to regulate this software, which may include performing a clinical trial or secondary data analyses Analytical Validation (Category II): Evaluates whether your AI solution correctly processes input data to generate accurate, reliable, and precise output data. Part of the verification and validation phase that should be performed by the AI manufacturer. Provides objective evidence that the AI solution was correctly constructed and the data processing is reliable. May come as part of your good software engineering practices or from generating new evidence through use of curated databases or previously collected patient data Clinical Validation (Category III): Does the use of your AI's output data achieve your intended purpose in your target population in the context of clinical care? Related to the positive impact of an AI Solution on the health of an individual or population. Clinical Validation should be: Measurable, patient-relevant clinical outcome(s) Including outcome(s) related to the function of the model[Image]

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 A positive impact on an individual or public health Prior to product launch of the AI product (pre-market), evidence must exist on the following: AI accuracy Specificity Sensitivity Reliability Usability Limitations Scope of use in the intended use environment with the intended user After launching the product (post-market) the product must: Continue to collect real world performance data to Further understand the healthcare needs to ensure the AI solution is meeting those needs Monitor the product's continued safety, effectiveness and performance in real-world use The IMDRF identifies that clinical validation is a necessary component of regulation and that it can be demonstrated through several paths, including: Referencing existing data from studies conducted for the same intended use Referencing existing data from studies conducted for a different intended use, where extrapolation of such data can be justified Generating new clinical data for a specific intended use The Clinical Evaluation Process is an important component of the regulatory environment. FDA APPLICATION In addition to the general control process, there are other regulatory control requirements (data requirements) that accompany an SaMD application that depend on the application's category of risk (1 - 4), which can include one of these regulatory components: Premarket Notification 510(k) De Novo Notification Premarket Approval (PMA)[Image]

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 Premarket Notification 510(k) is the simplest data requirement, if the SaMD is similar to a product already on the market. The intent is to inform the regulatory agencies that the AI solution is safe and effective, which is determined by demonstrating the AI solution is equivalent to a legally marketed device, often known as a "predicate". To determine equivalence, the AI solution must have: The same intended use as the predicate AND have the same technological characteristics, OR The same intended use as the predicate and a different technology that will not raise safety or efficacy questions AND the AI solution is at least as safe and effective as the predicate As more and more applications become approved, pre-market notifications will become an easier and efficient pathway towards regulation. De Novo Notifications: Submitted when there is no "predicate". Limited to Category I and Category II SaMDs and are a risk-based classification process. The de Novo notification should include: Clinical data (if applicable) that are relevant to support the assurance of the safety and effectiveness of the AI solution Non-clinical data including bench performance testing Description of the probable benefits of the AI solution when compared to the probable or anticipated risks when it is used as intended Premarket Approval (PMA): Required for high risk SaMDs (Category III and IV). The most stringently regulated application required by the FDA. Includes rigorous technical studies, nonclinical laboratory studies, laboratory studies, and clinical investigations. Before PMA approval, the applicant must provide valid scientific evidence demonstrating reasonable assurances of safety and effectiveness for the device's intended use. [Image]

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Copyright 2020 © Stanford University After an AI solution receives regulatory approval, a modification may be required in certain circumstances. A modification request must be submitted if there is new risk or a change of an existing risk. A modification may be required if there is: A change to risk controls to prevent significant harm A change that significantly affects clinical functionality or performance. A change in clinical functionality could include o New indications for use o New clinical effects o Significant technology modifications that affect performance characteristics SaMD modifications generally fall into three categories: 1. Change in performance o Example: The incorporation of new training data or a change in AI architecture which could alter performance 2. Change to the model Input o Example: The incorporation of different sources of the same input or adding new inputs that were not previously considered 3. Change of the intended use of the output o Example: Change in disease (apply to new condition) Software modifications are common and essential in the total life cycle of the AI Solution. PRODUCT APPROVAL In line with the framework proposed by IMDRF, the FDA has developed the following diagram related to the total product life-cycle (TPLC) for an AI workflow. [Image]

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Copyright 2020 © Stanford University There are 4 distinct components in the total product life-cycle: 1. The culture of quality and organizational excellence, which is also referred to as Good Machine Learning Practices. This includes all components that must be considered when developing an AI solution, such as o Data selection and management o Model training and tuning o Model validation 2. Premarket assurance of safety and efficacy of the AI solution. It is expected that safety and effectiveness are continually monitored throughout the life cycle, including patient risks and patient safety. Regulators expect a manufacturer to perform a risk assessment and evaluate that the risks are reasonably mitigated throughout the TPLC. 3. Regular monitoring of safety and intended use, which is used to identify when a software modification is required. The regular monitoring of the deployed model is necessary and should include the ability to log and track model performance. 4. The Continuous Learning expected from an SaDM that Leverages Real World Data. “Continuous learning” is not “machine learning.” It refers to collecting post-market information to update and evaluate your existing AI solution. [Image]

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Copyright 2020 © Stanford University The TPLC diagram demonstrates how regulators are thinking about using real world data for continuous learning, the basis of the learning healthcare system. Locked algorithms are AI solutions that provide the same results each time the same input is provided. Generally fixed functions to a given set of inputs, and may use a manual process for updates and validation. How to regulate "adaptive" AI solutions or machine learning applications? Following deployment, these types of adaptive models may provide a different output in comparison to the output initially approved for a given set of inputs. These automated changes use a well-defined process, which aim to improve outcomes based on new data or additional data that is taken as an input. There are two stages to an adaptive algorithm: o Learning stage: The algorithm "learns" how to change its behavior, based on the addition of new input types or new cases to an already existing training set o Update stage: The algorithm will update when the new version of the algorithm is deployed This is a paradigm shift in the regulatory process and requires a new total product lifecycle that allows these devices to continually improve while providing effective safeguards. Considerations for the regulatory environment for an AI model: [Image]

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Copyright 2020 © Stanford University 1. Developers should be aware of SaMD Risk Classifications, total product life-cycle (TPLC) and Good Machine Learning Practices (GMLP) 2. The population, performance and intended use are aspects of the model that can be regulated and any change to these characteristics after approval would require a notice of modification 3. The intended use and state of the healthcare situation of the AI model drives the level of regulation and risk category 4. Safety and effectiveness must be continuously monitored post-market using real-world data to ensure a learning health system Example - Arterys Indications for Use: Arterys is a software that uses cardiovascular images acquired from magnetic resonance (MR) scanners. It analyzes blood flow from the heart using the MR images. The output is intended to be used to support cardiologists, radiologists, and other healthcare professionals for clinical decision making. Risk Classification: Class II o Significance of information is to "inform clinical management" o State of healthcare situation or condition is “critical” Example - IDx-DR Indications for Use: IDx-DR is a retinal diagnostic software device that incorporates an adaptive algorithm to evaluate ophthalmic images for diagnostic screening to identify retinal diseases or conditions Risk Classification: Class II o Significance of information is to “drive clinical management” o State of healthcare situation or condition is “serious” Example - Guardian Connect (Medtronic) Indications for Use: The Guardian Connect system is indicated for continuous or periodic monitoring of glucose levels in the fluid under the skin, in patients (14 to 75 years of age) diagnosed with diabetes. Risk Classification: Class II o Significance of information is to “drive clinical management o State of healthcare situation or condition is “serious” [Image]

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 The FDA does not regulate certain types of clinical decision support (CDS) tools under 21st century cures act. Three criteria determine whether CDS are regulated as an SaMD: 1. The software cannot receive, analyze or otherwise process a medical image or signal from an in vitro diagnostic device or from any other signal acquisition system 2. A healthcare professional must be able to understand the basis of its recommendations 3. The software cannot be intended as the sole source of recommendations regarding treatment, diagnosis or prevention of a disease The FDA doesn’t regulate AI solutions that are “laboratory-developed tests” designed, developed and deployed within a single health care setting. FDA’s Digital Health Software Precertification Program (Pre-Cert): Streamlines regulation of AI solutions Organizations may become “approved” which would allow them to bring products to market without a premarket review, provided the product is “lower risk” Once certified, organizations can make certain minor changes to its AI products without having to submit a modification request An organization must demonstrate the FDA’s five quality and organizational excellence principles in order to be considered for the Pre-Cert program: 1. Product quality 2. Patient safety 3. Clinical responsibility 4. Cybersecurity responsibility 5. Proactive culture GLOBAL ENVIRONMENT It is important to note that there are some differences across the globe regarding regulatory guidelines for AI in healthcare. EU’s General Data Protection and Regulation (GDPR) Outlines a comprehensive set of regulations for the collection, storage, and use of personal information which may be used in AI solutions [Image]

[Image]

 Describes the right of citizens to receive an explanation for algorithmic decisions. The implications would exclude the use of many types of algorithms used today in advertising and social networks - and eventually healthcare. Provides protection of data from EU citizens but given the global AI momentum, the laws may impact companies from the US and worldwide A critical component of this regulation is Article 22: “Automated individual decision making, including profiling.” Article 22 requires explicit and informed consent before any collection of personal data. The “right to explanation” o Requires that meaningful information about the AI logic as well as the potential significance and consequences of the data-driven system are provided upon request o Refers to the use of the black box algorithms o Could potentially limit the types of models that manufacturers are able to use in health-related applications o Holds the manufacturers of AI-based technologies more accountable While there are some differences between the US and EU regulations, a common theme from both entities is the protection of the individual, their data, and their right to information. China leads in the number of AI patents as a result of this favorable environment. AI governance in China is aimed at the development of “responsible AI” and focused on the societal beneficiary rather than the individual beneficiary. China has also put a focus on regulating education so that the nation produces more STEM workers. China requires businesses and private citizens to share their data with the government – almost the opposite of US and EU regulations. The incentives for data sharing and elimination of data silos could catapult China in clinically meaningful AI technologies. Principles for AI governance released by China’s Ministry of Science and Technology (MOST) include: 1. Harmony and friendliness 2. Fairness and justice 3. Inclusivity and sharing 4. Respect privacy 5. Secure/safe and controllable 6. Shared responsibility [Image]

[Image]

Copyright 2020 © Stanford University 7. Open collaboration 8. Agile governance China is rapidly developing important AI solutions in healthcare and these governing policies are essential to balance both the innovation of technology as well as the safety of healthcare delivery. The White House Office of Management and Budget (OMB) document provides Guidance for Regulation of AI Applications. The regulations are not specific to healthcare, but they provide the umbrella of regulations applied to AI solutions. The ten guiding principles are aligned with the FDA’s regulatory processes: 1. There must be public trust in AI o Public trust and validation is critical to the adoption and acceptance of these technologies in the healthcare sector o Privacy and other risks must be addressed with appropriate mitigation strategies that are well documented and transparently reported 2. There must be public participation in AI development o There should be ample opportunities for the public to provide information and participate at all stages possible of the “rule-making” process 3. AI solutions must be based on scientific integrity - clinical validation in the clinical evaluation process proposed by the IMDRF o AI in healthcare should be based on scientific and technical information and processes that are likely to have a clear and substantial influence on public policy or private sector decisions - and these standards should be held to the highest level of quality, transparency, and compliance o Best practices include clearly stating the strengths, weaknesses, intended optimizations or outcomes, bias mitigation, and appropriate uses of the AI application’s results o Data used to train the AI system must be of sufficient quality for the intended use 4. Every AI solution must have a Risk Assessment and Management component o A risk-based approach should be used to determine which risks are acceptable and which risks present the possibility of unacceptable harm, or harm that has expected costs greater than expected benefits o If an AI tool fails, the magnitude and nature of the consequences will inform the level and type of regulatory effort that is appropriate to identify and mitigate risks 5. Benefits and Cost [Image]

[Image]

Copyright 2020 © Stanford University o For an AI algorithm to be deployed, it must offer significant potential benefit o Before implementing an AI solution, agencies must consider the full societal costs, benefits, and effects related to the development and deployment of these applications 6. AI solutions must be flexible o Performance based and flexible approaches should be easily adapted and updated - this would include the continuous monitoring of real world evidence to improve upon the AI solution 7. AI solutions must be fair and non-discriminatory o Transparency regarding potential biases and discriminatory aspects of the algorithm is becoming more and more important as we see potential harm due to biases promoted or exacerbated through AI 8. AI solutions that provide Disclosure and Transparency will promote public trust o Healthcare systems should disclose when AI solutions are in use and how these applications can impact patients and decisions 9. All AI solutions must address Safety and Security issues o Safety and security should be considered throughout the design, development, deployment, and operation process o There should be controls in place to ensure confidentiality, integrity, and availability of the information processed, stored, and transmitted by AI systems 10. AI solutions must include multidisciplinary stakeholder involvement or Interagency Coordination o All sectors affected by the AI solution should coordinate and share experiences and challenges of AI solutions Safety, transparency and multidisciplinary aspects are key ingredients to a successful and wellthought out AI solution. CITATIONS AND ADDITIONAL READINGS FDA, U. 2019. “Proposed regulatory framework for modifications to artificial intelligence/machine learning (AI/ML)-based software as a medical device (SaMD).” FDA. US Department of Health and Human Services.

Software as a medical device (SAMD): Clinical evaluation. Guidance for industry and Food and Drug Administration Staff. 2017. [Image]

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3. EAA_StudyGuide Full.pdf

Summary

This Stanford University study guide provides a comprehensive roadmap for the **rigorous evaluation and implementation** of artificial intelligence within the medical field. The text moves beyond simple technical metrics like model accuracy, introducing a specialized framework centered on **outcome-action pairing (OAP)** to ensure that technological predictions result in meaningful clinical improvements. Structurally, the manual spans the entire lifecycle of an AI solution, addressing its role in **biomedical research, translational research, and medical practice** while emphasizing the necessity of navigating **regulatory environments and ethical challenges** such as algorithmic bias. Ultimately, the source serves to bridge the gap between data science and patient care, advocating for a focus on **clinical utility, feasibility, and stakeholder engagement** to achieve better health outcomes.

Keywords: AI in Healthcare, AI Model Evaluation, Clinical Utility, AI Deployment Challenges, Bias and Fairness

Content

[Image]

Evaluations of AI Applications in Healthcare Study Guide Study Guide Stanford University [Date] [Course title] [Image]

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Copyright 2020 © Stanford University 3 MODULE 1: AI IN HEALTHCARE LEARNING OBJECTIVES Recognize why we need AI in healthcare and understand what are the right questions we can ask AI to help solve Describe the breadth of AI in healthcare, from the molecular level to the population level Recognize the different categories of AI in medicine (biomedical research, translational research and medical practice) Explain why AI evaluations need to move beyond model accuracy AI IN HEALTHCARE Artificial Intelligence (AI): Involves the development of computer algorithms to perform tasks typically associated with human intelligence AI spectrum of learning: Machine learning Representation learning Deep learning Natural language processing Algorithms: A mathematical technique, generally developed by statisticians and mathematicians for a particular task, such as unsupervised learning or reinforcement learning Models: Well-defined computations formed as a result of an algorithm that takes some value, or set of values, as input and produces some value, or set of values as output AI solution: Refer to an evaluated and validated model AI has the potential to provide high performance data-driven medicine, optimize care trajectories, suggest the right therapy for the right patient and improve the process of clinical assertions and decision making. [Image]

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[Image]

Copyright 2020 © Stanford University 4 The best way to understand AIs potential use in healthcare is to think about its applications in three separate categories: Biomedical research: AI is assisting in automated experiments, automated data collection, gene function annotation, literature mining Translational research: AI is assisting in areas such as biomarker discovery, drug-target prioritization, and genetic variant annotation Medical practice: AI used for disease diagnosis, treatment selection, patient monitoring, and risk stratification models Reasons why AI is needed in medicine: AI and Data Synthesis- AI can make sense of the overwhelming amount of data associated with a single disease or a single patient to highlight the relevant information needed to best guide the treatment and care for each individual patient. AI can improve clinical reliability and be used to help identify relevant information AI tools can tackle tedious, mundane tasks because they don't suffer from fatigue, distractions, or moods like their human counterparts. Therefore, AI tools can be useful for reducing errors related to human fatigue. Patient-clinician engagement is another important area where we see AI solutions emerging. AI can improve patient outcomes by prioritizing patients in more urgent need and by recommending individualized treatments that account for a patient's unique characteristics, which often fall outside of the selective clinical trial evidence. AI has the potential to improve resource utilization and efficiency, thereby reducing overall healthcare costs. AI is being used to improve diagnostics. AI can quickly and more accurately spot signs of disease in medical images such as MRIs, CT scans, ultrasounds and x-rays.[Image]

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Copyright 2020 © Stanford University 5 AI can help physicians understand patients' values and goals. AI can assist by analyzing structured and unstructured patient data and present insights for physicians' consideration and to aid in shared decision making. We can think about AI solutions in medicine across the domains of biomedical research, translational research and medical practice. A non-exhaustive list of some recent AI applications in medicine that demonstrate how AI algorithms can benefit patients, doctors, and researchers through making diagnosis, treatment, discovery, and the practice of medicine faster, more accurate, and more efficient: Image analysis - AI is being used in clinical practice for disease diagnosis, AI solutions can reduce errors related to human fatigue and improve diagnostics. Drug discovery - AI development under the translational research branch. Key classification features are gene essentiality, mRNA expression, DNA copy number, mutation occurrence and protein–protein interaction network topology. Patient risk stratification (or Population level segmentation) - AI is used in clinical practice for risk stratification. Risk of 30-day readmissions – Another example of AI used in clinical practice for risk stratification. Basic biomedical research - AI is being used in basic research for automated experimentation. Home videos for autism diagnostics. These examples provide a glimpse of the broad range of AI applications we see emerging in medicine and the opportunities to improve diagnostics, care delivery, access to care, and patient outcomes. GROWTH OF AI IN HEALTHCARE [Image]

[Image]

Copyright 2020 © Stanford University 6 Both literature and media show that there is a hype around AI and this is coupled with an explosion of AI applications in healthcare. Recently, there has been a drastic increasing trend in medical literature related to “artificial intelligence” and “healthcare” or “medicine”. Most research: Provides an overall description of the clinical problem Includes the architecture of the proposed model Provides the accuracy or precision of the model Provides thoughts about how their AI solution COULD be used in the healthcare setting Think beyond simple accuracy or precision as a measure for AI models because these measures do not tell you anything about the impact of your model in the healthcare setting. Often researchers focus on a model without considering if the model output can be mitigated or will have an impact on clinical care. A systematic review identified articles predicting hospital readmission risk: Identified about 8,000 citations Included 30 studies that met the inclusion criteria Most models relied on retrospective administrative data Had poor discriminative ability (with a c-statistic less than 0.8) Only 1 study specifically addressed preventable readmissions The ability to predict who is at high risk for hospital readmission may not be clinically useful if the model does not convey anything that the team does not already know. However, if the model contained data on discriminatory actions during the in-patient stay that were associated with low risk of readmission then this information could be taken into account to improve the utility of the AI solution. However, most algorithms are developed without considering the clinical utility, feasibility, or the overall impact of implementing the AI solution. HOW TO KNOW IF AN AI MODEL IS GOOD [Image]

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Copyright 2020 © Stanford University 7 An example of a longitudinal data for a patient – There will be a window of observation, where you are collecting and aggregating data. This observation window could be the inpatient hospital stay. Then you set a time where you will start predicting an event. In our example, this could be time of hospital discharge. You make a prediction and that event can happen within the next hour or in the next N years, depending on what you are predicting. If you are predicting patient deterioration, it may happen within minutes or hours after the observation period. However, if you are predicting Alzheimer’s Disease based on genotyping and family history, your event might not occur for years, maybe even decades. So, this gives you your action window. And depending on the prediction, this action window can be categorized as acute or long-term. One must think about whether the action to my output has to happen immediately, or is it something that's going to happen over the next year, five years, or 10 years? Studies have shown that, regardless of acute or long-term actions, early warning lead times give more opportunities for action. Interestingly, almost no effort goes into the action side of the equation. You make a prediction; you publish the paper and then you are done. What we need to think about is how do we move beyond predictions and start thinking about the action side of the evaluation: How will the prediction be used and by whom - so that we can build better models and make sure they get integrated into clinical care. Start thinking beyond predictions and to do this, we will start with a framework that allows us to conceptualize outcome-action pairing (OAP). In OAP, the outcome is the purpose of the AI model: disease diagnosis, risk stratification, or event prediction. The action is the step that can be taken based on the outcome that will improve medical care. A different treatment pathway based on risk, a new treatment based on a diagnosis, or scheduled follow-up with a primary care physician to prevent a readmission. To evaluate an AI solution beyond its predictive value, we need to think about a framework that can be systematically applied across the broad range of AI solutions emerging in healthcare. We need a framework that provides criteria to evaluate the utility, feasibility, and overall clinical impact of an AI solution. [Image]

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Copyright 2020 © Stanford University 8 Utility: Refers to the purpose of the model and it is what matters to the patient most Feasibility: Refers to what is needed to implement the AI solution in the healthcare setting Clinical Impact: The overall effect that the AI solution can have on clinical care, patient outcomes, and care standards Artificial Intelligence spans the breadth of healthcare and there are so many opportunities to use AI to improve upon or augment healthcare delivery. AI can identify patterns in the expanding, heterogeneous data sets in healthcare to create models that accurately classify, predict or recommend actions. However, realizing the potential benefit of AI solutions for patients in the form of better care requires rethinking how we evaluate AI solutions. A framework for rigorously evaluating the performance of a model in the context of the subsequent actions it triggers is necessary to identify AI solutions that are clinically useful. CITATIONS AND ADDITIONAL READINGS Coquet, J., S. Bozkurt, K. M. Kan, M. K. Ferrari, D. W. Blayney, J. D. Brooks, and T. Hernandez-Boussard. 2019. "Comparison of orthogonal NLP methods for clinical phenotyping and assessment **of bone scan utilization among prostate cancer patients." J Biomed Inform 94: 103184.** Hao, S., Y. Wang, B. Jin, A. Y. Shin, C. Zhu, M. Huang, L. Zheng, J. Luo, Z. Hu, C. Fu, D. Dai, D. S. Culver, S. T. Alfreds, T. Rogow, F. Stearns, K. G. Sylvester, E. Widen, and X. B. Ling. 2015. "Development, Validation and Deployment of a Real Time 30 Day Hospital Readmission Risk **Assessment Tool in the Maine Healthcare Information Exchange." PLoS One 10(10): e0140271.**[Image]

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Copyright 2020 © Stanford University 9 Jeon, J., S. Nim, J. Teyra, A. Datti, J. L. Wrana, S. S. Sidhu, J. Moffat, and P. M. Kim. 2014. "A systematic approach to identify novel cancer drug targets using machine learning, inhibitor design **and high-throughput screening." Genome Med 6(7): 57.** Kansagara, D., H. Englander, A. Salanitro, D. Kagen, C. Theobald, M. Freeman, and S. Kripalani. 2011. "Risk Prediction Models for Hospital Readmission: A Systematic Review." Matheny, M. E., D. Whicher, and S. Thadaney Israni. 2019. "Artificial Intelligence in Health Care: **A Report From the National Academy of Medicine." JAMA.** Noack, M. M., K. G. Yager, M. Fukuto, G. S. Doerk, R. Li, and J. A. Sethian. 2019. "A Kriging-**Based Approach to Autonomous Experimentation with Applications to X-Ray Scattering." Sci Rep** 9(1): 11809. Rajpurkar, P., J. Irvin, R. L. Ball, K. Zhu, B. Yang, H. Mehta, T. Duan, D. Ding, A. Bagul, C. P. Langlotz, B. N. Patel, K. W. Yeom, K. Shpanskaya, F. G. Blankenberg, J. Seekins, T. J. Amrhein, D. A. Mong, S. S. Halabi, E. J. Zucker, A. Y. Ng, and M. P. Lungren. 2018. "Deep learning for chest radiograph diagnosis: A retrospective comparison of the CheXNeXt algorithm to practicing **radiologists." PLoS Med 15(11): e1002686.** Tariq, Q., J. Daniels, J. N. Schwartz, P. Washington, H. Kalantarian, and D. P. Wall. 2018. "Mobile detection of autism through machine learning on home video: A development and **prospective validation study." PLoS Med 15(11): e1002705.** Yu, K. H. and I. S. Kohane. 2019. "Framing the challenges of artificial intelligence in medicine." **BMJ Qual Saf 28(3): 238-41.** MODULE 2: EVALUATIONS OF AI IN HEALTHCARE LEARNING OBJECTIVES 1. Describing a framework for evaluating AI applications in healthcare 2. Understanding clinical utility and outcome-action pairing in AI solutions 3. Recognizing the many different aspects of an action to an AI solution. [Image]

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Copyright 2020 © Stanford University 10 A FRAMEWORK FOR EVALUATION Two important aspects of evaluation: stakeholders and beneficiaries. These are important attributes to consider in the development, design, and deployment AI solutions. Stakeholder Involvement: Important to understand what stakeholders should be involved throughout the design, development, evaluation, validation and deployment of an AI solution. Involves knowledge experts, decision makers, and end-users Beneficiary: Involves understanding who the AI solution is made for or who it will be used by Could be a provider, patient, hospital, payer Stakeholders and beneficiaries are components that need substantial thought and consideration in the development, design, and deployment AI solutions. Clinical Utility: Relates to its applicability and impact on the healthcare system It requires identifying the beneficiary of the AI solution and understanding what action can be taken based on the model outcome that will improve for the beneficiary Two components (action and outcome) will allow you to better understand the problem addressed by the AI solution, and if it is a problem worth solving with AI Considerations when evaluating an AI solution: What action can be taken based on the model output? Does a mitigating action (or therapy) exist? What is the ability to intervene? Who needs to take the action? What is the lead time offered by the prediction? What are the logistics and costs of the intervention? What are the incentives for acting on the output? Start with the problem you are trying to solve with AI and then ask, is that problem worth solving. To help you answer this question for any AI solution, we will start by analyzing an action that can be paired with the model outcome - something we refer to as the "output - action pairing" or what the cool people will call the "OAP" [Image]

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Copyright 2020 © Stanford University 11 Utility Assessment: For every good AI solution (or prediction), there should be the ability to act upon or mitigate the output OUTCOME: ACTION PAIRING In OAP, the outcome (or output of the model) is the purpose of the AI solution, for example, a disease diagnosis, risk stratification, or event prediction. An action is a step that can be taken based on the outcome that will improve medical care. When we think about evaluating an AI solution, we must understand the outcome and know if there is a mitigating action that could change this outcome. This is the basis of the outcome-action pairing framework. It is important to remember that for every good AI solution there is the ability to act upon or mitigate the output. When evaluating an AI solution, it is also important to think about the lead time your action needs. One must think about whether the action to my output is acute (it has to happen immediately), or is it long-term?

Studies have shown that, regardless of acute or long-term actions, early warning lead times give more opportunities for action. So, the further in advance I can make my prediction the more time one would have to respond with the action to that output. Lead-Time provided by the AI solution can directly impact its clinical utility in the healthcare system. [Image]

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Copyright 2020 © Stanford University 12 Once the action is decided based on AI outcome, the next question to ask is: What type of action should you take? You must clarify the type of action to evaluate the feasibility to implement the AI solution at point of care. We need to understand if the action is operational or medical - and an AI solution can be developed to have either action. This is a simple representation of the concept, with action types on the y-axis and action lead-times on the x-axis. What if my AI solution predicts ICU transfer? Where would this fall on the grid? This could be an acute operational action, as the hospital care team would need to act on this prediction by possibly 1) arranging the transfer; 2) finding a bed in ICU; and 3) identifying the ICU care team. If I am predicting cardiac arrest, this would be an acute medical action. Alternatively, if my AI solution predicts a hospital readmission, this could be either an operational or medical action. If the action is to schedule an appointment with a primary care provider, the action is operational and long-term. A population-level action is related to AI solutions that predict public health events or other population outcomes, such as flu outbreaks or the coronavirus spread. If you are predicting flu outbreaks or vaccine efficiency, in addition to your local stakeholders, it is likely you would want regulatory agencies, and/or local, state and federal governments as stakeholders. It is important to keep in mind that depending on the type of action, different stakeholders will be needed. [Image]

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Copyright 2020 © Stanford University 13 For instance, your output from an AI solution can be the predicted risk of a 30-day hospital readmission and your action can be to provide a list of patients at risk. A minimal action is, you create a list of, for example 100 patients at highest risk for hospital readmission. Your Prediction: Risk of Hospital Readmission The Action: Provide a list of the 100 highest risk patients You can send a list to somebody and hope for the best. For example, you can sell your list to the healthcare system and then it’s up to the health system to do something with that information. And a lot of health systems and insurance companies will buy these lists and then provide that list to their disease management or care management teams. In industry, this is called a chase list. If you go one level up on the action side, you might recommend an action. You need to understand what is going on and what the features mean in your model. A black box algorithm might not work with this action. Here is where interpretability and explainability usually are invoked. The next step is that you figured out what action is needed to take and then you actually take the action - you execute and follow through. Finally, if you're really sure and confident about the efficacy of your action plan and your ability to execute it, you might start assuming financial risk. These are some examples of how we can think about outcome-action pairing and how this framework can be used to evaluate AI solutions. Outcome-Action pairing can be an effective way to evaluate how you or if you would deploy an AI solution. [Image]

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 14 CLINICAL UTILITY When we think about utility, we can also think about different measures that assess the number needed to benefit from the model. The value of the predictive model and its resulting actions can be conceptually divided into two components: number needed to screen and number needed to treat Number needed to screen: The number of people you need to screen to identify one "true positive" Number needed to treat (NNT): The number of patients needed to be treated for one patient to benefit or the number of true positives you would need to take action on or treat for 1 patient to benefit Number needed to harm (NNH): The number of people who received the intervention in question that would lead to just one person being harmed. With NNH, instead of looking at desirable outcomes, you are comparing the absolute risk increase of bad outcomes. To evaluate clinical utility is to define and characterize the problem to be addressed by the AI solution and to determine whether that problem can be solved (or is worth solving) using AI We often look at the Receiver Operator Characteristics (ROC) curve to evaluate an AI solution. The ROC curve is a plot of the true positive rate against the false positive rate at different threshold settings. The true-positive rate is also known as sensitivity or recall. The false-positive rate is also known as probability of false alarm, which is 1 − specificity. You need to get a cost and a utility for your true positives and you need the same information for your false positives.

From this, you can estimate a zone on your ROC curve that if you set a [Image]

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Copyright 2020 © Stanford University 15 threshold, you would make better cost decisions than if you were to set a decision threshold - a net benefit analysis. Generally, a 2-step process of selecting a best model and then evaluating whether the model is helpful is used. This can be misleading because in machine learning hundreds of computerized models are created from which 1 is selected during the process of learning. A decision curve analysis takes the threshold probability of an event where the relative costs of a false-positive and a false-negative prediction are taken into consideration. This theoretical relationship can be used to derive the net benefit of the model across different threshold probabilities - which generates a “decision curve.” A decision curve can be used to derive the net benefit of the model across different threshold probabilities. Given estimated costs and benefits of the actions possible to mitigate a readmission, there is a region of higher utility than what the best model allows to be achieved. The “blue” model actually has a smaller area under the ROC curve than the “yellow” model but the “blue” curve has a higher utility based on the net benefit of readmission-preventing actions based on the model’s predictions. We see this because the blue ROC curve crosses over into the region of higher net utility on the graph. [Image]

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Copyright 2020 © Stanford University 16 In fact, several other models could be created during the process of learning that have even higher utility, but usually those will never be considered because the choice of the best model is often based on measures such as the area under the ROC curve rather than utility. This example illustrates how a 2-step process to evaluate net benefit will fail to uncover that a model more useful than the best model, based on the area under the ROC curve, exists. There are people who are interested solely in the accuracy or precision or AI models and they generally ignore the fact that their methods have no clinical utility. A decision curve analysis can help understand which model likely has the highest clinical utility - which is not always the model with the highest accuracy. FEASIBILITY For the feasibility of the algorithm, one must consider: Data availability and quality Implementation costs Deployment challenges Clinical uptake Maintenance over time Data availability and quality. Data are an important aspect for any algorithm that learn from data. It is important when evaluating an AI model to look very closely at the data used for training, validation and testing. Includes data retrieval, preprocessing, and data cleaning Important to know if the population benefiting from the model is well represented in the training data Consider how the label values were assigned and who assigned these labeled values Transparency in the reporting of data and data processing is essential to evaluate any model Other questions to think about regarding data: What is the ability of the data? Is it current and up to date? Are we using data that is five years old and trying to assess a new surgical procedure that was not well implemented in the time period of your dataset? [Image]

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 17 How are missing data handled? Is the longitudinal data considered? If so, how is lost to follow up dealt with? Feasibility of the action: Necessary resources. If you decide to act, do you have the necessary resources and equipment to perform that act? Necessary work capacity. Think about the work capacity necessary to act. A component necessary to understand utility includes the Clinical Evaluation of the AI solution. The International Medical Device Regulator Forum (or the IMDRF) has developed a framework for clinical evaluation that was adopted globally, including by the US Food and Drug Administration (FDA). The framework is used to assess the risk and impact of AI solutions and to demonstrate assurance of safety, effectiveness, and performance. Clinical evaluation: 1. Valid clinical association: Refers to the extent to which the model output is clinically accepted or well-founded based on an established scientific framework or body of evidence, and corresponds accurately in the real world to the healthcare situation and conditions identified by the AI solution. It is important to have a valid clinical association between your output and feature(s) if you expect clinical acceptance or clinical uptake. 2. Analytical validation: Does the model correctly process input data to generate reliable, accurate, and precise output data? This type of evaluation requires an understanding of the clinical data used to develop the model and the transparency in the reporting of the processing of training data, as well as a clear understanding of the labeled input and output variables. 3. Clinical validation: Measures the ability of an AI solution to generate a clinically meaningful output in the target disease, situation, or condition intended. Clinically meaningful refers to the impact the AI solution may have on the health of an individual or population. Defining the accuracy and predictive value of an AI solution is needed, but the true evaluation of AI healthcare is not easy. Current AI in healthcare evaluation systems are limited - or non-existent - in their applicability for estimating the net utility of model. In addition, good examples of deployment AI solutions across systems are limited. As a result, healthcare teams have to rely on their personal experience and the collective experience of their colleagues to bridge the “gap” between available evidence and the needed evidence on AI evaluations.

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Copyright 2020 © Stanford University 18 CITATIONS AND ADDITIONAL READINGS FDA, U. 2019. "Proposed regulatory framework for modifications to artificial intelligence/machine learning (AI/ML)-based software as a medical device (SaMD)." FDA. Shah NH, Milstein A, Bagley, PhD SC. Making Machine Learning Models Clinically Useful. JAMA. 2019;322(14):1351–1352. doi:10.1001/jama.2019.10306 MODULE 3: AI DEPLOYMENT LEARNING OBJECTIVES 1. Understanding the four components of AI deployment 2. Describing the role of academics in the development and deployment of AI solutions in healthcare 3. Understanding the investments required to integrate AI solutions into the care setting, from the researcher to the clinician to the healthcare system 4. Knowing the major challenges, one needs to overcome to successfully deploy an AI solution into healthcare delivery, including consideration of foreseeable or intended harm AI DEPLOYMENT AI in healthcare has moved slower because human lives are at stake and therefore regulations, policies, and standards have set a higher bar for product development and deployment. Enthusiasm for AI in healthcare has been overshadowed by the challenging path to successfully implement these solutions into routine clinical care. Challenges to deployment: Lack of buy-in from necessary stakeholders Lack of evidence for clinical utility Lack of resources and expertise in AI deployment The secondary use of clinical data that was generated and managed in the medical system with a primary purpose to support care and generate claims rather than facilitate clinical research and secondary analyses [Image]

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Copyright 2020 © Stanford University 19 To date, most literature and evaluations of AI solutions in healthcare have focused on proof of concept and the predictive abilities and accuracy of the AI solution. More recently, there has been an emerging focus on the downstream evaluations of AI applications in healthcare, including bias and fairness concerns across populations. When we think about moving an AI solution into the healthcare setting, we need to think about a holistic framework for translation of the product into the delivery system - and multiple stakeholders interacting with the final product. The AI model is one component of the total product life cycle (TPLC) and all components are necessary to understand and evaluate for deployment. The TPLC includes: Data Selection and Management Model Training and Tuning Model Validation Once you have completed the first section “AI Model Development”, you move to the AI Production Model, or the “Deployed Model” which continuously receives new (or live) data and is continuously monitored for safety and performance evaluations. Some practical questions that you might (or should) ask prior to deployment: 1. Research Question, which addresses Clinical Utility. What is the clinical question and can I answer the question with an AI product? Is it a question that can be resolved with the data I have and the accuracy of the models I build? [Image]

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 2. Stakeholder Involvement - Intended User. Who will use the intended user of the model output and how would they want to see the results? Will the results from the model need to be displayed to the clinician at point of care, to the hospital management team to evaluate readmissions, or to an insurance company to identify high cost patients? Have all stakeholders involved evaluated both the input data and intended output data? 3. Training Data. What is the source data that I will use to train my models and where will I get this data? How recent are the data and what is the quality of data? Do your data need to be updated weekly, nightly, or hourly to make your prediction? What is the population distribution of this data and is the outcome of interest well represented in this data and does it represent the applied population? 4. Implementation Costs - System Setup. What is the system setup you will need to run the model? Is the system in which you developed your model interoperable with the system in which you will use your model? Will you be able to have all the necessary settings in the system you will use in the real application setting? 5. Feasibility - Clinical Uptake. How do you get the model output back into the workflow correctly and in real-time? Will your output be integrated into the EHR system or will you have to display the output alongside the EHR system? How will your model output interrupt the clinical workflow? Do you have significant buy-in from your end user and will they be your champion for deploying the system? 6. Maintenance over time. Who will be responsible for the continuous monitoring and updating of the AI product? How will new data be added or how and when will the model be re-trained? What happens if biases or unintended consequences are discovered? Whose

responsibility will it be to update and maintain the product? These questions must be evaluated prior to the deployment of the AI solution. Deployment pathway has four fundamental components. 1. Design and development of the AI product - identifying the right problem to solve. 2. Evaluation and validation of the AI product, which will include multiple iterations. The model will likely first be evaluated using retrospective data 3. Diffuse and scale the AI production 4. Continuous monitoring and maintenance of the AI solution for safety and effectiveness. [Image]

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Copyright 2020 © Stanford University 21 DESIGN AND DEVELOPMENT The design and development of the product begins with the problem: Is this an important problem and can it be answered with the data and AI model that is available? This step needs to be completed before deployment. It is important to define and characterize the problem to be addressed by an AI solution and to evaluate whether that problem can be solved (or is worth solving) using AI; meaning identifying a model output that is actionable and will impact either care delivery or patient outcomes. It is important to understand the outcome you are defining in the context of the healthcare system. It is also important to remember that medical definitions evolve over time. No medical definition is a static definition. When designing an AI solution, it is important to remember that the solution must be flexible, it must be able to accommodate a change in definition, a change in technology, a change in data type, or a change in the healthcare delivery system. The health care setting is not a static field; clinical care guidelines and standards of care are constantly evolving. AI algorithm must be flexible to the point that it can adapt these new functionalities or changes over time with minimal effort. Finally, it’s important to remember that with any change, there will need to be new code reviews, new model performance metrics, and new monitoring capacities setup. In the design and development component, stakeholder involvement must be assessed. Importantly, one must understand the intended user, their existing workflow, and how the output of the model will be delivered for action to this intended user. Each different intended user would need a unique pathway or dashboard to receive the model output. For the respiratory distress example, who are your stakeholders? The IT team - you need to get the data in real time The Clinician - you need to know how and where they want to receive the data Organizational leaders - do you have the buy-in to deploy an AI product in the healthcare system Financial leaders - who is going to cover the overhead costs UI experts - who will you design the interface for your model output EHR systems experts - how will you integrate your results with the existing system Any many more users[Image]

[Image]

Copyright 2020 © Stanford University 22 Identifying the right stakeholders and having them involved in the design and development of AI products is a key to successful deployment. Training Data - Data sources, data types and data availability are the crux of any AI product and knowing your data quality, reliability, representativeness and updates are imperative to AI deployment. AI products can utilize 3 different data sources: 1. Internal data sources 2. External data sources. This would be a team at one setting who is working with data from another setting. There is limited control on the quality and missingness of external data, unlike the internal data, unless one goes through a data agreement application - which can be laborious and difficult. 3. Public data sources. Many AI products are developed using public data sources, for example the MIMIC-III dataset, which is a freely accessible critical care database. An advantage of using public data sources is that the AI products developed and deployed with these datasets can be evaluated and reproduced by other scientists - increasing their reliability and impact in the field. Whether AI products are developed from internal, external, or publicly available datasets, each data source has opportunities and challenges related to its use. When designing and developing an AI product, it is important to consider the setting and funding (Implementation Costs - System Setup) of the original question, as this will drive or possibly hamper deployment. AI products that begin in industry are generally marketable and diffusible. Industry is often forced to partner with either academia or other industries in order to obtain the data needed for training and development of their models. Therefore, it is important to understand the data type and source from industry-lead products. Particularly their representativeness of the population to which the product will be applied. In academia, you have lead scientists who are thinking about innovative questions, innovative methodologies, and their focus usually begins with a specific research question. They have access to clinical data and clinical expertise. Given the collaborative academic environment, these teams are experts in developing multidisciplinary teams and these teams are easy to construct and maintain. However, academia has trouble recruiting and retaining [Image]

[Image]

Copyright 2020 © Stanford University 23 technical expertise - they generally cannot compete with industry salaries - and efficient scalability is generally not a strongpoint. AI products can also originate from the start-up setting. AI products from start-ups are usually self-funded, very efficient and focused on a single product or expertise. Often, when the startup companies have a good product, they are bought up by larger companies, who can then diffuse and scale their product. Philanthropic organizations generally work on target areas and partnerships. They might fund product development that addresses patient safety across settings. Sometimes philanthropy will partner with academics, industry, or government, as necessary, to develop their products of interest. The design and development of the AI product can begin in many different settings, including industry, academics, start-ups, philanthropy, or government settings. Each setting has a unique set of challenges and opportunities for deployment. EVALUATE AND VALIDATE The second component of AI Deployment is the evaluation and validation of the AI product. Prior to deployment, the initial AI product must undergo rigorous in silico evaluations which includes: Investigation of the clinical utility (or net utility) Statistical validity Economic utility of the AI model The utility of the AI solution (clinical utility) is likely the most important criteria to evaluate when considering the deployment of an AI solution in medical care. The utility of an AI solution relates to its applicability and impact on the healthcare system. Factors that can affect clinical utility: Who needs to take the action Lead time offered by the prediction The existence of mitigating action (or therapy) The ability to intervene The logistics and cost of the intervention, incentives, etc. To understand clinical utility for any AI solution you must ask: What is the primary task of the model and who are the main stakeholders, which is known as the outcome-action pair framework. [Image]

[Image]

Copyright 2020 © Stanford University 24 Clinical Utility is related to how well the AI product can demonstrate real-world workflow improvements or improve clinical care and patient outcomes. Clinical utility must be compared to baseline performance data - you have to show that the adoption of your product is useful, again in terms of clinical care (including clinical efficiency) and patient outcomes compared to current baseline performance data. Net utility is related to the usefulness of the AI solution given the prevailing constraints in the care environment. Methods such as decision curve analyses can quantify the net benefit of using a model to guide subsequent actions given the costs of alternative actions, their corresponding benefits, and the various measures of model performance. Net utility should be examined upfront, in order to have a useful model on the front line. One must consider the costs and benefits of the actions triggered by the AI product in order to form better decisions. The economic utility asks the question: Is there a real net benefit from the investment, or what is the cost of operational integration. For this, you must think about cost savings, increased reimbursement, or increased efficiency related to AI product. Work capacity refers to the ability of a system to respond to a prediction. Work capacity is an important component of AI evaluation that needs to be evaluated prior to considering the deployment of an AI solution in healthcare. During this evaluation, it is also important to consider optimal utility (taking action on people who will benefit the most). Optimal utility is extremely important when predicting the use of scarce resources. Net utility and work capacity are often ignored when AI products are reported in the scientific literature, yet they are essential to investigate prior to deploying an AI product in the healthcare setting. Another aspect of the evaluate and validate deployment component is statistical validity. This includes performance metrics, such as accuracy, reliability, precision, recall and calibration. The statistical validity of a model is essential and often reported as a marker of model performance. However, there are a lack of guidelines for discrete levels of performance. The most accurate algorithm is often not necessarily the best algorithm to deploy. Statistical validity is a component of deployment. Identifying standards and markers of algorithm performance is becoming more [Image]

[Image]

 25 important and an emerging concept suggests there be a compliance or conformance component to the performance evaluation of AI products. PRODUCT VALIDATION When deploying, one must ascertain human engagement. Will humans be involved in the loop or will the AI product work autonomously and define actionable insights? This is one of the most important aspects in the evaluation and validation of an AI product before live integration into clinical care. In silent mode, the AI product is deployed at point of care and predictions are made in real time but no action is taken on the predictions. The predictions are provided to the intended who then evaluates if the predictions are good or not or if they can be used to improve either the workflow or patient outcomes. This is crucial for finalizing workflows and product configurations, as well as the prospective, temporal validation of an AI product. For care integration, an important step in this pathway is to consider the human-machine interaction: 1. Defining the clinical problem. It is important to identify a problem that is suitable to be addressed by the AI algorithm. 2. Think about the human-machine interaction. The silent evaluation is very important to ensure the human - or intended user of the model output - is interpreting the model output correctly and the output is being applied appropriately and to the correct population. When we assess the human machine interaction, we need to think about how the clinical workflow is designed and how it will implement the AI product. In addition, you need to test the usability of the interface and the effect of your product on clinical decision making, including the legal and ethical issues of your AI product. Silent mode is an important, although often overlooked aspect of deployment. There is an enormous gap between AI developed for research and AI deployed into clinical care settings. Therefore, Clinical integration might be the most difficult part of the deployment process. Some key considerations for the clinical integration of an AI product includes (1) Structural Considerations, and (2) Partnership Considerations. Structural considerations:

[Image]

[Image]

Copyright 2020 © Stanford University 26 1. Organizational capabilities 2. Personnel capacity 3. Cost, revenue and value considerations a. Initial costs b. Anticipated return on investment c. The value related to the AI deployment 4. Safety and efficacy surveillance 5. Cybersecurity and privacy Partnership considerations: 1. Stakeholder consensus 2. Securing commitment from organizational leadership 3. Identifying leadership 4. Engaging stakeholders 5. Define milestones, metrics and outcomes to measure successful deployment Clinical integration, while only a small mark on our pathway, is likely the biggest hurdle to overcome for a successful AI deployment. Technology in the research environment greatly differs from the hospital environment. Significant effort and infrastructure investments are required to integrate AI products into real-time systems at point of care. One must consider the data platforms involved, the platform environment and the specific technology needed to get the data to run your models at point of care. Due to lack of interoperability and data standards, when another organization would like to implement a product already developed, they must also incur the same cost because they have to go through the same data gathering, cleaning, model evaluation and validation process as the original product development. Given the cost of real-world implementation of AI products, operational integration of the model should be considered carefully. DIFFUSE AND SCALE The third phase of deployment is to diffuse and scale the product. Diffuse and scaling the product comes after you solve local healthcare setting. Three different types of systems to consider when developing deployment modalities: [Image]

[Image]

Copyright 2020 © Stanford University 27 Fully integrated into the EHR system Partially integrated into the EHR system Standalone models In order to diffuse the AI products, it must be able to ingest different data from different systems and support on premise and cloud deployment. A well-designed product would be able to adapt to an epic system or a Cerner system, or any other homegrown EHR system. Most modules up to date are stand alone. It is important to understand that the majority of the products on the market, originally were developed in academics. Under the academic setting, products rarely get diffused and disseminated at scale, thus, they generally are coupled with industry partnerships to develop the full product. Products get externally licensed and scaled and diffused via commercial entities. Products are funded either through venture capitalists, or government in the healthcare set in academic setting. MONITORING AND MAINTENANCE Once an AI product is deployed a plan must exist to ensure the product will be continuously monitored and maintained. This will include regular architecture updates, addition of new training data, and perhaps yearly and/or irregular updates when industry reference files change. Deterioration of model performance can occur within the same healthcare system over time when, for example, clinical care environments evolve, patient populations shift, or rates of exposure or outcomes change. A new code to diagnostic code for a disease of interest or a new clinical definition for an outcome of interest might become available. This would require an update of the AI product to account for these changes. There are also minor model updates or bug fixes that will need to happen at irregular time frames. There are a number of approaches used to account for systematic changes to source data. These methods range from completely regenerating models on a periodic basis to recalibrating models using a variety of methods. However, the frequency and volume of these changes are not standardized. Major and minor model improvements or new functionalities to address evolving clinical deeds are important. [Image]

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 28 CHALLENGES OF DEPLOYMENT Deployment is complex and many issues need to be addressed before, during and after the implementation of an AI product in the healthcare system. An important challenge in AI deployment is the ethical challenge: Data security and patient privacy: patients may be unaware their data are being used, shared, or sold for AI product development. In some healthcare settings there is a waiver of consent. Training samples not being representative of the intended population: This issue is further amplified because most AI products are not transparent about their training samples and often the demographic distribution of the training data is not reported. Transparency: The details about the evaluation metrics and validation are often not reported. Interoperability: If one system would like to deploy a product that was developed at another setting, they will likely need to re-incur the same cost as due to interoperability, most systems cannot seamlessly exchange code. New standards are emerging, such as SMART and FHIR. o FHIR is a standard for health care data exchange, published by HL7 o The SMART App Framework connects third-party applications to EHR data, allowing apps to launch from inside or outside the user interface of an EHR system Lack of best practice standards for performance measures: There are no standard performance metrics for these models. Stealth science: Stealth science refers to science that is developed and disseminated without rigorous peer review. In industry, stealth science is common where companies may try to protect their trade secrets or avoid academic scrutiny. This is

a particular for AI and healthcare, particularly as many AI products are developed by industry. The models developed in research studies rarely translated into clinical care, hence, it is challenging to evaluate their real clinical and economical effect. Prediction of sepsis is a good example to go through to show how machine learning models for sepsis prediction can be translated into clinical care workflow: Sepsis Watch: The product was internally validated (prospectively) through a registered clinical trial and then licensed for commercial use in 2019 Dascena Insight: The product was externally validated in a prospective clinical trial and retrospectively across 6 institutes to access generalizability. [Image]

[Image]

Copyright 2020 © Stanford University 29 TREW Score: Developed using a publicly available dataset (MIMIC-II). TREW Score has been implemented in several hospitals. There are several more similar algorithms; One might ask, why are there so many algorithms performing the same predictions and what is the best algorithm to deploy? It is important to think about all of the challenges we have discussed regarding deployment and think about how one can evaluate or compare these like AI products. CITATIONS AND ADDITIONAL READINGS Gupta, A., T. Liu, and S. Shepherd. 2020. "Clinical decision support system to assess the risk of **sepsis using Tree Augmented Bayesian networks and electronic medical record data." Health Informatics J 26(2): 841-61.** Sendak, M. P., W. Ratliff, D. Sarro, E. Alderton, J. Futoma, M. Gao, M. Nichols, M. Revoir, F. Yashar, C. Miller, K. Kester, S. Sandhu, K. Corey, N. Brajer, C. Tan, A. Lin, T. Brown, S. Engelbosch, K. Anstrom, M. C. Elish, K. Heller, R. Donohoe, J. Theiling, E. Poon, S. Balu, A. Bedoya, and C. O'Brien. 2020. "Real-World Integration of a Sepsis Deep Learning Technology Into **Routine Clinical Care: Implementation Study." JMIR Med Inform 8(7): e15182.** Sendak, M.P., D'Arcy, J., Kashyap, S., Gao, M., Nichols, M., Corey, K., Ratliff, W. and Balu, S., 2020. A path for translation of machine learning products into healthcare delivery. European Medical Journal Innovations. Shah NH, Milstein A, Bagley, PhD SC. Making Machine Learning Models Clinically Useful. JAMA. 2019;322(14):1351–1352. doi:10.1001/jama.2019.10306 Topiwala, R., K. Patel, J. Twigg, J. Rhule, and B. Meisenberg. 2019. "Retrospective Observational Study of the Clinical Performance Characteristics of a Machine Learning Approach to Early Sepsis **Identification." Crit Care Explor 1(9): e0046.** [Image]

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Copyright 2020 © Stanford University 30 MODULE 4: DOWNSTREAM EVALUATIONS

OF AI IN HEALTHCARE: BIAS AND FAIRNESS LEARNING OBJECTIVES 1. Overview of Bias and Fairness in AI solutions in healthcare 2. Understand the different types of bias in AI healthcare solutions 3. Describe algorithmic fairness 4. Identify solutions to address bias and fairness in AI solutions BIAS IN AI SOLUTIONS Recently, reports have questioned whether AI solutions in healthcare might actually perpetuate discrimination if trained on historical data—which are often poorly representative of broader populations. Often, AI models are trained using historical or retrospective data which are often derived from affluent academic medical centers that likely do not contain all populations, particularly diverse populations for which the AI solutions will be applied. AI models exclusively trained on such data may further perpetuate disparities and fail to demonstrate external validity in broader patient communities. This is likely due to a lack of diversity represented in the training data, a lack of understanding how a disease may manifest and progress in different populations, and a lack of human understanding of the potential consequences and biases that may be inherent in AI solutions. Fairness and bias in AI solutions may be a larger problem in countries where important health disparities exist based on patient demographics, such as the United States. Therefore, as tools proliferate across clinical settings, it is important to think about and understand potential demographic biases underlying model development and deployment. Heart failure example: Not long ago we didn’t know that symptoms of heart attack look different in women compared to men, which led to differences in cardiovascular mortality rates between women and men. The problem here is that the model was developed based on male symptoms so the model might be very accurate for identifying males with heart attack symptoms, but it might not work well in females, who present with different symptoms. The predictive accuracy, when analyzed against the true clinical outcomes, will decline for women, but not men. Genomic database example: [Image]

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 31 Genomic databases used across the research community where we find major racial bias in the genomic samples collected. These databases are the center of precision medicine, where our ability to identify whether a genetic variant is responsible for a given disease or phenotypic trait depends in part on the confidence in labelling a variant as pathogenic.

Research suggests that these databases heavily reflect European ancestry, and in fact are missing major population-specific pathogenic information, particularly African-specific pathogenicity data. Therefore, genomic test results for persons of non-European ancestry could be less accurate, more challenging, or simply unattainable. Dermatology example: In general, patients with darker skin present with more-advanced skin disease and have lower survival rates than fair-skinned patients. It is possible that the only fair-skinned populations may benefit because of the lack of inclusion of darker skinned patients in model training and development. If the algorithm is basing most of its knowledge on how skin lesions appear on fair skin, then theoretically, lesions on patients of color are less likely to be diagnosed and therefore benefit from the AI solution. These examples provide you with an idea of how wide-spread the challenges are related to fairness and bias in AI solutions for healthcare. TYPES OF BIAS Bias can occur during almost any stage of AI model building and implementation - from data collection to model deployment. Types of bias: Historical bias Representation bias Measurement bias Aggregation bias Evaluation bias Deployment bias [Image]

[Image]

[Image]

 32 Historical bias occurs if the present or past state of the world influences a model in a way such that its predictions are considered unfair given societal values and norms. Historical bias refers to judgement based on preconceived notions or prejudices. AI algorithms are entirely data dependent and historical bias encoded in real-world data cannot even be overcome by perfect sampling and feature selection. Therefore, it is important to remember that historical healthcare data, in general, is extremely male and extremely white, and this has real-world impacts. Representation bias (also called sampling bias) arises when the sample collected to train an AI solution does not represent the actual distribution of the population it is intended to be applied to. It occurs when certain parts of the final use population are underrepresented in the training data. Measurement bias arises in situations if the noise is not randomly distributed but differs across groups, which leads to differential performance. Often the only available and measurable features as well as labels are only noisy proxies of the actual variable of interest. Usually, one cannot change the data, some historical biases might be indistinguishably linked to the data – but the awareness about the problem is important and mitigation strategies can be identified by taking preventive measures such as pre- and post-processing actions. Aggregation bias occurs while developing the model when we try to combine different populations whose underlying distribution of the outcome under study differs. This problem is

known as inframarginality and requires separate models for the different populations or including the demographic variable into the model to account for the systemic differences. In terms of model development, one size does not fit all. In order to identify aggregation bias, developers need to understand the meaningful distinct groups and reasons why they are different from each other. Evaluation bias occurs during the model validation and tuning. Evaluation bias arises if the testing data, which often includes external benchmark datasets, is not representative of the final population to which the AI solutions will be applied. This is the difference between the data used for model evaluation and the data used for model's real-world predictions. Since developers mostly use a benchmark dataset or a synthetic dataset for training, their evaluation often does not fit the realworld. A solution of this problem is external validation of the AI model on a different unseen data selected from the targeted population. Evaluation bias can arise if inappropriate performance metrics are used. Evaluation bias also refers to usage of evaluation metrics inefficiently and to avoid it, using granular and comprehensive evaluation metrics is suggested. Deployment bias arises during the implementation of the model. It refers to using the model inappropriately or misinterpreting its results. In other words, if the model's intended use is different from the way it is used, deployment bias occurs. Deployment bias is the interaction of society with the AI solution - how society or the medical community uses the AI solution and its output. [Image]

[Image]

 33 ALGORITHMIC FAIRNESS Ethical analysis of AI Solutions in healthcare demand that we take a view of fairness, or more appropriately, justice that centers on the health and lives of people, not the outputs alone. A lot of historical medicine has been influenced by white normativity, which is the basis of many medical facts. This is evident because of a lack of inclusion of diverse patients in clinical research. have knowledge that the insiders don't. Bringing diverse perspectives actually enhances the quality and accuracy of your scientific endeavor. A key difficulty in developing fair AI algorithms is the fact that no universal notion of fairness exists. Many different definitions have been proposed by researchers over the years and they can be broadly regrouped into three main classes: anti-classification, classification parity and calibration. These fairness definitions have been shown to all suffer from significant statistical shortcomings. Therefore, special caution and awareness about the notion's limitations and weaknesses is always necessary when applying these concepts in model evaluation settings. Anti-classification or "fairness through unawareness" 1. Requires the exclusion of any protected attributes in the outcome modeling 2. Requires the omission of any unprotected characteristics that are proxies of protected attributes

The main shortcoming of this fairness definition is that some clinical risk models need to explicitly include protected attributes for it to be equitable. In particular, this applies to situations where the true underlying risk distribution differs across subpopulations, known as the problem of inframarginality. Therefore, an accurate model must include protected attributes, but must also learn to avoid bias based on these attributes. AI solutions that use datasets which may be under-representative of certain groups, may need additional training data to improve accuracy in the decision-making and reduce unfair results. Anti-classification requires the exclusion of any protected attributes in the outcome modeling. Classification parity of fairness asks for equal predictive performance across any protected group. When selecting the metrics to examine, it is important to keep in mind the actionable insights resulting from a model output. Two measures have received particularly high attention by the machine learning community: [Image]

[Image]

 1. False positive rate: The probability of predicting a positive outcome when the true outcome is negative should be the same regardless of protected attributes of the patients 2. Proportion of positive decisions: The probability of predicting a positive outcome should not vary across different demographic groups given all else equal. It is also known as demographic parity as it requires the classifier's predictions to be independent of protected traits. These definitions are problematic when the risk distributions are different for different groups, a problem known as infra-marginality. Classification parity asks for equal predictive performance across groups. Calibration is when a model reaches a good agreement between model predictions and observed outcomes. Calibration means that when conditioning on risk estimates, outcomes should be independent of the protected attributes. Think of calibration as a comparison of the actual output and the expected output given by a system. Calibration is open to manipulation of risk distributions for different groups. Model calibration is an important aspect of development and must be evaluated before model deployment. Applying fairness measure: Anti-classification: Checking the requirement of not using any protected attributes in the decision rule. It gets more complicated when trying to impose the stricter condition to also not use any proxies of protected attributes. Classification parity: Once the 2-3 most relevant performance metrics are selected, we evaluate performance differences on the test-set across different demographic groups. It is always better to calculate empirical confidence intervals of this statistic to decrease the dependence on the test set. Calibration: Only look at the overall rate of predicted and observed outcomes across demographic groups. TRANSPARENCY It is important to think about whether the population captured in the EHRs system is representative of the broader community, particularly if these are coming for an

academic medical center. If AI solutions are being developed on non-representative populations, the utility and applicability of these advances for the broader patient community falls into question. This relates to an important [Image]

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 35 issue around transparency and reporting and many AI studies, particularly machine learning, do not report the demographic breakdown of the training data used to develop and train the models. Lack of reporting makes it very difficult to evaluate the bias and fairness of the AI solution and its applicability across populations. Studies that mentioned variables often did not report if they were included as model inputs. In the studies that reported demographics, the average populations included higher proportions of whites and Blacks yet fewer Hispanics compared to the general population. While each study might not be applicable to the general population, these findings emphasize the present lack of transparency in reporting details of training data used for development and evaluation by machine learning models in healthcare. To ensure the unbiased deployment and application of any AI model in healthcare, detailed information on the data used to develop and train the model are necessary. As a solution for transparent reporting and to identify best-practices for designing machine learning models to account for biases and fairness, MINIMAR template is suggested. (MINIMAR = The MINimum Information for Medical AI Reporting) MINIMAR Requirements: 1. Include information on the population providing the training data, in terms of data sources and cohort selection 2. Include information on the training data demographics in a way that enables a comparison with the population the model is to be applied to 3. Provide detailed information about the model architecture and development so as to interpret the intent of the model and compare it to similar models [Image]

[Image]

 36 4. Model evaluation, optimization, and validation should be transparently reported to clarify how local model optimization can be achieved and to enable replication and resource sharing You can understand that by providing these details of the training data and population, model design and intent, an end-user will have a great understanding of how to best deploy the model and in which populations. DOWNSTREAM EVALUATIONS While we have covered many topics in this lecture regarding bias and algorithmic fairness, there are still many more challenges and opportunities for Fair AI research: Defining

fairness: There are several definitions of AI fairness that have been proposed in the literature. It is nearly impossible to understand how one fairness solution could address all challenges. Identifying the correct or best definition is an ongoing debate. From equality to equity: Equity suggests that each group is given the amount of resources needed to have similar outcomes. Understanding how to develop and implement a model that provides both equality and equity presents a paradigm shift in the way to think about healthcare delivery and is an active area of research. Identifying biases in models, and particularly in datasets: Many biases are systematic and we are often unaware they exist. We still have a long way to go before we can systematically mitigate these biases and provide our professionals with the appropriate tools they need to address these issues at point of care. The perspective collection and reporting of AI outputs, clinical recommendations and patients decisions coupled with eventual outcomes is essential in being accountable as healthcare institutions and as clinicians. This work and transparency in reporting AI solutions is absolutely critical for populations who have difficulty trusting the medical establishment. The key is to use AI in a way that actually does benefit all groups, which requires thoughtful evaluations and human interpretations. CITATIONS AND ADDITIONAL READINGS Adamson, A. S. and A. Smith. 2018. “Machine Learning and Health Care Disparities in **Dermatology.” JAMA Dermatol 154(11): 1247-48.** [Image]

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Copyright 2020 © Stanford University 38 MODULE 5: THE REGULATORY ENVIRONMENT FOR AI IN HEALTHCARE LEARNING OBJECTIVES Understand why most AI solutions in healthcare have not received regulatory approval, to date Describe best practices for AI development, in particular good machine learning practices Recognize the risk framework used to classify AI solutions in healthcare that is used by the Food and Drug Administration (FDA) Know the 3 concepts of model properties that can be regulated Understand main differences between EU, China and US regulations on AI solutions OVERVIEW International Medical Device Regulators Forum (IMDRF): An international workgroup composed of AI regulators that come together to develop a path for standardized AI regulations. Software as a Medical Device (SaMD): Software intended to be used for one or more medical purposes that perform these purposes without being part of a hardware medical device. Medical purpose: Intended to treat, diagnose, cure, mitigate, or prevent disease or other conditions SaMDs are NOT part of hardware COMPONENTS OF REGULATION The SaMD can be described in 3 components: SaMD inputs, SaMD algorithms, and SaMD outputs. [Image]

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Copyright 2020 © Stanford University 39 This is the framework used by the US Food and Drug Administration to regulate AI Solutions / SaMD. The FDA's regulatory framework starts with a Market Application, which includes a definition statement and category (I, II, III, or IV). The category is based on the risk associated with the use of the proposed AI solution. Depending on the category defined, data requirements necessary for regulation may include: A premarket notification (or 510(k) statement of equivalency) De Novo request (no similar product exists on the market for comparison) Premarket Application (PMA), which is reserved for high risk AI applications Definition Statement is required for every SaMD application and is used to identify the submission category, which defines risk and subsequent data requirements. The definition statement must: Clearly identify the intended medical purpose of the model (treat, diagnose, drive clinical management, inform clinical management) State the healthcare situation or condition that the AI model is intended for, which includes critical, serious, and non-serious conditions Include the intended population for the application Identify the intended users (or stakeholders) of the model Using information from the definition statement, the SaMD Category is defined, which is based on a risk framework developed by the IMDRF. [Image]

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[Image]

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 40 In the framework, risk is set by the intended use of the SaMD and the state of the health situation it targets. The columns in the grid represent the Significance of information provided for a healthcare decision. This is the ACTION in the outcome-action pairing framework used for AI evaluation. The rows represent the State of healthcare situation or condition, which identifies the state of the healthcare situation or condition as critical, serious, or non-serious An example that demonstrates the definition statement and risk category: A SaMD AI application that "drives clinical management" in a "critical healthcare situation or condition" (Category 3): In a hypothetical situation, an AI application is developed for ICU patients that receives electrocardiogram, blood pressure, and temperature signals from a patient monitoring system. The physiologic signals are processed and analyzed to detect patterns that occur at the onset of patient instability and deterioration, a threshold set by a utility analysis. When physiologic instability is detected, an alarm is generated to indicate that immediate clinical action is needed to prevent potential harm to the patient. SaMD regulations place devices into four categories based on the risks associated with the use of the device. Category I being the lowest risk; Category IV being the highest risk. Category I devices require general controls Category II devices are medium to moderate risk and require the same general controls and some additional controls Category III and IV applications require "general controls" and pre-market approval which is the most stringent regulatory category. To date, there are very few Category IV SaMDs that [Image]

[Image]

[Image]

Copyright 2020 © Stanford University 41 are approved in the US. These are high risk, generally life-supporting, life-sustaining AI applications. All applications must include general controls. General controls require that all AI solutions comply with three components: 1. Quality system regulations 2. Current good manufacturing practices 3. Properly labeled - the label of the device follows FDA guidelines and regulations Any adverse event associated with the AI solution must be reported. Quality System Regulations: Manufacturers are required to have processes in place for controlled bug resolution, incident reporting, standardized design processes and overall risk management. CLINICAL EVALUATION PROCESS All AI applications needing regulatory approval must include general controls. As part of the general control, the Clinical Evaluation Process is a framework used by regulators to understand quality system regulations. The IMDRF defines the clinical evaluation process as ongoing activities conducted for the assessment and analysis of a SaMD's clinical safety, effectiveness and performance. The Clinical Evaluation Process includes three components. Valid Clinical Association (Category I): Is there a valid clinical association between your SaMD output and your SaMD's targeted clinical condition? Scientific validity of the AI solution or the extent to which the SaMD's output is clinically accepted or well-founded (based on an established scientific evidence), and accurately corresponds to the healthcare situation and condition identified [Image]

[Image]

Copyright 2020 © Stanford University 42 in a real-world setting. An indicator of the level of clinical acceptance and confidence one can have of the SaMD's output Minimum evidence to support the clinical association could include: Literature searches Original clinical research Professional society guidelines Examples of how your model can generate new evidence Secondary data analysis The performance of a clinical trial based on your AI solution Required for AI regulation and ensures the clinical acceptance and uptake of the AI solution in the healthcare setting Novel associations - associations that are newly discovered by your AI: As literature and existing randomized clinical trials do not exist for this association, there are other solutions to regulate this software, which may include performing a clinical trial or secondary data analyses Analytical Validation (Category II): Evaluates whether your AI solution correctly processes input data to generate accurate, reliable, and precise output data. Part of the verification and validation phase that should be performed by the AI manufacturer. Provides objective evidence that the AI solution was correctly constructed and the data processing is reliable. May come as part of your good software engineering practices or from generating new evidence through use of curated databases or previously collected patient data Clinical Validation (Category III): Does the use of your AI's output data achieve your intended purpose in your target population in the context of clinical care? Related to the positive impact of an AI Solution on the health of an individual or population. Clinical Validation should be: Measurable, patient-relevant clinical outcome(s) Including outcome(s) related to the function of the model A positive impact on an individual or public health[Image]

[Image]

Copyright 2020 © Stanford University 43 Prior to product launch of the AI product (pre-market), evidence must exist on the following: AI accuracy Specificity Sensitivity Reliability Usability Limitations Scope of use in the intended use environment with the intended user After launching the product (post-market) the product must: Continue to collect real world performance data to Further understand the healthcare needs to ensure the AI solution is meeting those needs Monitor the product's continued safety, effectiveness and performance in real-world use The IMDRF identifies that clinical validation is a necessary component of regulation and that it can be demonstrated through several paths, including: Referencing existing data from studies conducted for the same intended use Referencing existing data from studies conducted for a different intended use, where extrapolation of such data can be justified Generating new clinical data for a specific intended use The Clinical Evaluation Process is an important component of the regulatory environment. FDA APPLICATION In addition to the general control process, there are other regulatory control requirements (data requirements) that accompany an SaMD application that depend on the application's category of risk (1 - 4), which can include one of these regulatory components: Premarket Notification 510(k) De Novo Notification Premarket Approval (PMA) [Image]

[Image]

Copyright 2020 © Stanford University 44 Premarket Notification 510(k) is the simplest data requirement, if the SaMD is similar to a product already on the market. The intent is to inform the regulatory agencies that the AI solution is safe and effective, which is determined by demonstrating the AI solution is equivalent to a legally marketed device, often known as a "predicate". To determine equivalence, the AI solution must have: The same intended use as the predicate AND have the same technological characteristics, OR The same intended use as the predicate and a different technology that will not raise safety or efficacy questions AND the AI solution is at least as safe and effective as the predicate As more and more applications become approved, pre-market notifications will become an easier and efficient pathway towards regulation. De Novo Notifications: Submitted when there is no “predicate”. Limited to Category I and Category II SaMDs and are a risk-based classification process. The de Novo notification should include: Clinical data (if applicable) that are relevant to support the assurance of the safety and effectiveness of the AI solution Non-clinical data including bench performance testing Description of the probable benefits of the AI solution when compared to the probable or anticipated risks when it is used as intended Premarket Approval (PMA): Required for high risk SaMDs (Category III and IV). The most stringently regulated application required by the FDA. Includes rigorous technical studies, nonclinical laboratory studies, laboratory studies, and clinical investigations. Before PMA approval, the applicant must provide valid scientific evidence demonstrating reasonable assurances of safety and effectiveness for the device’s intended use. [Image]

[Image]

 45 After an AI solution receives regulatory approval, a modification may be required in certain circumstances. A modification request must be submitted if there is new risk or a change of an existing risk. A modification may be required if there is: A change to risk controls to prevent significant harm A change that significantly affects clinical functionality or performance. A change in clinical functionality could include o New indications for use o New clinical effects o Significant technology modifications that affect performance characteristics SaMD modifications generally fall into three categories: 1. Change in performance o Example: The incorporation of new training data or a change in AI architecture which could alter performance 2. Change to the model Input o Example: The incorporation of different sources of the same input or adding new inputs that were not previously considered 3. Change of the intended use of the output o Example: Change in disease (apply to new condition) Software modifications are common and essential in the total life cycle of the AI Solution. PRODUCT APPROVAL In line with the framework proposed by IMDRF, the FDA has developed the following diagram related to the total product life-cycle (TPLC) for an AI workflow. [Image]

[Image]

[Image]

Copyright 2020 © Stanford University 46 There are 4 distinct components in the total product life-cycle: 1. The culture of quality and organizational excellence, which is also referred to as Good Machine Learning Practices. This includes all components that must be considered when developing an AI solution, such as o Data selection and management o Model training and tuning o Model validation 2. Premarket assurance of safety and efficacy of the AI solution. It is expected that safety and effectiveness are continually monitored throughout the life cycle, including patient risks and patient safety. Regulators expect a manufacturer to perform a risk assessment and evaluate that the risks are reasonably mitigated throughout the TPLC. 3. Regular monitoring of safety and intended use, which is used to identify when a software modification is required. The regular monitoring of the deployed model is necessary and should include the ability to log and track model performance. 4. The Continuous Learning expected from an SaDM that Leverages Real World Data. “Continuous learning” is not “machine learning.” It refers to collecting post-market information to update and evaluate your existing AI solution. The TPLC diagram demonstrates how regulators are thinking about using real world data for continuous learning, the basis of the learning healthcare system. [Image]

[Image]

[Image]

[Image]

[Image]

Copyright 2020 © Stanford University 47 Locked algorithms are AI solutions that provide the same results each time the same input is provided. Generally fixed functions to a given set of inputs, and may use a manual process for updates and validation. How to regulate “adaptive” AI solutions or machine learning applications? Following deployment, these types of adaptive models may provide a different output in comparison to the output initially approved for a given set of inputs. These automated changes use a well-defined process, which aim to improve outcomes based on new data or additional data that is taken as an input. There are two stages to an adaptive algorithm: o Learning stage: The algorithm “learns” how to change its behavior, based on the addition of new input types or new cases to an already existing training set o Update stage: The algorithm will update when the new version of the algorithm is deployed This is a paradigm shift in the regulatory process and requires a new total product lifecycle that allows these devices to continually improve while providing effective safeguards. Considerations for the regulatory environment for an AI model: 1. Developers should be aware of SaMD

Risk Classifications, total product life-cycle (TPLC) and Good Machine Learning Practices (GMLP) [Image]

[Image]

Copyright 2020 © Stanford University 48 2. The population, performance and intended use are aspects of the model that can be regulated and any change to these characteristics after approval would require a notice of modification 3. The intended use and state of the healthcare situation of the AI model drives the level of regulation and risk category 4. Safety and effectiveness must be continuously monitored post-market using real-world data to ensure a learning health system Example - Arterys Indications for Use: Arterys is a software that uses cardiovascular images acquired from magnetic resonance (MR) scanners. It analyzes blood flow from the heart using the MR images. The output is intended to be used to support cardiologists, radiologists, and other healthcare professionals for clinical decision making. Risk Classification: Class II o Significance of information is to “inform clinical management” o State of healthcare situation or condition is “critical” Example - IDx-DR Indications for Use: IDx-DR is a retinal diagnostic software device that incorporates an adaptive algorithm to evaluate ophthalmic images for diagnostic screening to identify retinal diseases or conditions Risk Classification: Class II o Significance of information is to “drive clinical management” o State of healthcare situation or condition is “serious” Example - Guardian Connect (Medtronic) Indications for Use: The Guardian Connect system is indicated for continuous or periodic monitoring of glucose levels in the fluid under the skin, in patients (14 to 75 years of age) diagnosed with diabetes. Risk Classification: Class II o Significance of information is to “drive clinical management o State of healthcare situation or condition is “serious” The FDA does not regulate certain types of clinical decision support (CDS) tools under 21st century cures act. Three criteria determine whether CDS are regulated as an SaMD: [Image]

[Image]

[Image]

Copyright 2020 © Stanford University 49 1. The software cannot receive, analyze or otherwise process a medical image or signal from an in vitro diagnostic device or from any other signal acquisition system 2. A healthcare professional must be able to understand the basis of its recommendations 3. The software cannot be intended as the sole source of recommendations regarding treatment, diagnosis or prevention of a disease The FDA doesn’t regulate AI solutions that are “laboratory-developed tests” designed, developed and deployed within a single health care setting. FDA’s

Digital Health Software Precertification Program (Pre-Cert): Streamlines regulation of AI solutions Organizations may become “approved” which would allow them to bring products to market without a premarket review, provided the product is “lower risk” Once certified, organizations can make certain minor changes to its AI products without having to submit a modification request An organization must demonstrate the FDA’s five quality and organizational excellence principles in order to be considered for the Pre-Cert program: 1. Product quality 2. Patient safety 3. Clinical responsibility 4. Cybersecurity responsibility 5. Proactive culture GLOBAL ENVIRONMENT It is important to note that there are some differences across the globe regarding regulatory guidelines for AI in healthcare. EU’s General Data Protection and Regulation (GDPR) Outlines a comprehensive set of regulations for the collection, storage, and use of personal information which may be used in AI solutions Describes the right of citizens to receive an explanation for algorithmic decisions. The implications would exclude the use of many types of algorithms used today in advertising and social networks - and eventually healthcare. [Image]

[Image]

Copyright 2020 © Stanford University 50 Provides protection of data from EU citizens but given the global AI momentum, the laws may impact companies from the US and worldwide A critical component of this regulation is Article 22: “Automated individual decision making, including profiling.” Article 22 requires explicit and informed consent before any collection of personal data. The “right to explanation” o Requires that meaningful information about the AI logic as well as the potential significance and consequences of the data-driven system are provided upon request o Refers to the use of the black box algorithms o Could potentially limit the types of models that manufacturers are able to use in health-related applications o Holds the manufacturers of AI-based technologies more accountable While there are some differences between the US and EU regulations, a common theme from both entities is the protection of the individual, their data, and their right to information. China leads in the number of AI patents as a result of this favorable environment. AI governance in China is aimed at the development of “responsible AI” and focused on the societal beneficiary rather than the individual beneficiary. China has also put a focus on regulating education so that the nation produces more STEM workers. China requires businesses and private citizens to share their data with the government – almost the opposite of US and EU regulations. The incentives for data sharing and elimination of data silos could catapult China in clinically meaningful AI technologies. Principles for AI governance released by China’s Ministry of Science and Technology (MOST) include: 1. Harmony and friendliness 2. Fairness and justice 3. Inclusivity and sharing 4. Respect privacy 5. Secure/safe and controllable 6. Shared responsibility 7. Open collaboration 8. Agile governance

[Image]

[Image]

 51 China is rapidly developing important AI solutions in healthcare and these governing policies are essential to balance both the innovation of technology as well as the safety of healthcare delivery. The White House Office of Management and Budget (OMB) document provides Guidance for Regulation of AI Applications. The regulations are not specific to healthcare, but they provide the umbrella of regulations applied to AI solutions. The ten guiding principles are aligned with the FDA's regulatory processes: 1. There must be public trust in AI o Public trust and validation is critical to the adoption and acceptance of these technologies in the healthcare sector o Privacy and other risks must be addressed with appropriate mitigation strategies that are well documented and transparently reported 2. There must be public participation in AI development o There should be ample opportunities for the public to provide information and participate at all stages possible of the "rule-making" process 3. AI solutions must be based on scientific integrity - clinical validation in the clinical evaluation process proposed by the IMDRF o AI in healthcare should be based on scientific and technical information and processes that are likely to have a clear and substantial influence on public policy or private sector decisions - and these standards should be held to the highest level of quality, transparency, and compliance o Best practices include clearly stating the strengths, weaknesses, intended optimizations or outcomes, bias mitigation, and appropriate uses of the AI application's results o Data used to train the AI system must be of sufficient quality for the intended use 4. Every AI solution must have a Risk Assessment and Management component o A risk-based approach should be used to determine which risks are acceptable and which risks present the possibility of unacceptable harm, or harm that has expected costs greater than expected benefits o If an AI tool fails, the magnitude and nature of the consequences will inform the level and type of regulatory effort that is appropriate to identify and mitigate risks 5. Benefits and Cost o For an AI algorithm to be deployed, it must offer significant potential benefit [Image]

[Image]

 52 o Before implementing an AI solution, agencies must consider the full societal costs, benefits, and effects related to the development and deployment of these applications 6. AI solutions must be flexible o Performance based and flexible approaches should be easily adapted and updated - this would include the continuous monitoring of real world evidence to improve upon the AI solution 7. AI solutions must be fair and non-discriminatory o Transparency

regarding potential biases and discriminatory aspects of the algorithm is becoming more and more important as we see potential harm due to biases promoted or exacerbated through AI 8. AI solutions that provide Disclosure and Transparency will promote public trust o Healthcare systems should disclose when AI solutions are in use and how these applications can impact patients and decisions 9. All AI solutions must address Safety and Security issues o Safety and security should be considered throughout the design, development, deployment, and operation process o There should be controls in place to ensure confidentiality, integrity, and availability of the information processed, stored, and transmitted by AI systems 10. AI solutions must include multidisciplinary stakeholder involvement or Interagency Coordination o All sectors affected by the AI solution should coordinate and share experiences and challenges of AI solutions Safety, transparency and multidisciplinary aspects are key ingredients to a successful and wellthought out AI solution. CITATIONS AND ADDITIONAL READINGS FDA, U. 2019. "Proposed regulatory framework for modifications to artificial intelligence/machine learning (AI/ML)-based software as a medical device (SaMD)." FDA. US Department of Health and Human Services. Software as a medical device (SAMD): Clinical evaluation. Guidance for industry and Food and Drug Administration Staff. 2017. [Image]

[Image]

[Image]

Copyright 2020 © Stanford University 53 BEST ETHICAL PRACTICES BEST ETHICAL PRACTICES – PROBLEM FORMULATION In this section, we will learn about specific actions that have been recommended as best ethical practices in the development and deployment of machine learning-based AI in health care applications. Our thinking about best ethical practices will start at the stage of problem formulation and determining the purpose of the AI system that you want to develop. First, is the intent ethical, is its purpose to enhance the health and well-being of patients? Even if the intended purpose seems ethical, could there be negative consequences if misused? Increasingly we are seeing data scientists challenging the purpose and the uses of the products they are developing, whether the purpose is to target advertising to individuals using social media, or to identify individuals through facial recognition algorithms. For example, the Association of Computing Machinery, one of the leading professional organizations of computer scientists, has urged a suspension of the use of facial recognition technologies because of their potential for prejudicial impact on human and legal rights. The Association has also asserted that developers and operators, as well as users of facial recognition technology are accountable for these systems' use and misuse. While these might seem like questions that don't need to be asked in the health care AI context because the answers are obvious, they are especially important to ask for AI developed by teams that have conflicting or competing interests. Health care settings are rife with such competing interests because they are operating under resource constraints, including constraints on finances, personnel, equipment and supplies, not to mention financial incentives to improve care. Together, these interests and incentives all create pressure to avoid certain kinds of patients, or avoid providing certain kinds of treatments. AI developers have to be vigilant about mitigating conflicts of interest. But how? We'll talk about this more specifically, but first, let's consider the issue of problem formulation. Formulating the AI problem involves translating a high-level goal such as "improving patient care" into actionable questions that can be answered by available data. The challenge is that actionable questions and available data are usually limited, so a lot can get lost in translation, and practical constraints on problem formulation can lead to undetected errors and bias. A fairly common and seemingly straightforward task such as risk stratification is often fraught because risk can be defined in many different ways, and usually in ways that are not measured directly but through proxy variables. We have seen that the use of health care costs as a proxy for risk or health care needs has led to racial bias because risk scores generated from predictive models based on costs underestimated [Image]

[Image]

 the health care needs of Blacks as compared to Whites with the same risk score. Fortunately, such bias can be tested for, at least for variables for which you suspect underlying bias exists such as race or gender. In this example you would be looking at whether the relationship between risk scores generated by the model and the variable of interest, health needs, differed by race. Similarly, risk stratification on the basis of cost as a proxy for health needs tends to be biased towards older people with complex chronic conditions at the end of life, when most health care costs are incurred, and against children with acute, potentially fatal but treatable conditions. If the question to be answered is "who needs the most care" it is important to remember that this question is not merely quantitative but also a values question that demands nuance. How is "care" defined? Does the question distinguish between acute and chronic care? How is "need" defined? If patients have untreatable conditions and therefore typically do not receive costly care, does that mean that a model should assign them low risk scores, or that they do not need care? Questions to ask in problem formulation Is the purpose of the AI system to enhance the health and well-being of patients? Could there be unintended consequences if misused? What is the intent of the system – to identify, classify, predict? What is being identified, classified or predicted? o E.g. risk, disease, prognosis, costs, health care utilization, admission or readmission, treatment efficacy, decompensation How are these terms defined? Closely related to problem formulation is the choice of data. Of course, data that are already available are the easiest to use. But that doesn't mean that these are the right data. For example, electronic health records might be plentiful but they lack a lot of information that could be important to predictive models, such as environmental exposures, diet, or socio- economic factors, all of which we know are highly determinative of health and disease. EHRs from one hospital or health system, or insurance claims data also often do not provide a longitudinal timeline of data points for a patient over time because patients often move between health providers and insurers. And relying on data from a single time point could be misleading. On the other hand, grouping data together can mask important patterns. In our example of risk stratification, a model trained on data from people of a mix of age ranges could be masking patterns in data from a subpopulation, such as younger people. Of course, detailed knowledge of differences between subpopulations is necessary to know whether customized models are required. How does one know this in advance? [Image]

[Image]

 55 Questions to ask about data How well are the variables in the model represented by available data? Are proxy measures being used? Are there important variables that are likely to be associated with main outcome measures that are not represented in available data? Are there likely to be differences between subpopulations in main outcomes, especially by legally protected characteristics such as age, race, ethnicity, or gender, or socially important characteristics such as income and education? Best practices point to the need to include practicing clinicians who are intimately familiar with the relevant patient populations and conditions, and the data that are necessary for modeling They need to know what data are actually available, and the limitations of the data, especially in terms of likely biases. It is important to have team members with an understanding of the source of systematic error, such as whether error is introduced because of lack of data from specific subpopulations, physician biases (such as the lower likelihood of women with heart disease to get a diagnosis of heart disease as compared to men), or broader social inequities such as differential access to care. That is because understanding the source of systematic error indicates whether and how models can account for bias. Clinician team members can be critical at the problem formulation stage in particular, because deep clinical knowledge is so important to understanding the implications of asking questions in a particular way. BEST ETHICAL PRACTICES – IDENTIFYING CONFLICTS OF INTEREST In this section, I want to talk about best ethical practices that address the issue of conflicts of interest. In medicine and biomedical research, we deal with conflicts of interest all

the time, and have developed ways of mitigating their negative effects. So, how is that done, and how can we apply those strategies to the development and deployment of AI in health care settings? First, what is a conflict of interest? For our purposes, they exist only when there is a primary interest that is a duty. An example of a primary interest is the clinician's or hospital's duty to care for their patients. However, we all have multiple interests, some of which can compete or conflict with these interests. These other interests are called secondary interests, and can include, for example personal or institutional financial interests, duties to others such as people who are not the clinician's or hospital's own patients, or reputational interests, either positive or negative. [Image]

[Image]

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[Image]

Copyright 2020 © Stanford University 56 So, why are these secondary interests a problem? In the health care setting, the problem arises because in the course of carrying out duties, many decisions have to be made on behalf of patients, and decisions are often based on individual judgement. And judgement can be influenced by all of these secondary interests, in such a way as to subvert from the primary duty and cause harm to patients. An example of this in clinical care is the influence of pharmaceutical sales representatives who try to persuade doctors to use their drugs instead of the competitor's drugs, when in some cases the competitor's drug might be better for the patient. Influence might be exerted in the form of financial incentives or gifts. These secondary interests have been linked to significant changes to prescribing practices, so they have real effects. Although physicians think it is impossible to be influenced by accepting a slice of pizza paid for by a pharmaceutical company, there is a study demonstrating that prescriptions can be more than doubled specifically for drugs sold by the providers of meals over other similar drugs, with increases in prescribing seen even after just one Photo by Pinar Kucuk on Unsplash [Image]

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Copyright 2020 © Stanford University 57 meal and rising with every additional day of meals provided. So, part of the problem is that these secondary interests have effects without our being aware of them. In biomedical research, secondary interests can also include financial factors such as the promise of obtaining research grants or consulting fees from sponsors, or stock or royalties from companies whose products are being studied. Secondary interests can also be reputational or ideological, for example, a strong desire to gain fame or promote a specific hypothesis. In the case of research, the primary interest is usually the integrity of the research, although for research involving human subjects, the health and welfare of individual research participants is also of high priority. But what are the possible negative effects of influences of secondary interests? Again, let's look at all the decisions that are made in the course of research that could be subverted away from serving the primary interest of the integrity of the research. One is the formulation of the research question. Is the question in service of corporate interests or meeting a real patient need? For example, is a clinical trial designed to test a drug in order to serve patients who have no other therapeutic options, or a reformulation of an existing drug that will extend patent protection and a company's market position? Another way that secondary influences can have effects is through creating unconscious bias, leading to inaccurate measures of treatment effects. We know that, in general, systematic error tends to inflate effect sizes. Clinical researchers have long recognized this, and therefore use techniques such as randomization and blinding to reduce bias, or systematic error. Choosing to use such techniques represent design decisions. Other important design decisions that could be influenced by secondary interests are what populations and data you choose to study, and how you choose to analyze, report or share data and findings. Ways to Mitigate Conflicts of Interest: Source: DeJong C, Aguilar T, Tseng C, Lin GA, Boscardin WJ, Dudley RA. Pharmaceutical Industry–Sponsored Meals and Physician Prescribing Patterns **for Medicare Beneficiaries. JAMA Intern Med. 2016;176(8)** [Image]

[Image]

Copyright 2020 © Stanford University 58 Disclosure • Publicizing the secondary interest Mediation • Putting financial (secondary) interest in blind trust Putting some decisions about the primary interest in the hands of an independent body Recusal • Removing financial (secondary) interest or replacing the person with the primary interest So, what do we do to mitigate potential effects of conflicts of interest? We can focus mitigation strategies on protecting the primary interest, eliminating or reducing the secondary interest, or both. The most common strategy, which you have probably employed or been asked to employ, is disclosure, which entails making public your secondary interests, such as disclosing speaking fees or research sponsorships from companies whose products are related to the primary interests. Although it is the most common strategy, it is also the weakest because it does nothing to reduce or remove the secondary interest, but relies on those to whom the interest is disclosed to understand the implications of the disclosure and act in some way to counteract possible negative effects. Other strategies that are more robust involve mediation of the primary or secondary interest, such as placing a financial interest in a blind trust, or having an independent oversight committee such as a Data Safety Monitoring Board in place to oversee or make important decisions, essentially taking them out of the control of the person who has the conflict of interests. The strongest strategies are recusal from secondary interests, such as selling one's stock, or even recusal from the primary interests, such as replacing an investigator with another person who does not have conflicting interests. BEST ETHICAL PRACTICES – MITIGATING CONFLICTS OF INTEREST In this section, we'll try to answer the question: how do we translate conflict of interest mitigation strategies to the development and deployment of AI for health care? Let's look at each of the three general types of strategies: disclosure, mediation, and recusal. The principle underlying disclosure is transparency, or facilitating awareness of people who are impacted by AI and who are owed duties of care that other interests exist that could influence this care. For AI, this transparency is complicated by the fact that most of the people who could be impacted, such as patients and providers, are probably not even aware of the existence of AI or how it might be used in decision making about their health care. So first, some public notification of the use of AI may be warranted, especially if the application deviates substantially from commonly [Image]

[Image]

Copyright 2020 © Stanford University 59 understood or accepted uses of data in the health care context, or if the AI or data use poses more than usual risks or be especially sensitive. Examples might be using and storing video data from telehealth visits, or collecting and analyzing patients' social media data to predict or diagnose mental health conditions, or using EHR, sensor and claims data to guide decisions about offering palliative care at the end of life. This notification could also include disclosure of how AI is used in decisionmaking about patient care and who developed the AI system. Other ways of implementing the principle of transparency include thorough reporting of how models were built, as suggested by members of the data science community. This includes clear descriptions of data sources, participants, outcomes and predictors, the contexts in which the model was validated, limitations and contraindications for deployment and assumptions or conditions that must be satisfied. The Association of Computing Machinery has also recommended that for facial recognition technologies, or other uses of AI where racial or other biases are of concern, that error rates be reported disaggregated by race, gender and other context-dependent demographic features. The principle behind the strategy of mediation is independent review or oversight. For AI development, this could be achieved by auditing of algorithms and models by third parties. This process was suggested by the Obama administration in 2016 to mitigate discriminatory practices and civil rights violations. Employing algorithmic audits also enhances transparency to the extent that it encourages developers to make algorithms auditable in the first place. The strategy of recusal in the context of AI development is difficult to implement. If AI development is being conducted in an academic research setting, it might be possible for the developers to recuse themselves of financial interests, or replace people with financial interests with others who are more independent. However, in a corporate setting, that might be impossible, so developers would have to rely on disclosure or mediation-based strategies, keeping in mind that disclosure-based strategies are very weak and do little to engender or maintain trust of key stakeholders. The takeaway points here are that: Best ethical practices in developing AI for health care include careful formulation of the problem to be solved, ideally with input from people who have deep knowledge of the specific clinical settings and data relevant to the problem. Conflicting interests can have real impacts on design decisions, but there are strategies to mitigate the potential negative effects of conflicts of interests.

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4. Modle 5 - International Definitions Used for Regulatory Purposes.mp4

Summary

Software as a medical device functions through a structured flow where **patient data inputs** are transformed by **computational algorithms** into meaningful results used for clinical intervention. Because these digital tools directly impact human health, the government oversees them using a **risk-based regulatory framework** designed to guarantee safety and performance. Depending on the level of potential danger to the patient, developers must navigate specific **FDA pathways**, such as demonstrating equivalence to existing products or fulfilling rigorous requirements for entirely **novel, high-risk innovations**. Ultimately, this systematic oversight ensures that advanced medical software is both **technically sound and clinically reliable** before it reaches the public.

Keywords: SaMD components, AI healthcare regulation, FDA regulatory framework, Risk-based categorization, Market application types

Content

The software as a medical device can be described in three components. This includes SAMD inputs, SAMD algorithms and S AMD outputs. The software as a medical device inputs include for example patient data. The S AMD algorithm includes the specific algorithm, dictionaries, equations, etc. The pieces that are used to process the SAMD inputs. And finally, The SAMD outputs are processed data that are used to treat, diagnose, cure, mitigate, or prevent disease or other conditions. I will refer back to these later in the lecture. So, it's important to have a clear understanding of these definitions. We're going to talk about regulation today. And regulation is a long and tedious process. Yet, it is essential to ensure the safety and effectiveness of AI in healthcare. Let's take a closer look at the framework used by the Food and Drug Administration to regulate AI solutions or software as medical devices. The FDA's regulatory framework starts with a market application which includes a definition statement and a category which can be either category 1, 2, three or four. The category is based on the risk associated with the use of the proposed AI solution. Depending on the category defined, data requirements necessary for regulation may include a pre market notification or a 510K statement of equivalency, a denovo request, meaning no similar product exists on the market for comparison, or a pre-market application, also known as a PMA, which is reserved for high-risk AI applications.

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5. Moduel 4 - Aggregation Bias.mp4

Summary

This text explores the concept of **aggregation bias**, which arises when developers mistakenly treat distinct subgroups as a single, uniform population despite significant differences in their baseline characteristics. By examining healthcare examples like diabetes and skin cancer, the source illustrates how **average statistics can mask critical variations** between ethnic and demographic groups, leading to inaccurate or even harmful outcomes. To combat this issue, practitioners must move beyond a **"one size fits all" approach** by identifying unique strata and tailoring models to account for these specific nuances. Ultimately, the purpose of the text is to highlight that **true algorithmic fairness** requires a deep

understanding of human diversity to ensure that AI tools do not disadvantage vulnerable populations.

Keywords: Aggregation bias, Model development, Healthcare data stratification, Ethnic health disparities, Demographic variables

Content

Aggregation bias occurs while developing the model when we try to combine different populations whose underlying distribution of the outcome under study differs. This problem is known as infra marginality and requires separate models for the different populations or including the demographic variable into the model to account for the systematic differences. An example of the latter could be a differing base rate of skin cancer across ethnicities, especially in domains such as health care. where the data set contains several distinct stratifications. In terms of model development, one size does not fit all. In order to identify aggregation bias, developers need to understand the meaningful distinct groups and reasons why they are different from each other. A notable example in healthcare is diabetes, which is known to be associated with differential complications across racial and ethnic groups. As we discussed, as we mentioned, US Hispanics tend to have higher prevalence rates and complication rates for diabetes compared to whites. However, Hispanic Latino Americans make up a diverse group that includes people of Cuban, Mexican, Puerto Rican, South, and Central American descent. Each group has its own history and traditions, but all are more likely to have type 2 diabetes, about 17% compared to non-Hispanic whites, which is about 8%. But that 17% is just an average for Hispanic Latinoamerican groups. For example, if your heritage is Puerto Rican, you’re about twice as likely to have type 2 diabetes as someone whose background is South American. Therefore, an AI solution that focuses on diagnosing or monitoring of diabetes should be adapted to the ethnic background of the patient. Importantly, not accounting for these differences may result in models that harm members of all groups. Think back to our example of gender bias in heart attack symptoms where the AI solutions worked Well, for men, but were inaccurate for women.

6. Moduel 5 - Valid Clinical Association.mp4

Summary

To achieve regulatory approval for medical artificial intelligence, developers must demonstrate a **valid clinical association**, which proves that the software's output is grounded in **established scientific evidence** or real-world logic. This framework ensures that the AI's conclusions are not merely statistical flukes—like the absurd correlation between clergy visits and mortality—but are instead **clinically accepted** and reliable for patient care. When an algorithm discovers a **novel association** previously unknown to science, the burden of proof shifts toward generating new evidence through **secondary data analysis or clinical trials**. Ultimately, this process serves as a vital safeguard, guaranteeing that AI tools possess the **scientific validity** necessary for integration into professional healthcare settings.

Keywords: Valid clinical association, Regulatory approval, Clinical evaluation process, Scientific validity, SAMD output evidence

Content

All AI applications needing regulatory approval must include general controls as part of the general control. The clinical evaluation process is a framework used by regulators to understand quality system regulations. The clinical evaluation process includes three components which we have previously discussed in the context of AI evaluation but here we will take a regulatory focus. The first category is clinical association. Is there a valid clinical association between the SAMD output and your SAMD's targeted clinical condition? This is the scientific validity of the AI solution or the extent to which the SAMD's output is clinically accepted or wellfounded based on an established scientific evidence and if it accurately corresponds to the healthcare situation and condition identified in a real world set. Let's think about a hypothetical situation where I have developed an AI solution that identifies features associated with inpatient mortality for patients in the ICU. In this hypothetical situation, let's say I find the strongest predictor of inpatient mortality was a visit from a clergyman. Therefore, to reduce inpatient mortality, I want to stop clergymen visits in the ICU. Is that a valid clinical association? Is there any scientific evidence supporting this association? Of course not. A valid clinical association is an indicator of the level of clinical acceptance and confidence one can have of the SAMD's output. There are many types of evidence that constitute a valid clinical association. Regulatory agencies are looking to require a minimum of evidence to support the clinical association which could include literature searches, original clinical research, professional society guidelines, examples of how your model can generate new evidence, secondary data analysis, or the performance of a clinical trial based on your AI solution. So, how do we deal with novel associations? Associations that are newly discovered by your AI in a hypothetical situation, again, let's say that I've developed an AI solution to predict stroke following an admission for coronary heart failure. In my model, I might include new inputs such as mood, uh, levels of pollen in the environment and outside temperature at the time of discharge. In my hypothetical model, I might find that levels of pollen are associated with stroke following coronary heart failure. How would I prove this is a valid association? As literature and existing randomized clinical trials do not exist for this association, there are other solutions to regulate this software, which may include performing a clinical trial based on your a I or secondary data analysis. A valid clinical association is required for AI regulation and ensures the clinical acceptance and uptake of the AI solution in the healthcare setting.

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7. Module 4 - Anti-classification.mp4

Summary

The concept of **anti-classification** suggests that fairness is achieved by **omitting protected attributes**, such as race or gender, and their potential proxies from algorithmic decision-making. While this approach aims for **fairness through unawareness**, it often proves impractical because removing all correlated variables can strip a model of its predictive power. Furthermore, the text highlights the **problem of infra-marginality**, noting that in clinical settings, ignoring these characteristics can actually lead to inequity when biological risks differ across populations. Ultimately, the source argues that true equity requires **representative data** and sophisticated modeling that accounts for sensitive traits to improve accuracy without reinforcing systemic disparities.

Keywords: Anti-classification, Protected attributes, Proxy variables, Algorithmic bias, Clinical risk models

Content

Anti-classification or fairness through unawareness. This first definition of fairness requires the exclusion of any protected attributes such as gender or race in the outcome modeling. An even stricter notion of anti-classification would be to require the omission of any unprotected characteristics that are proxies of protected attributes such as attendance at a single sex school or height and weight as proxies for gender. This can be impractical as removing all possible proxies may leave very few predictively useful features. Also, it’s often hard to know whether a particular variable is a proxy for a protected characteristic without further data collection and analysis. Also, critics have pointed out that an algorithm may classify information based on proxies for the sensitive attributes, yielding a bias against a group even without making decisions directly based on one’s membership in that group. The main shortcoming of this fairness definition ition is that some clinical risk models need to explicitly include protected attributes for it to be equitable. In particular, this applies for situations where the true underlying risk distribution differs across subopuls known as the problem of infra marginality. A notable example in healthcare is diabetes which requires different diagnosis and monitoring protocols across ethnicities and gender. Therefore, an accurate model must include protected attributes. but must also learn to avoid bias based on these attributes. A very difficult challenge. If the goal is to avoid reinforcing inequalities, what then should developers and operators of algorithms do to mitigate potential biases? It is important that developers of algorithms first look for ways to reduce disparities between groups without sacrificing the overall performance of the model, especially whenever there appears to be a trade-off. AI solutions that use data sets which may be under representative of certain groups may need additional training data to improve accuracy in the decision-making and reduce unfair results. Anticlassification requires the exclusion of any protected attributes in the outcome modeling.

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8. Module 4 - Applying Fairness Measures.mp4

Summary

This source outlines the practical methodologies for auditing AI systems through the three primary lenses of **algorithmic fairness**. It begins by addressing **anti-classification**, noting that while excluding protected traits is simple, identifying **subtle proxies** like zip codes remains a significant challenge. To ensure **classification parity**, the text recommends using **bootstrapping** to establish confidence intervals, which helps determine if performance gaps between groups are statistically meaningful. Finally, it explores **calibration**, emphasizing that a model is fair only if its **predicted risks align with observed outcomes** consistently across all demographics, typically visualized through calibration curves.

Keywords: Algorithmic fairness, Anti-classification, Protected attributes, Performance metrics, Calibration analysis

Content

Now that we have learned about the most common concepts of algorithmic fairness, it is time to ask the question of how they can be applied to studying whether model predictions are fair. Let’s start with anti-classification. Checking the requirement of not using any protected attributes in the decision rule is very straightforward. However, it gets more complicated when trying to impose the stricter condition to also not use any proxies of protected attributes. A well-known example is zip code that shows a high correlation with race, but many other proxies might be more subtle to detect. Once the two or three most relevant performance metrics are selected, we can evaluate performance differences on the test set across different demographic groups. Just like with model selection and comparison, it is always better to calculate empirical confidence intervals of this statistic to decrease the dependence on the test set. One popular strategy to build these intervals is by using boot strapping non-over overlapping confidence intervals are an indication of a statistically significant difference in the performance among the different protected groups. Finally, calibration. The idea here is to only look at the overall rate of predicted and observed outcomes across demographic groups. We can again use bootstrapping to check whether the observed disparities between prediction and outcome rates are consistent. It gives us an idea of systematic risk under and overestimation. In the context of binary predictions with a continuous risk score function, calibration is most tricky to analyze as patients have to be grouped by risk in order to compute the observed incidence rate which can then be compared to the predicted risk with the goodness of fit test. The choice of the number of groups and how to do the splitting for example by risk score percentiles or quantiles adds variability to the analysis which can in certain cases impact the calibration analysis quite significantly. The standard choice is to use 10 groups group by deciles. One can then compute a kiquare statistic. What is more common though is to plot the observed rates as a function of the predicted risks. Ideally, these points lie close to the diagonal line for a good calibration. If the calibration curve looks very different across demographic groups, the fairness criterion is violated. In this section of algorithmic fairness, we have seen three different ways of how fairness can be conceptualized and applied to AI solutions. Anti-classification, classification parity, and calibration.

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9. Module 4 - Calibration.mp4

Summary

Calibration serves as a vital measure of algorithmic fairness by ensuring that a model's **predicted risk matches the actual frequency** of observed outcomes across all demographic groups. Rather than just focusing on accuracy, this process demands that a specific probability score—such as a 25% risk—remains **consistent and independent of protected attributes** like race or gender. Without this precise alignment, skewed data can create **misleading expectations** in high-stakes environments like healthcare, potentially causing emotional distress or physical harm to patients. Ultimately, developers must prioritize this **harmony between expectations and reality** to prevent clinical errors and ensure a system is ethically fit for deployment.

Keywords: Algorithm fairness, Model calibration, Risk prediction accuracy, Protected attributes, Clinical decision making

Content

Finally, calibration is the third form of algorithm fairness we will discuss. In general, we desire a model to not only have good discriminatory capabilities, but also reach a good agreement between model predictions and observed outcomes, which is known as calibration. This means that we want roughly 25 out of 100 patients with a risk score of 25 to indeed experience the event we are trying to predict. In the framework of fairness analysis, calibration means When conditioning on risk estimates, outcomes should be independent of the protected attributes. Therefore, given the estimated risk scores, calibration aims at making sure that there is good agreement between the model risk prediction and the probability of a positive outcome no matter what demographic group the patient belongs to. Therefore, we can think of calibration as a comparison of the actual output and the expected output given by a system. When we calibrate a model, we try to improve our model such that the distribution and behavior of the probability predicted is similar to the distribution and behavior of the probability observed in the training data. Calibration is open to manipulation of risk distributions for different groups. For example, by changing the way data is aggregated in feature representations, you can change the outcome. By adding or removing a grouping feature, you can cause a trend to reverse. If an AI model is used to triage high- risk patients, Poorly calibrated risk estimates will lead to false expectations with end users, either patients or healthcare professionals. This can lead to poorly defined decisions in anticipation of an event or the absence thereof that were in fact misguided. Let's think about in vitro fertilization treatment. Irrespective of how well the model can discriminate between treatments that end in live birth versus those that do not, it is clear that strong over or underestimation of the chance of a live birth makes the algorithms clinically unacceptable. For instance, a strong overestimation of the chance of a live birth after invitral fertilization would give false hope to couples going through an already stressful and emotional experience. Treating a couple who in reality has a favorable prognosis exposes the woman unnecessarily to possible harmful side effects, for example, ovarian hyper simulation syndrome. Model calibration is an important aspect of development and must be evaluated before model deployment.

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10. Module 4 - Deployment Bias.mp4

Summary

Deployment bias occurs when an AI system is applied to a real-world situation that differs from its original design or **intended purpose**. This phenomenon highlights a dangerous disconnect between a model's technical accuracy and its **societal implementation**, often resulting from the misinterpretation of data as a proxy for unrelated outcomes. As seen in the Obermyer study, using a tool designed for cost prediction to instead measure medical need created **systemic inequality**, requiring certain patients to be significantly more ill to receive care. Ultimately, this source illustrates that the **intersection of human decision-making and technology** can lead to harmful consequences if the model's outputs are used inappropriately.

Keywords: Deployment bias, Model implementation, Healthcare inequality, Intended model use, Predictive proxy errors

Content

Deployment bias arises during the implementation of the model. It refers to using the model inappropriately or misinterpreting its results. In other words, if the model's intended use is different from the way it is used, deployment bias occurs. Let's think back to our example from the Obermyer study. The AI model in question was developed to predict costs of care. The model had great accuracy and was able to discriminate between high and lowcost patients. However, the problem arose when this model was used to predict health care need instead of health care costs. The deployment of the model was biased. The model's results were used as a proxy for the prediction of healthcare need, which is different from the model's output intended. The deployment bias of this AI solution resulted in a black patient needing to be almost twice as sick as a white patient in order to qualify for the same beneficial care program. One can see how deployment bias is a ser concern in healthcare. Deployment bias is the intersection of society with the AI solution. How society or the medical community uses the AI solution and its output.

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11. Module 4 - Evaluation Bias.mp4

Summary

Evaluation bias emerges when the **testing protocols** used to judge an AI model fail to align with the complexities of the **real-world environment**. This discrepancy typically stems from using **non-representative benchmark data** or **narrow performance metrics** that overlook critical nuances within the target population. To ensure a system is truly fair and effective, developers must validate their models using **unseen data from the actual population** and employ a **comprehensive suite of granular metrics**. By prioritizing **external validation** over synthetic or generic benchmarks, creators can detect hidden flaws and prevent the model from failing once it is deployed.

Keywords: Evaluation bias, Testing data, Performance metrics, External validation, Benchmark data sets

Content

Model evaluation hinges on two important choices. The testing data and the metrics used. Evaluation bias occurs during the model validation and tuning. Let’s talk about the testing data. Evaluation bias arises if the testing data which often includes external benchmark data sets is not representative of the final population to which the AI solutions will be applied. This is the difference between the data used for model evaluation and the data used for models realworld predictions. Since developers mostly use a benchmark data set or a synthetic data set for training, their evaluations often do not fit with realworld settings. A solution of this problem is external validation of the AI model on a different unseen data set selected from the targeted population. Similarly, evaluation bias can arise if inappropriate performance metrics are used. For example, if there is significant class imbalance in your data set and you choose to only evaluate area under the curve. There may be evaluation bias because you neglected to look at other important metrics such as positive and negative predicted probabilities on subgroups. Therefore, evaluation bias also refers to the usage of evaluation metrics inefficiently and to avoid it using granular and comprehensive evaluation metrics is suggested.

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12. Module 4 - Historical Bias.mp4

Summary

Historical bias emerges when AI models replicate **deeply ingrained societal prejudices** found in real-world data, leading to outcomes that conflict with modern ethical standards. Because these algorithms are entirely dependent on existing information, even technically perfect systems will mirror **long-standing systemic inequities** in fields like hiring, criminal justice, and medicine. The source highlights how medical research has historically centered on **white male demographics**, resulting in diagnostic criteria and screening guidelines that fail to account for the unique symptoms and risk factors of diverse populations. Ultimately, this means that unless we actively address these **unrepresentative data legacies**, AI will continue to perpetuate health disparities for those who already carry a disproportionate burden of disease.

Keywords: Historical bias, Data dependency, Societal prejudices, Healthcare data disparities, Non-diverse clinical trials

Content

Historical bias occurs if the present or past state of the world influences a model in a way such that its predictions are considered unfair given societal values and norms. A significant difference in the target distribution for different groups can be due to underlying human prejudices in the data such as we see for historical differences in hiring, incarceration or even in receipt of certain types of medical treatment. So historical bias refers to judgment based on preconceived notions or prejudices. AI algorithms are entirely data dependent and historical bias encoded in real world data cannot even be overcome by perfect sampling and feature selection. Therefore, it is important to remember that historical healthcare data in general is extremely male and extremely white and this has realworld impacts. Let’s think back to our example of heart attack symptoms used to train AI models for triage. The symptoms used to train the model were male specific because we once did not know that there was a difference in symptoms between men and women. As another example, we can look at past guidelines for lung cancer screening where historical bias was apparent. The guidelines for lung cancer screening stated that you must be 55 to 80 years of age and you must have smoked 30 packs per year. These guidelines were based on a study that included over 53,000 people. However, 4% of the study population were African-American. There are important differences in smoking habits between whites and African-Americans, including the use of menthol cigarettes. So, lung cancer screening in the community may not identify all those in need of screening because the research that led to the guidelines were not precise for the people who carry the greater burden of the disease, but rather more precise for white males. This is inherent historical bias in the healthcare data that has been ingrained Due to decades of data collection and clinical trials that were dominated by non-diverse participants,

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13. Module 4 - Lack of Transparency.mp4

Summary

This source highlights how a pervasive **lack of transparency regarding training data** undermines the safety and fairness of artificial intelligence in the medical field. By examining studies based on electronic health records, the author reveals that researchers frequently **fail to report essential demographic variables** like race, age, and socioeconomic status, making it impossible to detect hidden biases. Because many models are built using **non-representative patient populations**, their clinical utility for the broader community remains unproven and potentially discriminatory. Ultimately, the text argues that **detailed data disclosure is a prerequisite** for healthcare providers to evaluate, trust, and safely implement these technological advancements.

Keywords: AI healthcare discrimination, Data transparency, Population representativeness, Demographic reporting, Algorithmic bias mitigation

Content

We have raised several questions about whether AI solutions in healthcare will perpetuate discrimination if they are trained on historic retrospective data or data from an electronic medical record system. In thinking about EHRs of which many AI solutions are developed and trained on, it is important to think about whether the population captured in the EHR system is representative of the broader community, particularly if these are coming from an academic medical center. If AI solutions are being developed on non-representative populations. The utility and applicability of these advances for the broader patient community falls into question. This relates to an important issue around transparency and reporting. And many AI studies, particularly machine learning studies, do not report the demographic breakdown of the training data used to develop and train their models. Lack of reporting makes it very difficult to evaluate the bias and fairness of the AI solution and its applicability across population. s. In a study conducted by my team, we investigated the reporting and demographics and population representativeness in machine learning models using electronic healthcare records. Across the studies reviewed, demographic variables were inconsistently reported or included as model inputs. For example, race and ethnicity were not reported in 64% of the articles and gender and age were not reported in three/arters of the studies and social economic status of population was not reported in over 90% of the studies. In addition, studies that mentioned these variables often did not report if they were included as model inputs. In the studies that reported demographics, the average populations included higher proportions of whites and blacks, yet fewer Hispanics compared to the general population. While each study might not be applicable to the general population, these findings emphasize the present lack of transparency in reporting details of training data used for development and Evaluation of machine learning models in healthcare to ensure the unbiased deployment and application of any AI model in healthcare. Detailed information on the data used to develop and train the model are necessary. Without such information, no human, be it a healthare worker or patient or insurer, can understand and mitigate any inherent bias in the model.

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14. Module 4 - Measurement Bias.mp4

Summary

Measurement bias occurs when the data used to train algorithms consists of **imperfect proxies** that do not represent all demographic groups with equal accuracy. This source illustrates the concept through a medical study where **healthcare spending** was used as a substitute for actual medical need, mistakenly assuming that lower costs equated to better health. Because systemic factors caused Black patients to have lower expenditures despite being sicker, the model **disproportionately allocated resources** to white patients who appeared more “expensive” to treat. To combat these skewed results, developers must recognize that historical data often contains **non-random noise** and implement specific technical interventions to ensure equitable outcomes.

Keywords: Measurement bias, Noisy proxies, Racial bias, Healthcare resource allocation, Data mitigation strategies

Content

For many problems, the only available and measurable features and labels in a data set are noisy proxies of the actual variables of interest and additional noise is added through the measurement process itself. Measurement bias arises in these situations if the noise is not randomly distributed but differs across groups which leads to differential performance. As an example of measurement bias, we can discuss the Obermeer study. Dr. Obermyer and colleagues examined a widely used algorithm to predict patients in need of additional care support and identified racial bias in the algorithm. They showed that the algorithm computed black patients needed to be significantly sicker with more coorbidities to achieve the same predictive score as their white counterpart patients for potentially beneficial care programs. This misinterpretation resulted in allocating health care resources disproportionately among different racial groups. This is a prime example of measurement bias as often the only available and measurable features as well as labels are only noisy proxies for the actual variable of interest. In this case, the algorithm was developed to predict healthcare costs. Yet, it was deployed to predict healthcare need. The labeling bias was due to using health care costs as a flawed proxy for medical illness. Usually, one cannot change the data. Some historical biases might be indistinguishably linked linked to the data. But the awareness about the problem is important and mitigation strategies can be identified by taking preventative measures such as pre and post-processing actions.

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15. Module 4 - Minimal Reporting Standards.mp4

Summary

To enhance transparency and fairness in healthcare technology, the authors introduce **MINIMAR**, a standardized framework designed to ensure comprehensive reporting of medical artificial intelligence models. This protocol requires researchers to disclose specific details regarding **demographic data, model architecture, and validation methods** to facilitate the replication of results and the identification of potential biases. By applying these criteria to a study on postoperative pain prediction, the text demonstrates how documenting **data sources and optimization parameters** allows clinicians to understand a tool’s intent and limitations. Ultimately, this structured approach aims to improve **model**

generalizability and provide end-users with the necessary context to deploy AI solutions safely across diverse patient populations.

Keywords: Transparent AI reporting, MINIMAR standards, Training data demographics, Model development replication, Algorithmic fairness validation

Content

As a solution for transparent reporting and to identify best practices for designing machine learning models to account for biases and fairness, we propose a template or a checklist to assist with the reporting of AI solutions. The minimum information for medical AI reporting or miniar. Minimar serves for the following requirements. One, to include information on the population providing the training data in terms of data sources and cohort selection. Two, the inclusion of information on the training data demographics in a way that enables comparison with the population the model is being applied to. Three, providing detailed information about the model architecture and development so as to interpret the intent of the model and compare it to similar models. This will permit replication. Four, model evaluation, optimization, and validation should be transparently reported to clarify how local model optimization can be achieved and to enable replication and resource sharing. Let's walk through one example to check the reporting according to miniar criteria. Here is an example study from our lab in which we present a machine learning approach aimed to predict postoperative pain among patients taking a combination of certain anti-depressants and prodrug opioids. The hypothesis was that certain anti-depressants inhibit the activation of common prodrug opioids. So the targeted population was patients undergoing elective surgery and we declared that the study was conducted at an academic hospital. We used EHRs as a data source and included only adult patients while excluding patients who died during hospitalization. In terms of demographic variables, the age, gender, race, and ethnicity breakdown of the patients were reported in the study. As a proxy for social economic status, we reported insurance types. The targeted output of the model was pain score, which was intended to be used by health care workers to triage patients at higher risk of uncontrolled pain following discharge from surgery. For the train test splits, we did a 10-fold data splitting. As the gold standard, we used 100 manually annotated notes and also pain scores recorded as structured data in the EHRs. Our model's task was prediction and we used a regression. We also reported in detail all the features used and any transformation applied and missing data presence with its imputation methods. We provided our model optimization parameters and included details on how we validated our model. We shared our code and deidentified data via GitHub. The aspect of transparency missing was external validation of our model using unseen data that are entirely separate from the data used for model development. Therefore, at this stage, we cannot verify the model’s generalizability. However, you can understand that by providing these details of the training data and population, Model design and intent. An enduser will have a great understanding of how to best deploy the model and in which populations.

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16. Module 4 - Opportunities and Challenges.mp4

Summary

This text examines the complex landscape of developing ethical artificial intelligence in medicine, highlighting the fundamental tension between **equality and equity** when defining what makes a system “fair.” The author explains that while identifying **hidden systematic biases** within datasets remains a primary obstacle, the industry is currently undergoing a significant shift toward more **transparent reporting** and accountability. Ultimately, the goal is to foster **trust among marginalized populations** by ensuring that data-driven tools are interpreted by humans to produce truly beneficial outcomes for every patient group.

Keywords: Defining AI fairness, Equality versus equity, Identifying systematic biases, Healthcare data mitigation, Transparency and accountability

Content

While we have covered many topics in this lecture regarding bias and algorithmic fairness, there are still many more challenges and opportunities for fair AI research. Let’s walk through a few of the main challenges. Defining fairness. As we discussed, there are several definitions of AI fairness that have been proposed in the literature. The definitions cover a wide range of use cases and therefore their view of fairness is broad. Because of this, it is nearly impossible to understand how one fair EST solution could address all challenges. Identifying the correct or best definition is an ongoing debate from equality to equity. While it is important that an AI model focuses on equality, meaning that each individual or group is given the same amount of resources, attention or outcomes. Think about the Obermyer example. There is also the question of equity. Equity suggests that each group is given the amount of resources needed to have similar out outcomes. Understanding how to develop and implement a model that provides both equality and equity presents a paradigm shift in the way we think about healthcare delivery and is an active area of research. Identifying biases in models and particularly in data sets. We have provided several examples of biases in AI and explained how the training data is a key to the mitigation of these biases. However, many of these biases are systematic and we are often unaware they exist. Think back to the blue lips example. While we can identify biases in data, we still have a long way to go before we can systematically mitigate these biases and provide our professionals with the appropriate tools they need to address these issues at point of care. While these are some of the challenges, there are also so many opportunities. The soaring number of papers discussing bias and AI solutions related to healthcare is evidence that we are forging a new path for fair and unbiased datadriven healthcare. The Espective collection and reporting of AI outputs, clinical recommendations and patients decisions coupled with eventual outcomes is essential in being accountable as healthc care institutions and as clinicians. This work and transparency in reporting AI solutions is absolutely critical for populations who have difficulty trusting the medical establishment, particularly where AI is involved. The key is to use AI in a way that actually does benefit all groups, which requires thoughtful evaluations and human interpretations.

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17. Module 4 - Parity Classification.mp4

Summary

This source explores the concept of **classification parity**, a framework that requires machine learning models to maintain **consistent predictive performance** across different demographic groups. It highlights essential metrics like the **false positive rate** and **demographic parity**, which aim to ensure that outcomes remain independent of an individual's protected traits. However, the text warns that these mathematical ideals often conflict with reality—a challenge known as **inframarginality**—where inherent differences in group risk distributions make equal error rates difficult to achieve. Ultimately, the material emphasizes that while these fairness standards are popular, they must be applied with a deep understanding of the **specific medical or social context** to be truly effective.

Keywords: Classification parity, Predictive performance, Protected demographic groups, False positive rate, Demographic parity

Content

Classification parity of fairness asks for equal predictive performance across any protected group. Depending on the nature of the problem you are trying to address, certain performance metrics are more relevant than others. When selecting the metrics to examine, it is therefore important to keep in mind the actionable insights resulting from the model output. In general, two measures have received particularly high attention by the machine learning community. False positive rate and the proportion of positive decisions. The parody of false positive rates means that the probability of predicting a positive outcome when the true outcome is negative should be the same regardless of the protected attributes of the patient. In some applications, it is however the false negative rate that is more relevant to survey. Parity in the proportion of positive decisions implies the probability of predicting a positive outcome should not vary across different demographic groups given all else equal. It is also known as the demographic parody as it requires the classifier's prediction to be independent of the protected traits which makes it also closely related to anti-classification. Classification parody includes many of the most popular mathematical definitions of fairness. However, these definitions are problematic when the risk distributions are different for different groups. Again, a problem known as infa marginality. This is particularly important in medicine because the risk distributions of protected groups will in general differ. This is particularly important in medicine because the risk distributions of protected groups will in general differ. Threshold-based decisions will typically yield error metrics that also differ by group. Therefore, classification par asks for equal predictive performance across groups.

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18. Module 4 - Representation Bias.mp4

Summary

This source examines **representation bias**, which occurs when the data used to train artificial intelligence fails to accurately mirror the **diverse characteristics** of the real-world population. By failing to include specific subgroups, developers risk creating tools that are ineffective or even harmful to the very people they are meant to serve. A practical example is found in **medical machine learning**, where researchers discovered that many diabetes models completely **omitted Hispanic populations** despite their high risk for the disease. Ultimately, the text serves as a warning that **sampling errors** in data collection can lead to systemic healthcare inequalities if the training sets lack **proportional demographic diversity**.

Keywords: Representation bias, Training data collection, Sampling bias, Population representativeness, Demographic data reporting

Content

The process of defining and collecting training data is vulnerable to representation bias. Representation bias arises when the sample collected to train an AI solution does not represent the actual distribution of the population it is intended to be applied to. In other words, it occurs when certain parts of the final use population is under represented in the training data. Representation bias might also be called sampling bias. Let’s talk about an example of representation. bias. My group at Stanford performed a systematic review of demographic data and representativeness in machine learning models using electronic health records. In an attempt to appraise the AI model sample representation of the population to which it would be applied, we evaluated type 2 diabetes models. In the papers reviewed, we found only 25% included population demographics. And in these papers, white and black populations included in the training data were sim similar to the US type 2 diabetes population. However, no study reported Hispanics in their training data. This raises concerns giving Hispanics higher prevalence and complication rates for type 2 diabetes compared to whites. It is unclear how type 2 diabetes models trained without representation of Hispanics would fare more broadly in nationwide health care systems to benefit diverse populations. This is an example of representation bias.

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19. Module 4 - Summary.mp4

Summary

This text outlines a comprehensive framework for identifying and mitigating **algorithmic bias** to ensure that healthcare AI promotes genuine **social justice** rather than merely technical accuracy. The author categorizes bias into six distinct stages of the model’s lifecycle and argues that “neutral” algorithms are an illusion; instead, developers must prioritize **problem reformulation** by choosing proxies that reflect actual patient needs rather than historical inequities like healthcare spending. Central to this approach is the **MINIAR framework**, which advocates for radical transparency through the reporting of disaggregated data across diverse demographics. Ultimately, the source shifts the focus from mathematical outputs to a **human-centered lens of equity**, demanding that AI evaluation remains rooted in the lived experiences and health outcomes of marginalized populations.

Keywords: Healthcare AI bias, Algorithmic fairness, Problem formulation, Reporting transparency, Ethical justice frameworks

Content

We have covered a lot of material in this lecture that is essential for the evaluation of AI solutions in healthcare. First, there is increasing recognition that bias exists in datadriven healthcare solutions, particularly machine learning models that are based on historical data with known reporting and labeling problems. We discussed the six pillars of bias. Historical bias, representation bias, measurement bias, aggregation bias, evaluation bias, and deployment bias. We also discussed algorithmic fairness, including anti-classification, classification parody, and calibration, and how to apply these concepts to AI solutions. However, there are limitations to algorithmic fairness as we discussed. For this reason, I suggest you resist the tendency to view algorithms as able to be rendered neutral. What is likely to be far more successful and accurate from an empirical perspective, as well as from an ethical one, is to just be specific and focused on the learning objective. Ive in the Obermeer example they changed the problem formulation such that instead of using cost as the proxy for total healthcare needed that you can use number of hospital visits symptom burden or other metrics that are closer to the patients own experience and referral patterns which appears to render the algorithm much more even between black and white patients. A step to mitigate bias is transparency in reporting. We reviewed the current state of reporting in literature and provided a framework for increasing transparency in reporting. Miniar miniar comes in at several different points in a model’s life cycle. It starts with knowing the demographic composition of your data sets and reporting disagregate performance metrics. A lot of fairness metrics are excellent for parsing out the contribution of a sensitive variable like ethnicity to the predictions. If a model reaches the point of care, having information about how the model performs across iverse populations can help clinicians interpret the outputs. More important, ethical analysis demands that we take a view of fairness or more appropriately justice that centers on the health and lives of people, not the outputs alone. I believe that in this lecture we have provided you with a framework that will allow you to review AI solutions through a lens of equality and equity. Most important, ethical analysis demands that we take a view of fairness or more appropriately, justice. that centers on the health and lives of people not the outputs alone. I believe that in this lecture we have provided you with a framework that will allow you to review AI solutions through a lens of equity and equality.

20. Module 4 - What is algorithmic Fairness.mp4

Summary

True algorithmic fairness in healthcare requires a shift away from purely technical metrics toward a philosophy of **justice that prioritizes human lives** and lived experiences. The text illustrates how medical standards often suffer from **white normativity**, using the example of "blue lips" to show how clinical signs tailored to light skin can lead to dangerous inaccuracies for diverse populations. To correct these systemic biases, researchers must embrace **epistemic privilege**, valuing the unique insights of marginalized groups to improve scientific accuracy. While technical frameworks like **anti-classification, classification parity, and calibration** exist to measure fairness, they all possess **statistical limitations** that necessitate a cautious and critical approach during model evaluation. Ultimately, the source argues that building equitable AI depends on recognizing the **flaws in universal definitions** and incorporating diverse perspectives to overcome historical medical biases.

Keywords: Algorithmic fairness, Medical bias, Epistemic privilege, Healthcare equity, Fairness definitions

Content

Next we move to algorithmic fairness. Ethical analyses of AI solutions in healthcare demand that we take a view of fairness or more appropriately justice that centers on the health and lives of people not the outputs alone. A lot of historical medicine has been influenced by white normativity which is the basis of many medical facts. This is evident because of the lack of inclusion of diverse patients in clinical research. As an example, let's talk about blue lips as a clinical sign of lack of oxygen. When we think about this, it is not an objective clinical sign. It's actually biased toward light skin because dark-kinned people do not get blue lips with lack of oxygen. However, as this clinical sign is embedded in our medical education, it wouldn't occur to many that this clinical sign is biased. This ignorance stresses the importance of epistemic privilege, which is the idea that those outside of the majority or dominant group group have knowledge that the insiders don't and that bringing diverse perspectives actually enhances the quality and accuracy of your scientific endeavor. So let's talk about algorithmic fairness. A key difficulty in developing fair AI algorithms is the fact that no universal notion of fairness exists. Many different definitions have been proposed by researchers over the years and they can be broadly regrouped into three main classes. Anti-classification classification par and calibration. However, these three classes are by no means exhaustive of all the work done in this field. They rather regroup some of the most prominent concepts suggested over the last few years. Moreover, these fairness definitions do not come without any flaws. In fact, they have been shown to all suffer from significant statistical shortcomings. Therefore, special caution and awareness about the notions, limitations, and weaknesses is always necessary when applying these concepts in model evaluation settings. Now let's discuss in detail the three forms of algorithmic fairness. Anti-classification, classification parody, and calibration.

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21. Module 5 - Chinese Guidelines.mp4

Summary

China is aggressively positioning itself as a global leader in medical artificial intelligence by fostering a **favorable regulatory environment** that prioritizes **societal progress over individual autonomy**. Unlike Western models, the Chinese approach mandates **state-access to private data**, effectively dissolving data silos to accelerate the deployment of diagnostic tools for diseases like cancer. This strategy is underpinned by a formal framework of **ethical principles** designed to balance rapid technological innovation with systemic safety. Ultimately, the government's focus on **STEM education** and collaborative governance aims to create a highly efficient, data-rich ecosystem that could redefine the future of healthcare delivery.

Keywords: AI startup funding, Clinical AI implementation, Responsible AI governance, Mandatory data sharing, STEM education regulation

Content

China has also implemented favorable policies and funding for AI startup companies and implementation efforts. Also, China leads in the number of AI patents as a result of these favorable environments they've set. For example, in China, AI based screening tools have already been deployed in real practice, including AI based tools for the diagnosis of lung cancer, esophageal cancer, diabetic retinopathy, and general diagnostic assistance in pathology exam. ations. AI governance in China is aimed at the development of responsible AI. Unlike policies in the EU and US, AI governance in China is focused on the societal benefit rather than the individual beneficiary. And China has put focus on regulating education so that the nation produces more STEM workers. But more importantly, China requires businesses and private citizens to share their data with the government. Almost the opposite of US and EU regulations, the incentives for data sharing and the elimination of data silos could catapult China into the lead of clinically meaningful AI technologies. China's Ministry of Science and Technology has released principles for AI governance in China which include one harmony and friendliness, two fairness and justice, three inclusivity and sharing Four, respectful of privacy. Five, secure, safe, and controllable. Six, shared responsibility. Seven, open collaboration. And eight, agile governance. China is rapidly developing important AI solutions in healthcare. And these governing policies are essential to balance both the innovation of the technology as well as the safety of the healthcare delivery system.

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22. Module 5 - Clinical Evaluation.mp4

Summary

This text outlines the process of **clinical validation**, which serves as the final pillar of regulatory oversight for artificial intelligence in medicine. The primary objective is to confirm that a tool's output generates **measurable, patient-relevant improvements** in health while maintaining rigorous standards for safety and accuracy. Manufacturers must provide evidence of effectiveness during both the **pre-market phase** to establish reliability and the **post-market phase** to monitor real-world performance. Ultimately, the framework ensures that the **scientific rigor** applied to an AI model scales with its potential risk, ensuring that technology-driven healthcare decisions are both dependable and beneficial for the target population.

Keywords: Clinical validation, Patient relevant outcomes, Pre-market evidence, Postmarket performance data, Regulatory compliance standards

Content

The last category of the clinical evaluation process is clinical validation. Under clinical validation, we can ask the question, does the use of your AI's output achieve your intended purpose in the target population in the context of clinical care? There are many ways in which clinical validation can be measured including sensitivity, specificity, number needed to treat, number needed to harm, etc. This aspect of regulation is related to the positive impact of the AI solution on the health of an individual or a population. It should be a measurable patient relevant clinical outcome or outcomes that includes outcomes related to the function of the model or a positive impact on an individual or public health. Clinical validation is provided by the manufacturer both pre-market and postmarket. Prior to the launch of the AI product during the pre-market phase, ev idence must exist on the AI accuracy, specificity, sensitivity, reliability, usability, limitations, and scope of use in the intended use environment with the intended user. After launching the product or in postmarket, the product must continue to collect realworld performance data, which includes complaints, patient safety data, etc. to further understand the health care needs and to ensure the AI solution is meeting those needs and to monitor the product's continued safety, effectiveness, and performance in real world use. The IMDRF identifies that clinical validation is a necessary component of regulation and that it can be demonstrated through several paths, including referencing existing data from studies conducted for a different intended use, where extrapolation of such data can be justified, or finally generating new clinical data for a specific specific intended use. As healthcare decisions are increasingly dependent on information provided by the outcome of AI models, it is expected that performance metrics for software as medical devices or AI solutions will have the scientific level of rigor that corresponds to the risk and impact of the model to demonstrate assurance of safety, effectiveness, and performance. The clinical evaluation process is an important component of the regulatory environment.

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23. Module 5 - Definition Statement & Risk Framework.mp4

Summary

To regulate software as a medical device (SaMD), developers must first craft a **definition statement** that identifies the model's medical purpose, the severity of the targeted health condition, and the specific user population. This statement serves as the foundation for a **standardized risk framework** which categorizes applications into four levels of regulatory oversight based on the impact of the information provided and the criticality of the healthcare situation. While lower-risk categories are subject to **general controls**—such as quality system regulations and proper labeling—higher-risk devices that diagnose or treat critical conditions require the most **stringent pre-market approval**. Ultimately, this structured approach ensures that AI solutions are evaluated through a consistent lens of **patient safety and clinical significance** before they are integrated into medical practice.

Keywords: Definition statement, Risk framework, Medical device categories, Intended use, Regulatory controls

Content

So let’s look closer at the definition statement which is required for every software as a medical device application and is used to identify the submission category which defines the risk and subsequent data requirements. The definition statement must clearly identify the intended medical purpose of the model which again is to treat, diagnose, drive medical or inform clinical management and it must state the healthc care situation or condition that the AI model is in intended for which includes critical, serious or non-s serious conditions. The definition statement must also include the intended population for the application. For example, is your model intended to work in the adult population or pediatrics in the outpatient setting or inpatient or among dermatology patients. The statement must also identify the intended users or stakeholders of the model. Using information from the definition state, the software as a medical device category is defined which is based on a risk framework developed by the IMDRF that combines the state of the healthcare situation or condition and the significance of the information provided by the software as a medical device to the health care system. A major deliverable from the IMDRF forum in 2017 was a risk framework to define the software as a medical device category which sets the level of regulation for an AI application. The SA and D category is determined by this standardized framework. In the proposed framework, risk is set by the intended use of the SAMD and the state of the health situation it targets. In the grid, the columns represent the significance of information provided for a health care decision which identifies the intended use of the information provided by the software as a medical device. Again, which is to treat or diagnose, to drive clinical management, or to inform clinical management. This is the action in the outcome action pairing framework used for AI evaluation. If we look at the rows in the grid, they represent the state of the healthcare situation which include critical, serious or non-serious conditions. Now we can walk through some examples to demonstrate the definition statement and risk category. In a hypothetical situation, let’s say I have developed an AI application intended for U patients that receives electroc cardiogram, blood pressure and temperature signals from a patient monitoring system. The physiological signals are processed and analyzed to detect patterns that occur at the onset of patient instability and deterioration, a threshold set by the utility analysis. When physiological instability is detected, an alarm is generated to indicate that immediate clinical action is needed to prevent potential harm to the patient. So, let’s think about this. This software as a medical device application will drive clinical management in a critical healthcare situation or condition. This would be defined as a category 3 based on the risk stratification framework. In another example, let’s say I’ve developed an AI solution intended for patients seeking consultation at a dermatology clinic that receives dermoscopic images of skin lesions. The images are processed using a deep learning algorithm. and the output of the model is the likelihood that the lesion is a melanoma. When the likelihood is above a threshold set by the developers, an alert is sent to the dermatologist. So, how about this example? In this example, the software as a medical device will diagnose melanoma in a serious healthcare situation or condition. Again, this is a category 3 risk. As a last example, if I were developing an AI solution that will provide a list of hospitalized patients who have a high probability of a 30-day readmission. The algorithm would inform clinical management and the health condition would be labeled as a serious healthcare situation or condition. Here I am not providing any treatment or diagnosis. I am simply providing a list of high-risisk patients to the hospital management team. From our grid we see that this application would be a category one. So to recap this important aspect of regulation software as a medical Device regulation places devices into four categories based on risk associated with the use of the device with category one being the lowest risk and category 4 being the highest risk. Category 1 devices require general controls and I'll talk about the general controls later. Category 2 devices are medium to moderate risk and require the same general controls and some additional controls. Finally, category 3 and four applications require general controls and a pre-market approval, which is the most stringent regulatory category. To date, there are very few category 4 software as medical devices that have been approved in the US. Those are high-risk, generally life-supporting, life sustaining AI applications. As I just mentioned, all applications must include general controls. General controls require that all AI solutions comply with three components of reg. regulation. First, the quality systems regulations. Second, current good manufacturing practices. And third, that they are properly labeled, meaning the label of the device follows FDA guidelines and regulations. In addition, any adverse events associated with the AI solution must be reported. So, let's talk about the quality system regulations. This refers to the fact that manufacturers developing software as a medical device must have in place processes for controlled bug resolution, incident reporting, standardized design processes, and an overall risk management.

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24. Module 5 - EU Regulations.mp4

Summary

The European Union's GDPR framework establishes a rigorous standard for digital ethics by prioritizing **individual autonomy and transparency** in the age of artificial intelligence. At its core, the regulation mandates **explicit informed consent** for data collection and grants citizens the **right to an explanation** for

automated decisions, a move that challenges the use of opaque "black-box" algorithms in sensitive sectors. By shifting **data ownership and control** back to the patient, these rules aim to foster greater **accountability and trust** within the healthcare landscape. Ultimately, while these policies impose strict technical requirements on manufacturers, they serve the essential purpose of ensuring that AI systems remain **reliable, ethical, and focused on human well-being**.

Keywords: EU AI Regulations, GDPR, Automated Decision-Making, Informed Consent, Right to Explain

Content

It is important to note that there are some differences across the globe regarding regulatory guidelines for AI and healthcare. In the European Union or EU, the General Data Protection and Regulation, the GDPR, outlines a comprehensive set of regulations for the collection, storage, and use of personal information which may be used in AI solutions, which was adopted by the European Parliament in 2016 and became effective in 2018. A critical component of this regulation is Article 22, automated individual decision-making, including profiling. This article of regulation describes the right of citizens to receive an explanation for algorithmic decisions. The implications would exclude the use of many types of algorithms used today in advertising, in social media, and eventually in the healthcare system. The GDPR provides protection of data for EU citizens, but given the global AI momentum, the laws may be impacted. companies from the US and worldwide. Let's talk a little bit more about this article 22. Specifically, article 22 requires explicit and informed consent before any collection of personal data. Informed consent has been a long-standing component of medical practice, but having to obtain informed consent for any collection of data still represents a higher bar than obtaining consent for specific items such as procedures or surgical interventions. This article provides a good foundation on how to track what data is being collected and the process to request removal of data at any time from an algorithm. Shifting the power to the patients in healthcare could provide better trust, transparency, and ultimately better patient satisfaction. The attributes of patient satisfaction highlight the importance of ongoing work needed to protect the patients right to privacy and to create the appropriate policies and procedures regarding data ownership. Another important component of the GDPR is the right to explain. This is of course referring to the use of blackbox algorithms and this could potentially limit the types of models that manufacturers are able to use in healthcare related settings. The right to explain requires that meaningful information about the AI logic as well as the potential significance and consequences of the datadriven systems are provided upon request. While Model interpretability is important in an AI based healthcare application. Given the high stakes of dealing with human health, it is possible that this obligation may help AI applications become more reliable and trustworthy. The decision for the right to an explanation for an output of an algorithm will hold the manufacturers of AI based technologies more accountable. While there are some differences between the US and EU regulations, a common theme from both entities is the protection of the individual, their data and their right to be informed.

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25. Module 5 - Examples.mp4

Summary

Navigating the regulatory landscape for healthcare AI requires understanding how the **intended use and clinical context** of a tool determine its **risk classification** and oversight requirements. This source outlines the essential framework for developers, emphasizing that **safety and effectiveness** must be maintained through **good machine learning practices** and rigorous **postmarket monitoring** of real-world data. By examining specific FDA-approved examples—ranging from clinical decision support for radiologists to **autonomous diagnostic systems** for diabetic retinopathy—the text illustrates how the transition from informing to driving clinical management impacts a device’s status. Ultimately, the material demonstrates that while AI regulations are constantly **evolving**, the core focus remains on managing the **total product life cycle** to ensure these technological solutions provide reliable medical insights.

Keywords: FDA regulations, Medical device classification, Clinical decision-making, Software as device, AI healthcare applications

Content

We talked about many aspects of FDA regulations of AI solutions in healthcare and we saw which processes or actions need to meet specific regulations. So when considering the regulatory environment for an AI model, we should keep in mind that one developers should be aware of software as a medical device risk classifications, total product life cycles and good machine learning practices. Second, the population performance and intended use are aspects of the model that can be regulated and any change to these characteristics after approval would require a notice of modification. Third, the intended use and state of the health care situation of the AI model drives the level of regulation and risk category. And finally, four, safety and effectiveness must be continuously monitored postmarket using real world data to ensure a learning health system. Although AI regulation is an evolving process. Of course, there are hundreds of AI applications in healthcare that have received approval. Let’s walk through a few AI solutions that have received FDA approval. First is Arteries. Arteries is a software that aids in finding lesions in pulmonary CT scans and liver CT and MRI scans using AI to segment legions and nodules. This is the first FDA approved deep learning clinical platform. The indications for use for our arteries is it is a software that uses cardiovascular images acquired from MR scanners. It analyzes blood flow from the heart using the MR images. The output is intended to be used to support cardiologists, radiologists, and other health care professionals for clinical decision-making. It is not intended to be a source of medical advice nor to determine or recommend a course of action. So, the significance of the information is to inform clinical management and the state of the healthcare situation is critical. Therefore, according to the FDA, the risk classification for this model would be class 2. Let’s talk about a second example. For the second example, let’s look at the IDXDR. This is a software that provides automatic detection of more than mild diabetic retinopathy in adults 22 years of age or older diagnosed with diabetes who have not been previously diagnosed with diabetic retinopathy. This is meant to be used in primary care settings with subsequent referral to an eye specialist if indicated and is the first autonomous AI diagnostic system as no clinician interpretation is needed before a screening result is generated. So what are the indications of use for this software? The IDXDR is a retinal diagnostic software device that incorporates an adaptive algorithm to evaluate images for diagnostic screening to identify al diseases and conditions. So the significance of the information is to drive clinical management and the state of the healthcare situation or condition is serious. This has been classified as class 2. For our third example, let’s talk about the Guardian Connect which is a metronic device. Guardian Connect continuously monitors glucose and sends collected data to a smartphone app. This uses IBM Watson’s technology to predict major fluctuations in blood glucose levels. 10 to 60 minutes in advance. So according to the software, the indications for use are the Guardian Connect system is indicated for continuous or periodic monitoring of glucose levels in the fluid under the skin in patients 14 to 75 years of age diagnosed with diabetes. So the significance of the information is to drive clinical management and the state of the healthcare situation is serious. This has been classified as a class 2 algorithm.

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26. Module 5 - General Control.mp4

Summary

When seeking regulatory clearance for medical software, the specific requirements and data submissions are determined by the product's **assigned risk category**. A primary pathway for approval is the **pre-market notification (510K)**, which is used when a developer can prove their technology is **substantially equivalent** to a device already available on the market. To succeed with this streamlined approach, the new software must share the **same intended use** as the existing "predicate" device and demonstrate that any technological differences do not compromise **safety or efficacy**. Ultimately, this process allows innovators to leverage previously approved applications to create a more **efficient and predictable route** toward federal regulation.

Keywords: Regulatory control requirements, Pre-market notification, 510K pathway, Substantial equivalence, SaMD risk categories

Content

In addition to the general control process, there are other regulatory control requirements, data requirements that accompany an S AMD application that depend on the applicant's category of risk, which was 1 to4, which can include a pre-market notification, the 510K, a denovo notification, or a pre-market approval, a PMA. Next, we will walk through each of these regulatory components in detail. First, the Simplest data requirement is the pre-market notification or the 510K. If the software as a medical device is similar to a product already on the market, you would submit a pre-market notification. The intent of this notification is to inform the regulatory agencies that the AI solution is safe and effective, which is determined by demonstrating the AI solution is equivalent to a legally marketed device, often known as the predicate. To determine equivalence here. The AI solution must have the same intended use as the predicate and have the same technological characteristics or the AI solution must have the same intended use as the predicate and a different technology that would not raise safety or efficacy questions and the AI solution is at least as safe and effective as the predicate. Let's walk through an example. If I develop an AI model that uses MR scans that analyze blood flow patterns, that will be used to notify clinicians which patients are at high risk of hospitalization due to blood flow velocity. I could use the existing artery software as a predicate as it has already received FDA approval, has a similar intended use, and is based on the same technology. As more and more applications become approved, pre-market notifications will become an easier and efficient pathway towards regulation.

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27. Module 5 - Learning Objectives.mp4

Summary

This module transitions from the technical creation of medical AI to the critical **regulatory and ethical landscapes** that govern its real-world implementation. Students will explore how major global powers, including the **United States, China, and the European Union**, balance the need for technological innovation with the necessity of **patient safety and clinical effectiveness**. By examining specific **FDA-approved applications**, the lesson clarifies the complex boundaries between standard software and regulated medical devices. Ultimately, the material provides a comprehensive framework for understanding the **legal opportunities and challenges** involved in bringing an artificial intelligence product to the healthcare market.

Keywords: AI healthcare solutions, Global regulatory environment, Product distribution considerations, FDA approval process, Safe innovation practices

Content

By now you should have an understanding of how to develop and evaluate an AI solution in healthcare. Here we will talk about additional considerations for the use of AI solutions in healthcare particularly focusing on considerations for product development and distribution in terms of population performance and intended use. Today I will provide an overview of the regulatory environment of AI solutions in healthcare and share a framework used to understand when an AI model can and cannot be regulated. Globally, regulators are actively following AI developments and engaging in discussions to think about how to best promote innovation while facilitating safe and effective use in the healthcare setting. We will examine how the US Food and Drug Administration, the FDA, the European Union, and China are currently regulating AI solutions. Then we will discuss a few AI applications that have received FDA approval. So, by the end of this lecture, You should have an understanding of the current state of AI regulations, opportunities, and of course the challenges.

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28. Module 5 - Locked vs Adapted AI solutions.mp4

Summary

This text explores the fundamental shift in healthcare technology from **locked algorithms** to **adaptive machine learning systems**. While traditional models are defined by **fixed functions** that yield consistent, repeatable results, adaptive solutions possess the unique ability to **continuously learn and evolve** after they are deployed. Because these modern tools can produce **different outputs over time** as they incorporate new data, they challenge existing safety standards. Consequently, the author argues for a **new regulatory paradigm** based on a total product life cycle that balances the benefits of **constant improvement** with rigorous medical safeguards.

Keywords: Locked algorithms, Healthcare AI regulation, Adaptive AI solutions, Machine learning applications, Product life cycle

Content

So far we have discussed the regulation of AI solutions in healthcare but we have focused on locked algorithms. Locked algorithms are AI solutions that provide the same result each time the same input is provided. These are generally fixed functions. For example, static lookup tables, decision trees or complex classifiers. These locked algorithms may use a manual process for updates and validation. For example, updating the lookup table or adding to the lookup table. So how do we regulate adaptive AI solutions or machine learning applications? The power of these adaptive algorithms lies within the ability to continuously learn after deployment. Following deployment, these types of adaptive models may provide a different output in comparison to the output initially approved for a given set of inputs. That is to say that for a given set of inputs, the output may differ before and after the learning. changes are implemented. These automated changes use a well-defined process which aim to improve outcomes based on new data or the addition of data that is taken as an input. There are two stages to an adaptive algorithm. The learning stage and the updating stage. In the learning stage, the algorithm learns how to change its behavior based on the addition of new input types or new cases to an already existing training set. In the The update stage, the algorithm will update when the new version of the algorithm is deployed. Stated differently, given the same set of inputs at time A before the update and time B after the update, the output of the algorithm may differ. This is really a paradigm shift in the regulatory process and requires a new total product life cycle that allows these devices to continuously improve and learn while providing effective safeguards.

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29. Module 5 - Non-Regulated Products.mp4

Summary

This text outlines the specific conditions under which health-related artificial intelligence avoids strict oversight, focusing on the **exemption of certain clinical decision support tools** from FDA regulation. For software to remain non-regulated, it must function transparently so providers can **understand the underlying logic**, avoid processing direct medical signals, and serve only as a secondary resource rather than the primary diagnostic authority. The material also highlights the **Pre-Cert program**, an innovative regulatory shift that prioritizes the **trustworthiness of the organization** over individual product reviews to accelerate the deployment of low-risk technologies. By adhering to core principles like **patient safety and proactive culture**, certified developers gain the flexibility to iterate on their tools without the burden of constant federal re-approval.

Keywords: Clinical decision support, FDA regulation criteria, Laboratory developed tests, Digital health precertification, Software medical devices

Content

So far we have focused our discussion on AI solutions in healthcare. It is important to note that the FDA does not regulate certain types of clinical decision support tools under the 21st century cures act. CDS tools help providers make care decisions by for instance providing suggested drug drug interactions or drug dosage. There are three criteria that determine whether clinical decision support software are regulated as a software as a medical device. The first First is that the software cannot receive, analyze, or otherwise process a medical image or signal from an in vitro diagnostic device such as a DNA sequencing machine or from another signal acquisition system. Second, a health care professional must be able to understand the basis of its recommendation. And third, the software cannot be intended as the sole source of recommendations regarding treatment, diagnosis, or prevention of disease. Additionally, the FDA doesn't regulate AI solutions that are laboratory developed tests, meaning they have been designed, developed, and deployed within a single healthcare setting. More recently, the FDA announced the development of the digital health software pre-certification program, the pre-ERT, that they hope will help streamline regulation of AI solutions. Under this program, organizations become approved, which would allow the organization to bring products to the market without appre market review provided the product is low risk. Another advantage of the presert program is that once certified, organizations can make minor changes to their AI products without having to submit a modification request. To be considered for the presert program, the organization must demonstrate the FDA's five quality and organizational excellent principles, which include one, product quality, two, patient safety, three, clinical responsibility, four, cyber security responsibility, and five, proactive culture. This is an exciting advancement to address previously long approval times, and the program is still work in progress.

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30. Module 5 - OMB Guidelines.mp4

Summary

The White House Office of Management and Budget has established a framework of ten guiding principles designed to oversee the deployment of artificial intelligence, ensuring these tools are integrated safely and ethically into society. This regulatory "umbrella" prioritizes **public trust and participation**, requiring that developers remain transparent about how algorithms function while actively involving stakeholders to mitigate risks to privacy and civil liberties. To ensure clinical and technical excellence, the guidelines mandate **scientific integrity and rigorous risk assessment**, where the potential societal benefits of an AI application must clearly outweigh its costs and possible errors. Finally, the framework emphasizes that AI must be **fair, flexible, and secure**, necessitating a multidisciplinary approach to monitoring bias and maintaining data safety throughout the technology's entire lifecycle.

Keywords: OMB AI Guidelines, Public Trust, Risk Management, Scientific Integrity, Transparency and Fairness

Content

We will wrap up our conversation by discussing the released document from the White House Office of Management and Budget that provides guidelines for the regulation of AI applications. Now, these regulations are not specific to healthcare, but they will provide the umbrella of regulations applied to AI solutions. There are 10 guidance points provided by the Office of Management and Budget, which are aligned with the FDA's regulatory processes. Let's walk through the 10 guiding principles. The first, there must be public trust in AI. AI solutions in healthcare should have a positive impact across sectors of social and economic life. However, we know that AI applications pose risks to privacy, individual rights, autonomy, and civil liberties that must be carefully assessed and appropriately addressed. Public trust and validation is critical to the adoption and acceptance of these technologies in The healthcare sector privacy and other risks must be addressed with appropriate mitigation strategies that are well documented and transparently reported. There must be public participation in AI development. Throughout this course, we have talked about stakeholder involvement in the development of AI to improve accountability and regulatory outcomes. AI solutions should encourage public participation. There should be ample opportunities for the public to provide information and participate at all stages possible of the rulemaking process. Furthermore, good machine learning practices and AI standards should be used to create the models and be disseminated and promoted to the public. Third, AI solutions must be based on scientific integrity. Clinical validation in the clinical evaluation process proposed by the IMDRF. AI and healthcare should be based on scientific and technical information. and processes that are likely to have a clear and substantial influence on public policy or private sector decisions. And these standards should be held to the highest level of quality, transparency, and compliance. Best practices such as good machine learning practices include clearly stating the strengths, weaknesses, intended optimizations or outcomes, bias mitigation, and appropriate uses of the AI application's results. The intended use and the intended population. For AI applications to produce predictable, reliable, and optimized outcomes, data used to train the AI system must be of sufficient quality for the intended use. The fourth article, every AI solution must have a risk assessment and management component. Risk assessment and risk management should be a fundamental aspect of every application. Of course, no one can mitigate every foreseeable risk. However, a riskbased approach should be used to determine which risks are acceptable and which risks present the possibility of unacceptable harm or harm that has exceeded the cost greater than the expected benefits. Risk assessment must be transparent, including assumptions and conclusions about risk to foster accountability. If an AI tool fails, the magnitude and nature of the consequences will inform the level and type of regulatory effort that is appropriate. to identify and mitigate risks. Five, benefits and costs must be part of the AIA deployment decision. For an AI algorithm to be deployed, it must offer significant potential benefit. Before implementing an AI solution, agencies must consider the full societal costs, benefits, and effects related to the development and deployment of these applications such as potential benefits and costs of employing AI compared to the existing systems. Will AI change the type of errors created by the system? If so, what are they and how do we compare them to the existing risks? And finally, what are the critical dependencies when evaluating AI costs and benefits? This should include technical factors such as data quality and changes in the magnitude of the risks and benefits. Six, AI solutions must be flexible. AI solutions should be performance based and flexible approaches should be easily adopted and updated. This would include the continuous monitoring of real world evidence to improve upon the AI solution. Seven, AI solutions must be fair and non-discriminatory. AI applications should have transparency regarding potential biases and discriminatory aspects of the algorithm. This topic is becoming more and more important as we see the potential harm due to biases promoting through AI. Eight, AI solutions should provide disclosure and transparency that will promote public trust. Healthcare systems should disclose when AI solutions are in use and how these applications can impact patients and decisions. Nine, AI solutions must address safety and security issues. AI solutions should be safe, secure, and operate as intended. Safety and security should be considered Throughout the design, development, deployment, and operational process, there should be controls in place to ensure confidentiality, integrity, and availability of the information processed, stored, and transmitted by the AI systems. Finally, AI solutions must include multiddisciplinary stakeholder involvement or inter agency coordination. All sectors affected by the AI solution should coordinate and share experiences and challenges of AI solutions. These guidelines set out by the OM office are clearly important considerations to understand and are well aligned with the FDA's regulatory processes. Safety, transparency, and multiddisciplinary aspects are key ingredients to a successful and well-th thoughtout AI solution.

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31. Module 5 - Software Modification.mp4

Summary

This module outlines the regulatory requirements for **updating healthcare AI** after its initial market approval, focusing on when developers must notify the FDA of changes. A formal modification request is triggered whenever a software update introduces **new patient risks**, alters critical safety controls, or fundamentally shifts the **clinical functionality and performance** of the device. These necessary revisions generally fall into three pillars: enhancing the underlying **AI architecture**, incorporating **new data inputs** from different sources, or expanding the **intended clinical use** to new diseases and populations. Ultimately, the text illustrates that software modification is an essential, regulated stage in the **total life cycle** of medical technology to ensure continuous safety and efficacy.

Keywords: Software modification, Regulatory approval, Risk assessment, Clinical functionality, AI model inputs

Content

So we have reviewed the concepts used by regulators and the process for approval of an AI solution in healthcare or a software for a medical device from the FDA point of view. So what's next? After regulation, software modification is a necessary consideration. After an AI solution receives regulatory approval, a modification may be required in certain circumstances. Generally, if there is new risk or a change of an existing risk, then a modification request must be submitted. If there is a change to risk controls to prevent significant harm, then a modification may be required. Finally, if there is a change that significantly affects clinical functionality or performance, then a modification may be required. A change in clinical functionality would include new indications for use, new clinical effects, or significant technology modifications that affect performance characteristics. Software as a medical device Modifications generally fall into three categories. The first is a change in performance such as significant improvement or deterioration of performance metrics that were submitted with the application of the AI solution. The second is a change to the model input. And finally, the third category is a change of the intended use of the output. So let's go through an example of each of these changes that would require a modification. A change in performance could include clude the incorporation of new training data or a change in AI architecture. Such changes could alter performance. Therefore, a modification request is necessary. A change to the AI inputs also requires a modification. A change to inputs could include, for example, the incorporation of different sources of the same input, for example, from a new manufacturer, or adding new inputs that were not previously considered, for example, including electroc cardiogram data. in addition to pulse and body temperature. Finally, if the intended use of an AI solution changes, a modification is required. Changes of intended use may include a change in application, clinical management to diagnosis, for example, a change in population, applying your population that was intended for adults to pediatrics, or a change in disease, applying your application to a new condition. Software modifications are common. and essential in the total life cycle of the AI solution.

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32. Module 5 - Summary.mp4

Summary

This text examines the evolving landscape of medical AI oversight, emphasizing the **regulatory shift** from static, “locked” software to **adaptive algorithms** that learn in real-time. To ensure patient safety without stifling progress, governing bodies are developing risk-based frameworks that categorize tools by their clinical impact and intended purpose. However, the path forward is complicated by the **ethical tension** between protecting data privacy and the need for transparent, unbiased training sets that prevent algorithmic discrimination. Ultimately, the source highlights that establishing **clear accountability** and universal standards is essential for balancing rapid technological innovation with the fundamental rights and safety of patients.

Keywords: Healthcare AI Regulation, Adaptive Learning Algorithms, Patient Safety, Data Privacy, Risk Categorization Framework

Content

The regulation of AI solutions in healthcare is complicated and global efforts are emerging to standardize the safe and efficient regulation of AI in healthcare. Most algorithms that are currently regulated are locked algorithms which are algorithms that get the same output for a given input. Learning algorithms or adaptive algorithms are a new challenge for regulators. Current practices require a notification of modification if the AI solution has a change to the intended population, model performance or intended use. However, there are several examples of AI applications that are currently in the market with approval. The current regulatory environment is rapidly being developed with regulatory leaders engaging with diverse stakeholders to develop a framework that both promotes innovation while ensuring safety, privacy, and good intent. In addition to the numerous opportunities, there are several challenges that we high highlighted throughout this lecture. First, developing a regulatory environment for AI applications that are continuously learning from new training data is difficult because the output and performance of the models is continuously changing. Guaranteeing safety and effectiveness on moving output and performance presents a paradigm shift for regulators. Second, there is a global consensus that AI solutions must be fair and non-discriminatory. However, Restrictions around data sharing and data privacy, particularly regarding training data, promote data silos rather than an environment that can promote unbiased and fair AI solutions. The notion of explanability and informed consent, including the right to have your data removed from an AI application, is a new challenge for the rapidly evolving AI tech field and identifying a balance between innov Ovation and patients rights is an enormous challenge. Finally, the notion of patient safety and accountability are important aspects that AI regulators must address. Safety and risk is the primary concern of all regulations. However, AI solutions in healthcare are deployed in complex environments where learning and uncertainty of human machine interactions may result in significant variation in the AI performance and If there is a patient safety event due to an AI solution, who is responsible? The physician or the machine? These are challenges that currently hamper the regulatory environment and explain in part the limited regulations we see in healthcare AI. So, let’s review the topics we’ve covered in this lecture. AI solutions, also referred to as software as a medical device, must clearly state an intended purpose. For example, to treat, diagnose, or form and they must include the health care situation or condition that it is intended for critical serious or non-serious. The framework for risk categorization ranges from one to four with the higher risk requiring higher regulatory requirements. Monitoring the performance of AI models and creating a continuous learning system and the regulations required for them is essential. Finally, there are global efforts driving standards for AI regulation in healthcare. For example, in the EU, informed consent is necessary for data sharing and right of explanation is required. Stay tuned. We will bring this and the other topics covered throughout this course together and walk through a few examples in our upcoming lectures.

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33. Module 5 - TPLC.mp4

Summary

The FDA regulates medical artificial intelligence by overseeing a **total product life cycle** that ensures safety from development through long-term use. This framework begins with **good machine learning practices** to establish a high-quality foundation for data management and model validation, followed by rigorous **pre-market assessments** to mitigate patient risk. Once a tool is deployed, manufacturers must perform **regular monitoring** to track performance and identify when software modifications become necessary. Finally, the system relies on **continuous learning from real-world data**, which uses post-market evidence to refine the device’s functionality and clinical impact within a broader learning healthcare environment.

Keywords: Total Product Lifecycle, AI Healthcare Regulation, Machine Learning Practices, Risk Management, Real World Data

Content

So we have gone through the basic components used for the regulation of AI and healthcare using the frameworks proposed by the IMDRF which have been adopted by the FDA. In line with this framework, the FDA has developed the following diagram related to the total life cycle for an AI workflow. There are four distinct components in the total life cycle. Let's go through these different components to understand how the regulatory environment thinks about the total product. life cycle. The first component in the total product life cycle is the culture of quality and organizational excellence, which they also refer to as good machine learning practices. This includes all components that must be considered when developing an AI solution, such as data selection and management, model training and tuning, and model validation, including performance and clinical evaluation. The second component is part of the pre-market application, which assures the safety and effectiveness of the AI solution. This is an important aspect of regulation and it is expected that safety and effectiveness are continually monitored throughout the life cycle including patient risks and patient safety. This is the riskmanagement approach and regulators expect a manufacturer to perform a risk assessment and evaluate that risks are reasonably mitigated throughout the total product life cycle. The third component of the TPLC is the regular monitoring of safety and intended use. use which is used to identify when a software modification is required. The regular monitoring of the deployed model is necessary and should include the ability to log and track model performance. It is also important to acknowledge the fourth component of the total product life cycle that regulatory agencies are also monitoring. This is the continuous learning expected from a software as a medical device that leverages real world data. It is important to understand that here Continuous learning is not machine learning. Meaning we are not talking about models that automatically learn from their training data. Rather, it refers to collecting post-market information to update and evaluate your existing AI solution. Of course, in a learning healthcare system, a hypothesis may be developed based on the continuous collection of data and monitoring of post-market AI solutions. Ideally, we want the future functionality and intended use of a software as a medical device. to be informed by continuously collecting and analyzing data on use of the software as a medical device in a postmarket setting. These real world data may include a variety of information such as safety data, performance studies, ongoing clinical evidence generation, new research publications and results that support and strengthen the clinical association of the software as a medical device output to a clinical condition or direct inuser feedback. The postmarket collection of real world data can speak to both the superiority or inferiority of the original valid clinical association. The monitoring of real world evidence can be used to update and modify the original software as a medical device definition statement as well as the risk classification statement. This figure demonstrates how regulators are thinking about using real world data for continuous learning, the basis of a learning healthcare system.

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34. Module 5 - The Problem.mp4

Summary

The International Medical Device Regulators Forum (IMDRF) works to create a global standard for overseeing artificial intelligence by focusing on the specific classification of **Software as a Medical Device (SaMD)**. To qualify under this regulatory framework, a digital tool must function **independently of medical hardware** and possess a clear **medical purpose**, such as diagnosing, treating, or preventing specific health conditions. While an algorithm that predicts patient no-shows is considered administrative, a model designed to **mitigate disease complications** or manage clinical risks falls under these strict guidelines. Ultimately, this distinction ensures that **regulatory oversight** is applied only to software that directly impacts **clinical outcomes and disease management**.

Keywords: AI model regulations, IMDRF, Medical device software, Clinical purpose, Risk-based framework

Content

So let’s take a moment to review the key concepts and terms used when discussing AI model regulations. Many of the terms are used globally. Therefore, it’s important to have an understanding of their specific definitions. Let’s start with the international medical device regulators forum or the IMDRF. The IMDRF is an international work group composed of AI regulators that come together to develop a path for standardized AI regulations. Their current efforts focus on the development of standard terminology, a riskbased framework, quality management principles, and an approach to making AI solutions clinically meaningful to users. The IMDRF defines software as a medical device, SAMD, as I quote, software intended to be used for one or more medical purposes that perform these purposes without being part of a hardware medical device. So a medical purpose as they define it is one that is intended to treat, diagnose, cure, mitigate or prevent disease or other conditions. There are two very important components here. First, software as medical devices are not part of hardware. For example, a model that uses signals from a pacemaker to alert an arhythmia would not be considered a software as medical device.

Second, the Purpose of the software must be to treat, diagnose, cure, mitigate or prevent disease. Previously we spoke about actions for an AI solution and some of the actions discussed would not qualify as a software for a medical device. For example, if I have an AI model that identifies patients likely to not show up for an MRI appointment, this would not be considered as a software as a medical device because the output of this model does not treat diagnose, cure, mitigate, or prevent disease. However, if I have a model that generates a list of people at high risk for 30-day hospital readmission following a hospitalization for coronary heart disease, for example, this would be considered a software as a medical device because I am trying to mitigate a readmission which is highly correlated with disease management. Therefore, it is important to remember that regulations we will discuss in this lecture apply only to software as a medical device. And software as a medical device must intend to treat, diagnose, cure, mitigate, or prevent disease or other conditions and must not be part of a hardware.

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35. Module 5 - de novo Notifications.mp4

Summary

When a developer creates medical software that lacks a direct legal predecessor, they must navigate a specific **risk-based classification process** to gain regulatory clearance. Moderate-risk applications typically submit a **de novo notification**, which balances clinical evidence with an analysis of **probable benefits versus potential harms**. In contrast, the most dangerous or complex tools are subject to the **pre-market approval (PMA)** pathway, the FDA’s most **rigorous regulatory standard** requiring exhaustive scientific validation. Ultimately, these demanding protocols serve as an **essential safeguard** to ensure that artificial intelligence functions reliably and safely within a clinical environment.

Keywords: De novo notifications, Risk-based classification, Clinical data, Pre-market approval, Safety and effectiveness

Content

If there is no predicate, then the software as a medical device application may submit a denovo notification. Denovo notifications are limited to category 1 and category 2 SMDs and are a riskbased classification process. The denovo notification should include clinical data if applicable that are relevant to support the assurance of the safety and effectiveness of the AI solution. Nonclinical data including bench performance testing, a description of the probable benefits of the AI solution when compared to the probable or anticipated risks when it is used as intended. Information that may be generated from the clinical evaluation process just described. Finally, for high-risk SMDs, category 3 and category 4, a pre-market approval or PMA is required. Pas are the most stringent regulatory applic ations required by the FDA. The PMA includes rigorous technical studies, nonclinical, laboratory studies, laboratory studies, and clinical investigations. Before PMA approval, the applicant must provide valid scientific evidence demonstrating reasonable asurances of the safety and effectiveness for the devices intended use. Again, regulation is a long and tedious process. Yet, it is essential to ensure the safety and effectiveness of AI I in healthcare.

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36. Module 5- Analytical Evaluation.mp4

Summary

Analytical validation serves as a rigorous internal assessment to confirm that an artificial intelligence tool **transforms raw data into precise outputs** with consistent reliability. By utilizing curated datasets and established ground truths, manufacturers provide **objective evidence** that the software's underlying architecture and data-labeling processes are technically sound. This stage functions as a critical bridge within **good software engineering practices**, ensuring the system is built correctly before it is applied to real-world scenarios. Ultimately, the process focuses on the **technical integrity of data processing** to guarantee that the machine's mathematical results are both accurate and reproducible.

Keywords: Analytical validation, Clinical evaluation, Data processing, Software engineering practices, Performance evaluation

Content

The second category in the clinical evaluation process is analytical validation. Analytical validation evaluates whether your AI solution correctly processes input data to generate accurate, reliable and precise output data. This is generally part of the verification and validation phase that should be performed by the AI manufacturer. Analytical validation provides objective evidence that the AI solution was correctly constructed and the data processing is reliable. This includes for example the process to label data, how the ground truth was developed, the performance evaluation. Analytical validation may come as part of your good software engineering practices or from generating new evidence through the use of curated databases or previously collected patient data.

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37. Module 6 - 1 Problem Formulation.pdf

Summary

Establishing **best ethical practices** in healthcare AI begins with the critical stage of **problem formulation**, where developers must ensure that their primary intent is to promote **patient well-being** while mitigating potential misuse. This process requires a careful translation of broad goals into specific questions, yet it remains vulnerable to **systematic error and bias**, particularly when **proxy variables** like healthcare costs are used to estimate actual medical needs. To safeguard against these pitfalls, the text emphasizes the necessity of **multidisciplinary teams** that include clinicians who can identify gaps in **available data** and recognize how social inequities might distort algorithmic outcomes. Ultimately, the source serves as a guide for developers to maintain **accountability** by questioning the definitions of their metrics and the inclusivity of their datasets to ensure equitable care for all **subpopulations**.

Keywords: Ethical AI practices, Problem formulation, Algorithmic bias mitigation, Data selection challenges, Clinical team collaboration

Content

[Image]

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Copyright © Stanford University 1 EVALUATIONS OF AI APPLICATIONS IN HEALTHCARE WRITTEN BY MILDRED CHO BEST ETHICAL PRACTICES – PROBLEM FORMULATION In this section, we will learn about specific actions that have been recommended as best ethical practices in the development and deployment of machine learning-based AI in health care applications. Our thinking about best ethical practices will start at the stage of problem formulation and determining the purpose of the AI system that you want to develop. First, is the intent ethical, is its purpose to enhance the health and well-being of patients? Even if the intended purpose seems ethical, could there be negative consequences if misused? Increasingly we are seeing data scientists challenging the purpose and the uses of the products they are developing, whether the purpose is to target advertising to individuals using social media, or to identify individuals through facial recognition algorithms. For example, the Association of Computing Machinery, one of the leading professional organizations of computer scientists, has urged a suspension of the use of facial recognition technologies because of their potential for prejudicial impact on human and legal rights. The Association has also asserted that developers and operators, as well as users of facial recognition technology are accountable for these systems' use and misuse. While these might seem like questions that don't need to be asked in the health care AI context because the answers are obvious, they are especially important to ask for AI developed by teams that have conflicting or competing interests. Health care settings are rife with such competing interests because they are operating under resource constraints, including constraints on finances, personnel, equipment and supplies, not to mention financial incentives to improve care. Together, these interests and incentives all create pressure to avoid certain kinds of patients, or avoid providing certain kinds of treatments. AI developers have to be vigilant about mitigating conflicts of interest. But how? We'll talk about this more specifically, but first, let's consider the issue of problem formulation. Formulating the AI problem involves translating a high-level goal such as "improving patient care" into actionable questions that can be answered by available data. The challenge is that actionable questions and available data are usually limited, so a lot can get lost in translation, and practical [Image]

[Image]

 constraints on problem formulation can lead to undetected errors and bias. A fairly common and seemingly straightforward task such as risk stratification is often fraught because risk can be defined in many different ways, and usually in ways that are not measured directly but through proxy variables. We have seen that the use of health care costs as a proxy for risk or health care needs has led to racial bias because risk scores generated from predictive models based on costs underestimated the health care needs of Blacks as compared to Whites with the same risk score. Fortunately, such bias can be tested for, at least for variables for which you suspect underlying bias exists such as race or gender. In this example you would be looking at whether the relationship between risk scores generated by the model and the variable of interest, health needs, differed by race. Similarly, risk stratification on the basis of cost as a proxy for health needs tends to be biased towards older people with complex chronic conditions at the end of life, when most health care costs are incurred, and against children with acute, potentially fatal but treatable conditions. If the question to be answered is "who needs the most care" it is important to remember that this question is not merely quantitative but also a values question that demands nuance. How is "care" defined? Does the question distinguish between acute and chronic care? How is "need" defined? If patients have untreatable conditions and therefore typically do not receive costly care, does that mean that a model should assign them low risk scores, or that they do not need care? Questions to ask in problem formulation Is the purpose of the AI system to enhance the health and well-being of patients? Could there be unintended consequences if misused? What is the intent of the system – to identify, classify, predict? What is being identified, classified or predicted? o E.g. risk, disease, prognosis, costs, health care utilization, admission or readmission, treatment efficacy, decompensation How are these terms defined? Closely related to problem formulation is the choice of data. Of course, data that are already available are the easiest to use. But that doesn't mean that these are the right data. For example, electronic health records might be plentiful but they lack a lot of information that could be important to predictive models, such as environmental exposures, diet, or socio- economic factors, all of which we know are highly determinative of health and disease. EHRs from one hospital or health system, or insurance claims data also often do not provide a longitudinal timeline of data points for a patient over time because patients often move between health providers and insurers. [Image]

[Image]

Copyright © Stanford University 3 And relying on data from a single time point could be misleading. On the other hand, grouping data together can mask important patterns. In our example of risk stratification, a model trained on data from people of a mix of age ranges could be masking patterns in data from a subpopulation, such as younger people. Of course, detailed knowledge of differences between subpopulations is necessary to know whether customized models are required. How does one know this in advance? Questions to ask about data How well are the variables in the model represented by available data? Are proxy measures being used? Are there important variables that are likely to be associated with main outcome measures that are not represented in available data? Are there likely to be differences between subpopulations in main outcomes, especially by legally protected characteristics such as age, race, ethnicity, or gender, or socially important characteristics such as income and education? Best practices point to the need to include practicing clinicians who are intimately familiar with the relevant patient populations and conditions, and the data that are necessary for modeling They need to know what data are actually available, and the limitations of the data, especially in terms of likely biases. It is important to have team members with an understanding of the source of systematic error, such as whether error is introduced because of lack of data from specific subpopulations, physician biases (such as the lower likelihood of women with heart disease to get a diagnosis of heart disease as compared to men), or broader social inequities such as differential access to care. That is because understanding the source of systematic error indicates whether and how models can account for bias. Clinician team members can be critical at the problem formulation stage in particular, because deep clinical knowledge is so important to understanding the implications of asking questions in a particular way. [Image]

[Image]

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38. Module 6 - 2 Identifying Conflicts of Interest.pdf

Summary

In this instructional module, Mildred Cho explores how **conflicts of interest** arise in healthcare when **primary duties**, such as patient welfare and research integrity, are compromised by **secondary interests** like financial gain or professional reputation. The text illustrates how even small incentives, such as industry-sponsored meals, can exert **undue influence** on a provider's clinical judgment and lead to biased medical outcomes. To preserve ethical standards, the author outlines a hierarchy of solutions ranging from **simple disclosure** to more robust methods like **independent mediation** and **recusal**. Ultimately, the source serves as a guide for identifying and neutralizing these hidden biases to ensure that medical and AI-driven decisions remain focused on **patient-centered care**.

Keywords: Conflict of interest, Healthcare AI ethics, Primary versus secondary, Mitigation strategies, Physician prescribing patterns

Content

[Image]

[Image]

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Copyright © Stanford University 1 EVALUATIONS OF AI APPLICATIONS IN HEALTHCARE WRITTEN BY MILDRED CHO BEST ETHICAL PRACTICES – IDENTIFYING CONFLICTS OF INTEREST In this section, I want to talk about best ethical practices that address the issue of conflicts of interest. In medicine and biomedical research, we deal with conflicts of interest all the time, and have developed ways of mitigating their negative effects. So, how is that done, and how can we apply those strategies to the development and deployment of AI in health care settings? First, what is a conflict of interest? For our purposes, they exist only when there is a primary interest that is a duty. An example of a primary interest is the clinician's or hospital's duty to care for their patients. However, we all have multiple interests, some of which can compete or conflict with these interests. These other interests are called secondary interests, and can include, for example personal or institutional financial interests, duties to others such as people who are not the clinician's or hospital's own patients, or reputational interests, either positive or negative. [Image]

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 So, why are these secondary interests a problem? In the health care setting, the problem arises because in the course of carrying out duties, many decisions have to be made on behalf of patients, and decisions are often based on individual judgement. And judgement can be influenced by all of these secondary interests, in such a way as to subvert from the primary duty and cause harm to patients. An example of this in clinical care is the influence of pharmaceutical sales representatives who try to persuade doctors to use their drugs instead of the competitor's drugs, when in some cases the competitor's drug might be better for the patient. Influence might be exerted in the form of financial incentives or gifts. These secondary interests have been linked to significant changes to prescribing practices, so they have real effects. Although physicians think it is impossible to be influenced by accepting a slice of pizza paid for by a pharmaceutical company, there is a study demonstrating that prescriptions can be more than doubled specifically for drugs sold by the providers of meals over other similar drugs, with increases in prescribing seen even after just one meal and rising with every additional day of meals provided. So, part of the problem is that these secondary interests have effects without our being aware of them. In biomedical research, secondary interests can also include financial factors such as the promise of obtaining research grants or consulting fees from sponsors, or stock or royalties from companies whose products are being studied. Secondary interests can also be reputational or ideological, for example, a strong desire to gain fame or promote a specific hypothesis. In the case of research, the primary interest is usually the integrity of the research, although for research involving human subjects, the health and welfare of individual research participants is also of high priority. But what are the possible Photo by Pinar Kucuk on Unsplash Source: DeJong C, Aguilar T, Tseng C, Lin GA, Boscardin WJ, Dudley RA. Pharmaceutical Industry–Sponsored Meals and Physician Prescribing Patterns **for Medicare Beneficiaries. JAMA Intern Med. 2016;176(8)** [Image]

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Copyright © Stanford University 3 negative effects of influences of secondary interests? Again, let's look at all the decisions that are made in the course of research that could be subverted away from serving the primary interest of the integrity of the research. One is the formulation of the research question. Is the question in service of corporate interests or meeting a real patient need? For example, is a clinical trial designed to test a drug in order to serve patients who have no other therapeutic options, or a reformulation of an existing drug that will extend patent protection and a company's market position? Another way that secondary influences can have effects is through creating unconscious bias, leading to inaccurate measures of treatment effects. We know that, in general, systematic error tends to inflate effect sizes. Clinical researchers have long recognized this, and therefore use techniques such as randomization and blinding to reduce bias, or systematic error. Choosing to use such techniques represent design decisions. Other important design decisions that could be influenced by secondary interests are what populations and data you choose to study, and how you choose to analyze, report or share data and findings. Ways to Mitigate Conflicts of Interest: Disclosure • Publicizing the secondary interest Mediation • Putting financial (secondary) interest in blind trust Putting some decisions about the primary interest in the hands of an independent body Recusal • Removing financial (secondary) interest or replacing the person with the primary interest So, what do we do to mitigate potential effects of conflicts of interest? We can focus mitigation strategies on protecting the primary interest, eliminating or reducing the secondary interest, or both. The most common strategy, which you have probably employed or been asked to employ, is disclosure, which entails making public your secondary interests, such as disclosing speaking fees or research sponsorships from companies whose products are related to the primary interests. Although it is the most common strategy, it is also the weakest because it does nothing to reduce or remove the secondary interest, but relies on those to whom the interest is disclosed to understand the implications of the disclosure and act in some way to counteract possible negative effects. Other strategies that are more robust involve mediation of the primary or secondary interest, such as placing a financial interest in a blind trust, or having an independent oversight committee such as a [Image]

[Image]

Copyright © Stanford University 4 Data Safety Monitoring Board in place to oversee or make important decisions, essentially taking them out of the control of the person who has the conflict of interests. The strongest strategies are recusal from secondary interests, such as selling one's stock, or even recusal from the primary interests, such as replacing an investigator with another person who does not have conflicting interests. [Image]

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39. Module 6 - 3 Mitigating Conflicts of Interest.pdf

Summary

To ensure the ethical integration of artificial intelligence in medicine, developers must utilize specific **mitigation strategies** to handle unavoidable **conflicts of interest**. The text outlines a tripartite framework consisting of **disclosure**, which promotes **transparency** through public notification and detailed technical reporting; **mediation**, which involves **independent oversight** and third-party algorithmic auditing to prevent bias; and **recusal**, though this remains difficult to implement within corporate environments. Ultimately, the source argues that maintaining **stakeholder trust** requires a combination of these rigorous ethical practices and the inclusion of diverse clinical expertise during the design process.

Keywords: Conflict mitigation, AI healthcare ethics, Transparency and disclosure, Independent oversight, Algorithmic auditing

Content

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Copyright © Stanford University 1 EVALUATIONS OF AI APPLICATIONS IN HEALTHCARE WRITTEN BY MILDRED CHO BEST ETHICAL PRACTICES – MITIGATING CONFLICTS OF INTEREST In this section, we'll try to answer the question: how do we translate conflict of interest mitigation strategies to the development and deployment of AI for health care? Let's look at each of the three general types of strategies: disclosure, mediation, and recusal. The principle underlying disclosure is transparency, or facilitating awareness of people who are impacted by AI and who are owed duties of care that other interests exist that could influence this care. For AI, this transparency is complicated by the fact that most of the people who could be impacted, such as patients and providers, are probably not even aware of the existence of AI or how it might be used in decision making about their health care. So first, some public notification of the use of AI may be warranted, especially if the application deviates substantially from commonly understood or accepted uses of data in the health care context, or if the AI or data use poses more than usual risks or be especially sensitive. Examples might be using and storing video data from telehealth visits, or collecting and analyzing patients' social media data to predict or diagnose mental health conditions, or using EHR, sensor and claims data to guide decisions about offering palliative care at the end of life. This notification could also include disclosure of how AI is used in decisionmaking about patient care and who developed the AI system. Other ways of implementing the principle of transparency include thorough reporting of how models were built, as suggested by members of the data science community. This includes clear descriptions of data sources, participants, outcomes and predictors, the contexts in which the model was validated, limitations and contraindications for deployment and assumptions or conditions that must be satisfied. The Association of Computing Machinery has also recommended that for facial recognition technologies, or other uses of AI where racial or other biases are of concern, that error rates be reported disaggregated by race, gender and other context-dependent demographic features. The principle behind the strategy of mediation is independent review or oversight. For AI development, this could be achieved by auditing of algorithms and models by third parties. This process was suggested by the Obama administration in 2016 to mitigate discriminatory practices and [Image]

[Image]

Copyright © Stanford University 2 civil rights violations. Employing algorithmic audits also enhances transparency to the extent that it encourages developers to make algorithms auditable in the first place. The strategy of recusal in the context of AI development is difficult to implement. If AI development is being conducted in an academic research setting, it might be possible for the developers to recuse themselves of financial interests, or replace people with financial interests with others who are more independent. However, in a corporate setting, that might be impossible, so developers would have to rely on disclosure or mediation-based strategies, keeping in mind that disclosure-based strategies are very weak and do little to engender or maintain trust of key stakeholders. The takeaway points here are that: Best ethical practices in developing AI for health care include careful formulation of the problem to be solved, ideally with input from people who have deep knowledge of the specific clinical settings and data relevant to the problem. Conflicting interests can have real impacts on design decisions, but there are strategies to mitigate the potential negative effects of conflicts of interests.

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40. Module 7 - A Deep Dive into Historical and Societal Dimensions.mp4

Summary

Artificial intelligence acts as a mirror that reflects the **embedded societal prejudices** and historical inaccuracies found within the data used to train it. The source highlights how medical errors are often exacerbated by a **race-centric approach**, such as the exclusion of minority groups from clinical trials or the use of **flawed correction factors** that lack a true biological basis. By hard-coding these biases into algorithms, technology risks **perpetuating stereotypes** and widening the gap in healthcare quality between different populations. Ultimately, the text serves as a critical examination of how **AI-driven decision support** must be restructured to move toward true health equity and away from systemic disparities.

Keywords: AI bias origins, Health equity, Racial disparities, Clinical trial representation, Medical AI ethics

Content

Recently, there's been significant movement in addressing AI biases and advancing health equity. To begin our dive into equitable AI, let's deconstruct a fundamental question. What shapes AI biases? AI is a tool neutral until it interacts with data. When data are embedded with societal prejudices, historical inaccuracies, or sample biases. The AI tool inevitably reflects these imperfections. It's evident that biases exist in medicine as well, particularly those rooted in racial disparities. A race ccentric medical approach often devoid of scientific rationale has led to numerous errors in medical history from skewed clinical trials to differential treatment protocols. But what are the realworld implications? Let me give you some tangible examples. First, certain groups have been at times under reppresented in clinical trials. Now, remember, clinical trials are an essential tool for advancing medical knowledge and developing new treatments. Evidence from clinical trials is used to inform clinical guidelines, and these recommendations are often included in AI models and sometimes used for clinical decision support systems. However, the historical underrepresentation of racial and ethnic minorities in clinical trials has resulted in drugs and treatments that may not benefit all populations. A second example is the use of race and ethnic correction factors in medical AI. This is a topic of significant debate and ethical concern. While there may be cases where accounting for race in AI models can appear to improve performance, there are several key issues and potential problems with this approach, including the reinforcement of biases, lack of B ological basis and the overgeneralization or perpetuation of stereotypes. Often the use of race and other protected attributes hard-coded in AI algorithms can lead to differential care and widen the disparities gap.

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41. Module 7 - Bias Mitigation Strategies.mp4

Summary

This text explores advanced methods for ensuring equity in artificial intelligence by introducing **counterfactual fairness** and **adversarial networks** as robust bias mitigation strategies. Counterfactual fairness evaluates justice through **"what-if" scenarios**, determining if a model's decision would remain identical if an individual's **protected attributes**, such as gender or race, were hypothetically altered. In contrast, adversarial training employs a **discriminator model** to identify and strip away sensitive information from the main system's internal data representations. By forcing these two components to work in opposition, the process successfully **minimizes information leakage** regarding sensitive traits. Ultimately, these techniques aim to produce outcomes based strictly on **relevant medical factors** rather than discriminatory data patterns.

Keywords: Bias mitigation strategies, Counterfactual fairness, Adversarial networks, Algorithmic fairness, Protected attributes

Content

In the previous course, we delved into some essential bias evaluation strategies such as removal of protected attributes, classification, parity, and calibration. Those are three fundamental aspects used to assess biases and algorithmic fairness in the context of AI and medicine. As the field of algorithmic fairness continues to evolve, other approaches are gaining traction. One such approach is counterfactual fairness. Counterfactual explanations are human interpretable explanations that provide insight into why a particular model made a specific prediction or decision. These explanations are counterfactual because they describe a hypothetical scenario in which the model's prediction would have been different. Counterfactual fairness focuses on the idea of fairness in terms of whatif scenarios. It aims to assess whether a decision would remain fair if the characteristics of an individual such as their protected attributes, race, gender, or ethnicity were different. Let's take an example of evaluating a classifier for severe depression following a cancer diagnosis. Counterfactual fairness involves ensuring the model's predictions would remain fair even if the protected attributes were different. Suppose you have a machine learning model Designed to predict the likelihood of severe depression in individuals following a cancer diagnosis. The model considers various features such as medical history, treatment plans, and social economic factors. Now, imagine two individuals, Alex and Bob, were both diagnosed with cancer at the same time. The model predicts that Alex has a higher likelihood of severe depression, while Bob is unlikely to experience severe depression following his cancer diagnosis. Counterfactual fairness would explore how the model's prediction might change if we were to alter certain features that were not directly related to the individual's depression, but might influence the prediction unfairly. Alex was identified as a female in the original prediction. If we hypothetically change Alex's gender to male while keeping all other features the same, does the model's prediction for severe depression following a cancer diagnosis remain the same? If the model's predictions remain the same after the change, then the model is considered counterfactually fair with respect to gender. However, if changing the gender leads to significant changes in predictions, it suggests that the model may be incorporating biases related to gender. A counterfactual explanation provides the smallest alteration required in the feature values to change the prediction. In this model agnostic approach, we work solely with the model's inputs and outputs and we do not delve into the model architecture. Another approach is adversarial networks. Adversarial networks are used to reduce bias by introducing a fairness adversary into the training process. This adversary tries to identify and rectify any discrimination in the model's predictions. The primary goal of adversarial fairness is to make models less sensitive to sensitive attributes such as race or gender when making predictions. thus promoting fairness and equity. In adversarial fairness, a specific type of adversarial neural network known as a discriminator or adversary is introduced during model training, which is a specific type of neural network. The discriminator's role is to assess how much information about sensitive attributes are present in the model's hidden representations and whether the model is inadvertently using sensitive attributes to make predictions. This is done by examining patterns, correlations, or data elements that may be indicative of sensitive attributes. The primary goal of this adversarial training is to minimize the information leakage regarding sensitive attributes by making the model’s hidden representations less reliant on the attributes. The model becomes less sensitive to them, which can help reduce biases and discriminations in its predictions. Adversarial fairness typically involves a two-step process. In the first step. The main model, for example, a classifier, makes predictions based on input data. In the second step, the discriminator model predicts sensitive attributes from the hidden representations produced by the main model. During training, the adversarial loss function is used to minimize information that can be used to predict sensitive attributes. The main model and discriminator are trained in opposition with the main model trying to minimize the adversarial loss. while still achieving good predictive performance. Through adversarial training, the model learns to make predictions based on medical factors rather than potentially sensitive data.

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42. Module 7 - Deploying AI into Healthcare Settings.mp4

Summary

This text explores the complicated transition of **generative AI from experimental technology to practical medical application**, highlighting its potential to revolutionize administrative and clinical efficiency. While large language models offer remarkable capabilities in areas like **documentation and decision support**, their integration is hindered by significant **ethical and regulatory hurdles** regarding data privacy, algorithmic bias, and the lack of clear legal accountability. The author argues for a **specialized regulatory framework** that treats generative AI as a unique category, distinct from traditional medical devices, to better manage its rapid evolution. Ultimately, the source emphasizes that AI should serve as a supportive tool rather than a replacement, ensuring that **humanistic care and the physician-patient relationship** remain the cornerstone of medicine.

Keywords: Generative AI deployment, Clinical documentation, Ethical framework, Regulatory challenges, Data privacy security

Content

We have discussed the array of breakthroughs in AI development over the past year with generative AI showcasing expertise in tasks ranging from writing poems to solving complex math problems. However, the transition of such cuttingedge AI from impressive tech demos to practical deployment in healthc care settings has proven challenging. Gener AI has gained widespread popularity, particularly in the medical community's interest in using off-the-shelf large language models provided by technology companies. These models, including chat bots, operate by learning word sequence probabilities on a massive scale with billions of parameters. While initially trained to predict the next word in a sentence, they demonstrate capacities beyond this, such as summarizing text and answering questions without explicit training. In medicine, models like chat GPT powered by generative pre-trained transformers show remarkable capacity in clinical documentation, decision support, information technology operations, medical coding, and patient physician communication. Generative AI impacts tasks like simplifying radiology reports, extracting information from medical records, answering patient queries, and summarizing complex medical dialogues. Despite its potential, stakeholders must consider ethical dimensions. Patient privacy, data security, and transparency and data utilization are paramount. Addressing biases and training data is crucial to prevent disparities in the impact of AI on different demographic groups. Continuous monitoring and mitigation of biases in real world applications are essential. Establishing an ethical framework prioritizing patient welfare Transparency and equity is indispensable for responsible use of AI in medicine. The regulatory challenges associated with generative AI, particularly large language models within the healthcare domain are intricate and multifaceted. The sheer scale and complexity of large language models trained on massive data sets with billions of parameters introduce difficulties related to interpretability, fairness, and unintended consequences. Adding to this complexity is the use of tokens in large language models which may not be adequately addressed by existing healthcare regulations. Additionally, the widespread applicability of large language models across diverse domains necessitates a regulatory framework adaptable to industry specific concerns with a particular emphasis on health care due to its high stakes. Real time adaption capacities, societal impact and the dynamic nature of the technology further complicate regulatory oversight demanding continuous monitoring and evaluation mechanisms. Issues related to data privacy and security amplified by the extensive training data requirements of large language models underline the need for robust frameworks to safeguard sensitive information. The legal implications of generative AI providing medical advice raise substantial concerns as AI lacks the legal status of a human. being leaving humans as the ultimate duty bearsers. The issue of accountability in cases of harm to patients introduces complexity. Should it lie with the patient, the treating hospital or open AI?

Comprehensive legal frameworks and guidelines are urgently needed to clearly define and allocate responsibility for AI use in healthcare. While open AI has taken initial steps with security standards and guidelines, the lack of mandatory measures combined with Open accessibility to many public generative AI models highlights the necessity for clear regulations to protect users. Privacy concerns in using generative AI within healthcare settings are paramount. The collection, storage, and processing of sensitive patient information raise concerns about unauthorized access, data breaches, and reidentification risk. To address these issues, healthcare organizations must implement robust security measur measures, including encryption, access controls, and regular vulnerability assessments. Transparent communication with patients about data usage, informed consent, and a clear privacy policy are essential to maintaining trust. Furthermore, regular monitoring and auditing of system performance and data processing methods are crucial to identify and address any issues regarding privacy. The integration of generative AI in healthcare demands a humanistic, ethical approach emphasizing compassion and individualized care. Health care professionals must prioritize patient well-being, ensuring that AI generated recommendations complement rather than replace empathetic support and tailored treatment plans. The physician patient relationship remains central, requiring health care professionals to use large language models as a tool to enhance their expertise while preserving human connection and trust. Upholding Integrity in disclosing AI involvement, accurately representing AI's capabilities and limitations, and maintaining honest communication with patients are essential components of humanistic ethics. Effectively addressing these challenges requires a thoughtful regulatory approach. It is imperative to establish a new regulatory category specifically tailored to generative AI, recognizing its distinct characteristics compared to traditional AI based medical technologies. Differentiating between large language models trained on medical data and those for non-medical purposes is essential to ensure tailored oversight. Adopting a regulatory approach akin to the FDA's digital health pre-certification program, which focuses on companies developing large language models rather than individual iterations, can streamline the oversight process, allowing it to keep pace with a rapid evolution of these techn technologies. Crucially, regulators must strike a balance between fostering innovation and ensuring the responsible and ethical use of generative AI in healthcare.

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43. Module 7 - Dismantling Race-Based Medicine.mp4

Summary

To dismantle the systemic bias inherent in race-based medicine, the text proposes a multi-layered roadmap centered on **diversifying clinical trials** and evolving beyond narrow demographic categories. This transformation moves through three stages: personalized medicine for group-based accuracy, precision medicine for genetic nuances, and finally, **precision care**, which integrates a patient's unique **social determinants of health** and personal values. By leveraging **AI-driven insights** to analyze complex environmental data rather than oversimplified racial assumptions, the medical field can shift toward a more equitable model of healing. Ultimately, this reform requires a combination of **standardized data collection** on social factors and robust professional education to address the root causes of health disparities.

Keywords: Clinical trial diversity, Personalized medicine, Precision care, Healthcare equity, Social determinants

Content

We have discussed different debiasing approaches for AI and healthcare. However, to address the deeply rooted issue of biased AI and race-based medicine in the US healthcare system, a comprehensive roadmap for reform is required. The first critical step involves increasing the diversity of clinical trial populations. Historically, certain populations, including racial and ethnic minority parties have been under reppresented in clinical trials leading to skewed guidelines and algorithms. To dismantle race-based medicine, it is imperative to ensure trial data includes diverse populations, providing insights beyond race and ethnicity. Actively involving minority communities in trial design, fosters trust and participation in medical research, contributing to more representative and equitable healthcare practices within the ever evolving landscape of healthcare. Fostering Equity becomes a cornerstone and the integration of three pivotal dimensions. Personalized medicine, precision medicine, and precision care become instrumental in advancing a health care system that prioritizes fairness and inclusivity. Personalized medicine addressing disease and demographic variations seeks to provide an accurate approach for the prevention, diagnosis, and treatment of diseases. This involves identifying which approaches or treatments will be effective for specific groups of individuals with similar characteristics. Precision medicine delves deeper, focusing on genetic distinctions for a more nuanced treatment strategy. Precision medicine aims to offer treatments adapted to the biological characteristics of the patient. Yet, to truly cultivate equity in health care and prioritize equitable AI, a paradigm shift is needed and precision care serves as the beacon for this transformation. Precision care adopts a holistic approach to care that considers not only the patients demographic and biological characteristics, but also the patients care goals given their multiple morbidities, functional status, values, and social contexts. Precision care emerges as a powerful instrument in dismantling health care disparities by aligning treatment options with the patients care goals and deliberately addressing social determinance of health. Precision care becomes a catalyst for reducing systemic inequalities. It aligns with a broader societal objective, eradicating race-based medicine by acknowledging and rectifying the broader factors that contribute to health disparities. The integration of AI into precision medicine further enhances its potential, enabling the analysis of vast data sets to uncover patterns and insights that may have been previously overlooked. AIdriven precision care not only facilitates more accurate diagnosis and treatment, but also contributes to a more comprehensive understanding of the diverse factors influencing health outcomes, ultimately aiding in the quest for health care equity. Promoting education and awareness is another crucial element of reform. Rejecting oversimplified assumptions about race and health requires a deep understanding of the complex social, economic, and environmental factors contributing to health disparities. Education efforts should target health care professionals, researchers, and the public, emphasizing the role of racism and discrimination in shaping health outcomes. Finally, to enable culturally responsive care, guidelines and policies are needed to enhance collection of comprehensive data on environmental factors affecting health outcomes. Both public and private sectors should focus on developing standardized data collection protocols and integrating social determinance of health. into medical records during routine patient encounters. Access to patient specific data on social determinants empowers healthcare providers to recognize and address the unique needs of diverse patient populations, fostering a more inclusive and equitable health care system.

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44. Module 7 - Exploring Potentials and Ethical Quandaries.mp4

Summary

This source examines the dual nature of **generative AI in healthcare**, highlighting how models trained on billions of parameters offer revolutionary capabilities while introducing significant **ethical risks**. While these systems can create synthetic data to protect privacy or act as **corrective tools to counteract historical biases**, they also threaten to amplify existing social inequities if their training data is flawed. To ensure these powerful tools remain **fair and transparent**, the text advocates for a rigorous framework of **human-in-the-loop evaluations** and strategic prompt engineering. Ultimately, the author emphasizes **adversarial training** as a proactive defense, using challenging inputs to penalize biased outputs and cultivate more **robust, equitable medical technologies**.

Keywords: Generative AI, Healthcare data privacy, Bias mitigation strategies, Human evaluation, Adversarial training

Content

Transitioning to generative AI, we delve into a domain of AI that learns data patterns to generate new data. In healthcare, imagine the potential of creating synthetic medical records for research without compromising patient privacy. These models represent a significant departure from traditional machine learning methods, marked by their unprecedented scale, complexity, and transformative potential. These generative AI models are trained on massive data sets with billions of parameters. That's right, billions. And this presents unique challenges for their use in the healthcare setting. The scale and complexity demand attention to issues of interpretability, fairness, and unintended consequences. In addition, these models necessitate substantial computational resources due to their large scale and the need for fine-tuning. While this holds promise for exciting opportunities, generative AI isn't without its challenges. It absorbs data holistically, biases included. If the foundational data harbors racial or historical biases, generative AI can inadvertently reinforce them. If our training data predominantly represent a certain group over others, these models may be less accurate or even potentially harmful for the underrepresented groups. Yet, there's an exciting opportunity here, and that is to use generative AI as a corrective tool. If we identify biases in our health care systems, we can train generative AI models to counteract these biases, effectively using AI to correct human errors of the past. By doing this, we can work towards a more equitable health care system where treatments and diagnoses are tailored for everyone irrespective of their racial or ethnic background. In general, we use the same bias mitigation strategies for generative AI as we do for other AI approaches. This includes validation on independent data sets, quality checks, diversity assessments, and human in the loop evaluations. Especially for platforms like chat, GPT, human evaluations are indispensable. By having real users interact with the model and then assess its outputs, we obtain qualitative insights that quantitative metrics might overlook. For generative AI, We must also consider prompt engineering which refers to the creation of specific interactions or queries prompts to guide the behavior of language models especially in the context of generative AI and tools such as chat GPT and MedPal. It is a technique used to elicit desired responses from the model by providing input in a structured way. While prompt engineering can be a powerful tool for obtaining desired results, it also raises ethical considerations. Biases can be inadvertently introduced through the language, choices, and prompts, and careful attention is needed to ensure fairness and avoid unintended consequences in the model's outputs. To mitigate biases in generative AI, adversarial training is necessary. While we have already learned about adversarial networks, in generative AI, adversarial training is the use of adversarial prompts to expose biases in AI responses. By Constructing adversarial prompts that target biases and vulnerabilities, we challenge the model during the training process. These adversarial examples are integrated into the training data set, enabling the model to learn from both standard and challenging inputs. The key lies in defining an adversarial loss function that penalizes biased outputs, guiding the model to minimize these undesired behaviors. As we iterate through training, regularly eval valuating performance on a mix of standard and adversarial examples, the model becomes more robust. This iterative process ensures the model's ability to handle diverse inputs, ultimately reducing its susceptibility to biases. Adversarial training stands as a proactive and effective approach, paving the way for AI models that are not only powerful, but also fair and ethically sound.

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45. Module 7 - Introduction Navigating the Intersections of AI and Medicine.mp4

Summary

Modern medicine is entering a transformative era driven by **vast data availability**, sophisticated **generative AI technologies**, and shifting public awareness. While these innovations offer immense potential, the text emphasizes that progress must be guided by a robust **regulatory framework** to ensure safety and legal compliance. Central to this discussion is the urgent need to address **ethical considerations**, specifically by mitigating biases and promoting **health equity** within clinical applications. Ultimately, the material serves as a roadmap for navigating this complex landscape, encouraging a **comprehensive understanding** of how to balance technological power with transparency and social responsibility.

Keywords: AI in healthcare, Generative AI, Health equity, Regulatory environment, Ethical considerations

Content

It’s an exciting time to be engaged in the world of artificial intelligence, particularly in the context of healthcare. We’ve witnessed remarkable transformations in this field and these changes can be attributed to several key drivers. Firstly, the availability and utilization of data have soared. Secondly, technological advancements have reshaped the landscape and among the notable contributors in our domain are generative AI technologies such as chat, GPT, MedPal and others. Thirdly, public opinion and awareness have played an integral role in shaping the direction of AI for healthcare. However, it is imperative that we establish the right metrics to assess these tools and equally important safeguard against perpetuating biases and reinforcing stereotypes. In this rapidly evolving landscape of AI for healthcare, a critical focus on fairness, transparency, and ethical considerations are central. Furthermore, the regulatory environment has significantly influenced the development, implementation, and utilization of AI in healthcare. In this context, the regulatory landscape plays a multifaceted role shaping the trajectory of AI powered solutions in healthcare. From ensuring complian With legal standards and quality benchmarks to safeguard against ethical concerns and promoting transparency, the regulatory environment forms an indispensable framework to ensure the responsible development and use of AI in healthcare. Today, I’m excited to bring you an update on current work in this space. This update material will illuminate the fascinating intersections of artificial intelligence and medicine as the realm of AI expands in healthcare. It offers a multitude of opportunities, but it’s imperative to critically engage with its implications, especially in the context of equity and medicine. This session aims to familiarize you with the evolving concepts of AI and health equity with a focus on generative AI. We will explore its potential to enhance health care while addressing the challenges associated with its safe deployment in the health care sector. As we walk through this material, I aim to provide you with a comprehensive understanding of the play between AI and healthcare, fostering critical thinking and engagement with the evolving landscape of this transformative field, providing you with valuable insights and skills.

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46. Module 7 - Lifecycle of AI.mp4

Summary

Advancing ethical healthcare technology requires a **holistic lens** that views the **AI life cycle** as a continuous journey through five distinct phases: data creation, acquisition, model development, evaluation, and deployment. Rather than focusing on isolated performance metrics, developers must prioritize **transparency and fairness** at every stage to prevent data biases from harming marginalized populations. The framework emphasizes that robust **generalizability** and the integration of **MLOps principles** are essential for moving algorithms from theoretical research into safe, large-scale clinical operations. Ultimately, the purpose of this comprehensive approach is to shift the industry toward **standardized reporting** and rigorous oversight, ensuring that medical AI is both equitable and effective across diverse real-world settings.

Keywords: AI life cycle, Ethical AI algorithms, Model generalizability, Data diversity, MLOps framework

Content

I want to start with an overview of the AI life cycle. A key aspect of advancing ethical AI lies in the comprehensive understanding of AI algorithms. It extends beyond mere performance metrics encompassing the complicated interplay of actors, processes, and objectives driving development and application. The AI life cycle has five phases. Data creation, data acquisition, model development, model evaluation, and model deployment. The AI life cycle framework offers a holistic lens to assess the ethical dimensions of clinical AI algorithms. In the past, efforts often focused on a single phase of the AI life cycle, but our approach underscores the imperative to examine their collective impact. Singularly focusing on a phase neglects the nuanced interactions and fails to address overarching issues like transparency, fairness, and potential harm that manifest differently across stages. Let’s walk through the different phases of the AI life cycle. In the data creation phase, a crucial aspect of developing clinical AI algorithms involves a comprehensive understanding of the origins and circumstances surrounding the generation of data used for training and evaluating the algorithm. The risk algorithms poorly serving diverse populations arises from unrepresentative data sets used for training algorithms. Therefore, it is essential to recognize that many data sources used for AI development such as academic medical records lack diversity in soio economic status, geography, and racial and ethnic representation. As the amount, quality, and depth of data can vary hugely depending on the source of origin, there must be a fun fundamental shift towards transparency in AI development. The metadata accompanying AI data sets should include information about data creation detailing the original purpose of the data, areas of representation, and initiatives undertaken to enhance diversity. The data acquisition stage involves the procurement and consolidation of data for model creation. During this stage, AI developers actively seek data sources that align with the requirements of their models. However, convenience often takes precedence over appropriateness, leading to the use of data that may not be the best fit for the intended purpose, incomplete data collection, redactions, and insufficient information on financial and privacy protections which hinder one's ability to assess the fitness of data sources. This calls for expanded regulatory and transparency requirements, which should include sharing essential demographic information on training data. and the logistical details of data acquisition to ensure the development of AI tools adhere to ethical standards. As we progress to model development, it is imperative to report algorithmic design details and optimization decisions which we discussed in earlier sessions. Optimization decisions, if not carefully managed, can inadvertently perpetuate discrimination and impede marginalized populations from benefiting from the algorithm. For example, Effectively managing imbalanced data sets and addressing skewed classes becomes a pivotal consideration in the process of model development. Imbalances occur when certain classes within a data set are under reppresented compared to others, potentially leading models to favor the majority class and neglect crucial patterns within the minority classes. Addressing this challenge involves employing sophisticated techniques such as oversampling the minority class, undersampling majority class or leveraging advanced algorithms designed to handle imbalances such as the synthetic minority oversampling technique. Smooch technical specifications of AIA models along with fairness considerations should be made public facilitating scrutiny and replication of the model development process. Therefore, we advocate for the establishment of standardized reporting criteria such as miniar that we described in the previous course. Moving through the AI life cycle. In the model evaluation phase, we assess a model's performance in the intended environment and understand its capabilities. Despite being an active area of research, AI bias and fairness evaluations currently take a backseat to the prevailing emphasis on measures for AI performance. In addition, there are limited assessments of how AI models might perform within clinical workflows, and this can mask potential societal harms once these models are deployed. Therefore, to ensure fairness and equity of AI models in healthcare, a comprehensive set of evaluation metrics is needed. The final stage of the AI life cycle is model deployment. And this phase remains significantly underststudied in the AI life cycle. Firstly, only a small fraction of AI models in healthcare progress to the deployment phase. And for those that do, the information about their success or failure in deployment is scarce. Secondly, there is a lack of incentives to systematically report practices in model deployment as this stage often moves the AI algorithm from research to operations. The scarcity of evidence contributes to a limited understanding of the ethical concerns that may arise once a model is deployed, particularly for low resource healthcare settings. Identifying standards and metrics to measure success becomes paramount in mitigating the potential for unintended consequences after Deployment machine learning operations or MLOps is a discipline concerned with the production, monitoring and maintenance of machine learning models at scale. The MLOps framework is widely adopted across the technology sector and derived from the established software development operations culture that embraces agile and continuous improvement. Principles of MLOps can inform the development of technical infrastructure to support the entire AI life cycle from feature selection to long-term model deployment and retraining and lead to better implementation. However, the consideration of MLOps principles is often lacking in an AI healthcare model deployment strategies in alignment with recent regulatory opportunities. Fore integration of MLOps principles can catalyze the adoption and scale of AI models for broad clinical impact. Model generalizability is paramount for deployment of AI model in the health care system. Yet, a critical gap often exists in assessing the generalizability of these models across different settings. While journals, funding bodies, and regulatory entities offer some guidance on generalizability requirements, a standardized and comprehensive definition of this concept remains elusive, leading to potential oversightes in evaluating a model's accuracy and safety in new contexts. To address this issue, I would like to point out levels of generalizability each associated with distinct objectives, validation methods, and stakeholders. First, internal generalizability focuses on the reproducibility within data not used in the training of the model. Internal validity checks the validity of the training process and tests for overfitting. Temporal generalizability ensures a model's performance over time at the development site. Next is external Generalizability. Here, geographical variation assesses a model's transferability to different institutions and locations. And domain generalizability gauges its adaptability to populations with diverse case mixes compared to the development population. Rather than universally aiming for broad generalizability, validation studies should tailor their approach based on the clinical prediction models characteristics and application scenarios. Researchers and developers should therefore adhere to established guidelines, justify their chosen validation strategy, report on a model's generalizability capacity, and provide disclaimers about the intended use and level of generalizability. This nuanced approach seeks to overcome vague notions of generalizability, facilitating the implementation of clinical prediction models while enhancing their comparability and minim izing risks associated with implementation.

47. Module 7 - Race-Based Medicine and Race-Aware Approach.mp4

Summary

The source examines the shift from **race-based medicine**—the problematic practice of using racial identity as a biological proxy in diagnostics—toward a more equitable **race-aware approach**. By analyzing clinical tools like the **EGFR for kidney function** and the **VBAC calculator for childbirth**, the text illustrates how outdated racial adjustments historically led to **delayed treatments and systemic disparities** for marginalized patients. Rather than simply ignoring race, the proposed solution advocates for **contextualizing health outcomes** by accounting for social determinants, structural inequities, and environmental factors. Ultimately, the material highlights that **revising medical algorithms** to remove racial bias is a critical step toward achieving **fairness and accuracy** in modern healthcare systems.

Keywords: Race-based medicine, Kidney disease diagnostics, VBAC risk assessment, Race-aware approach, Algorithmic fairness

Content

Let’s talk about race-based medicine. What is race-based medicine? Race-based medicine refers to medical practices that take a person’s racial or ethnic identity into account when diagnosing or treating medical conditions where there is no biological evidence to support these considerations. This approach has deep roots in the history of medicine with a focus on differences in disease prevalence among racial and ethnic groups. Here are two key examples. The first example is kidney disease and EGFR. EGFR or estimated glomeular filtration rate is a crucial metric in diagnosing and tracking kidney disease. Historically, it was adjusted for race, assuming that black patients naturally had higher creatinine levels. Unfortunately, this approach led to black patients being being diagnosed at later stages of the disease, often with more severe disease. Additionally, it had unintended consequences of reducing their chances of being placed on kidney transplant weight lists, highlighting the critical need to re-evaluate these racial adjustments in medical diagnostics. A second example is vaginal births after cescareian. The Vback calculator vaginal birth after cescareian while aiming to estimate the risks associated with VBAC had historically incorporated race into its risk assessment. It was based on the assumption that black patients face higher risk of adverse outcomes after a cacaian birth. As a result, the Vback calculator often overestimated the risk of adverse events following a cacaian birth for black women. This approach led to differential recommendations and limited treatment options for blacks and Hispanics, underscoring the need to reconsider the role of race in medical risk assessments and their consequences. Now, let’s turn our attention to emerging approaches in the field of ethical AI and algorithmic fairness in healthcare that address race-based medicine. A critical question arises, should we simply remove race as a variable from all algorithms? While the notion of raceblind algorithms may seem appealing, it poses challenges when accounting for legitimate differences in health outcomes among different racial and ethnic groups. So, it’s essential that we consider a more nuanced perspective, one that incorporates a raceaware approach. A raceare approach acknowledges the impact of social determinance, structural inequities, and systemic biases on health outcomes in AI algorithms. It doesn’t ignore race, but rather seeks to contextualize it. For example, it might mean that when considering healthcare outcomes, algorithms could incorporate information about social economic factors, access to health care, and environmental conditions which often correlate with race. So, it’s a careful balancing act. We want to ensure fairness and equity while still recognizing the reality of health disparities in our society. Let’s go back to our kidney function example. In a significant shift towards raceaware approach, the adjustment for EGFR for race was eliminated in 2021. fund. Simultaneously, the kidney allocation system was revisited to rectify the long-standing inequalities in kidney allocation. Notably, the revisions included the consideration of dialysis time, recognizing the historical reality that black and Hispanic patients spent more time on dialysis before transplantation. The impact was profound with decreased weight listing rates observed across all ethnic and racial groups. Most Significantly, this approach contributed to a substantial reduction in the disparity between black and white patients. This case serves as the compelling example of the potential of a raceaware approach to address systematic biases and promote equity in health care systems. In our second example, the VBA calculator recently underwent a substantial revision by removing race as an adjustment variable. The maternal fetal medicine network took an investigative approach to understand the inclusion of race within the Vback calculator. They found that race had been used as a proxy for chronic hypertension. The revised calculator free from racial adjustments has proven to provide more accurate risk assessments for all racial and ethnic groups, fostering a more equitable health care landscape. The ongoing commitment from the medical society to ensure that the calculator is widely and judiciously utilized helps to enhance healthare practices and reduce unwarranted disparities.